claude-windows / 05c32f87-3bfb-4a7f-861c-0050ac4169ac

Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac.jsonl`
- SHA-256: `fd93e95fc276693a13d9179742ea2317a0cddd4836b20791f6e04da2c70a792a`
- Source modified: `2026-07-07T17:39:21+00:00`
- Imported at: `2026-07-08T15:59:10+00:00`
- project: `C--Users-avidu-Projects-badminton-highlight-indexer`
- session_id: `05c32f87-3bfb-4a7f-861c-0050ac4169ac`

Transcript

1. user (2026-07-07T14:01:12.100Z)

On the hosted Cloud Run control-plane, a POST /api/process with compute="local" fails confusingly. Observed live 2026-07-07 (job f29d41d01fa54acb9e39784dc4b5cc98, "Video 1 - Badminton.MP4"): log showed `[COMPUTE] requested=local actual=in_process dispatched=None status=failed` and `WASB detector environment not ready: WASB weights not found at /opt/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar`.

Root cause: in backend/orchestration.py::_maybe_dispatch_remote (~line 448-449), `compute="local"` returns None ? the in-process segmenter runs on the CONTROL-PLANE. But the control-plane image has NO WASB weights (only the worker Cloud Run Job GCSFuse-mounts them at /gcs/models/...; WASB_WEIGHTS on the control-plane points at the image default /opt/models/... which is absent ? WASB-C3, weights are never baked). So wasb_hybrid in-process on the hosted control-plane can never succeed ? it's a guaranteed, confusing failure.

Two-part fix:
1. Backend guardrail: when the server is a hosted/cloud control-plane (deployment.compute_target=="cloud_run" and/or no readable local WASB weights) AND the request forces compute="local" for a detector that needs weights (wasb_hybrid / tracknet_hybrid / fusion_hybrid), fail FAST with a clear message ? mirror the existing compute="cloud"-not-available path (orchestration.py ~line 451-456: `_failed("cloud-serving was requested but this server isn't configured for it ...", False)`). Something like: "this hosted server can't run detection locally (no weights on the control-plane); choose 'Run in cloud'." Do NOT silently run in-process ? "weights not found".
2. SPA: confirm the compute-picker's DEFAULT on the hosted build (backend/ui). If it defaults to / offers "local" on a hosted deployment, change it so hosted defaults to cloud (leaving compute unset ? server default cloud_run, or explicit "cloud"). The pure default (ProcessRequest.compute=None) already dispatches correctly; the failure only happens when the SPA explicitly sends "local".

Acceptance: a compute="local" request on a cloud_run-configured server returns a clear, actionable failure (not "weights not found"), covered by a unit test in tests/test_api_cloud_dispatch.py; and the hosted SPA no longer sends compute=local by default. Repo pre-merge gate (full suite + ruff + mypy + coverage), branch from origin/master.

2. assistant (2026-07-07T14:01:19.602Z)

I'll start by getting oriented ? reading the session handoff and confirming the actual repo location, since memory notes the primary working dir is a stub.

3. user (2026-07-07T14:01:20.303Z)

```
1	---
2	name: actual-repo-location
3	description: "The badminton-highlight-indexer code lives under khelsutra-guru/, not the primary working dir stub"
4	metadata:
```

```
  node_type: memory
  type: project
  originSessionId: df5d5fb2-534a-447a-8181-b222f94655f3
---

The primary working directory `C:\Users\avidu\Projects\badminton-highlight-indexer` is a near-empty
**stub** ? it contains only `AGENTS.md`, an empty `.claude/`, and `output/nightly_regression/`
(a `LATEST_STATUS.txt`). The actual `khelsutra-guru/badminton-highlight-indexer` codebase lives at:

  `C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer`

(git remote `github.com/Khelsutra/badminton-highlight-indexer`, default branch `master`).

**How to apply:** when a task references this repo's code (backend/worker, pipeline/segmenters,
config.json, run_all_tests.py, etc.), `cd` to the khelsutra-guru path and work there ? the stub dir
has none of it. Confirmed 2026-07-06 while fixing the wf294 silent-detector-failure issue.

Repo conventions verified there: CI = ruff 0.15.16 (`ruff check .`), mypy 2.1.0 (`mypy backend
training`; `mypy.ini` quarantines several modules incl. `backend.worker.__main__`, `gemini`,
`yolo_hybrid` via `ignore_errors=True`), full suite via `python run_all_tests.py`, coverage floor
`--cov-fail-under=70` (actual ~86%). Segmenters use a `Success`/`Failure` Result contract
([[efficient-timeouts-and-retries]] is unrelated).

```

4. assistant (2026-07-07T14:01:23.651Z)

Now let me read the session handoff and pull the real repo. Let me do these in parallel.

5. user (2026-07-07T14:01:24.687Z)

```
---
name: session-handoff-gpu-resident
description: "Start here to pick up the GPU-resident WASB rollout (issue #506) ? implementation PR #507 is open; golden gate next"
metadata:
  type: project
---

# Session Handoff ? GPU-resident WASB inference (rollout)  ·  issue #506 · PR #507 MERGED

## SHIPPED 2026-07-07 ? GPU-resident NVDEC decode is LIVE in cloud-serving
Full arc done: PR #507 (feature) + PR #508 (cloud-serving preset default = nvdec_resident) both merged to master (44a2f47, 143dbb9). The `rally-worker` Cloud Run Job (asia-southeast1) runs the new image `rally-worker:latest` `sha256:465a9250?` (torch 2.6 + PyNvVideoCodec 2.1.0 + baked NVDEC libs) with `WASB_DECODE_BACKEND=nvdec_resident` set ? **nvdec is the live serving default**, validated on the real L4 (GX010128 served trajectory reproduced the VM: 26479?26481 visible; cpu+nvdec+canary smokes all ?). Record: `docs/COMPUTE_DECOUPLED_SERVING/runlogs/DEPLOY_REPORT_cloud-serving_gpu-resident-nvdec_2026-07-07.md`.
- **ROLLBACK levers:** `gcloud run jobs update rally-worker --region asia-southeast1 --remove-env-vars WASB_DECODE_BACKEND` (? torch-2.6 cpu path, same image); or `--image ?/rally-worker@sha256:3bfefaf6?` (? prior torch-1.11 image).
- **PRs:** #507 (feature) + #508 (preset default) + #509 (dispatch forwarding + release
```

hardening) all merged. master @ 66756ad.
14 - **DONE ? control-plane + worker both on image D2 `sha256:09d2edf7` (built from master 66756ad = #507+#508+#509).** Redeployed 2026-07-07: `gcloud run services update rally-control-plane --image ?:latest` ? rev **00017-qbd** (100% traffic, booted healthy: uvicorn up, cloud-serving profile ACTIVE, edge-auth on); `gcloud run jobs update rally-worker --image ?:latest`. So the #509 dispatch **forwarding** + #508 preset default are now LIVE in the control-plane image. NB the FIRST control-plane deploy this session used the stale `465a9250` (built from 44a2f47, pre-#508/#509) ? rev 00016; superseded by 00017. Rollback: `gcloud run services update-traffic rally-control-plane --to-revisions rally-control-plane-00015-kp9=100` (last pre-nvdec rev) or `--to-revisions ?-00016-?` .
15 - **`WASB_DECODE_BACKEND=nvdec_resident` env is still set on the worker Job** ? now REDUNDANT with the forwarded --set + preset (both say nvdec, env just wins), kept as belt-and-suspenders + manual override. Safe to remove ONLY after a real /api/process dispatch confirms the forwarding path live (needs a CF Access JWT ? not verifiable this session; forwarding is deployed + unit-tested but not live-dispatch-verified).
16 - `DEPLOY_REPORT_?gpu-resident-nvdec_2026-07-07.md` is current: PR #510 added the §5 "Update" (D2 redeploy of control-plane rev 00017 + worker) and corrected §6. All four PRs merged: #507 #508 #509 #510; master @ 3be0935.
17 - **Wrinkles fixed (#509):** dispatch now forwards indexing.wasb.{decode_backend,batch_size,prefetch_batches} (worker has no DEPLOYMENT_PROFILE ? never applied the preset; batch tuning was silently lost too); added `.gcloudignore` (Cloud Build had none ? relied on the .gitignore fallback to skip 8 GB output/scratch); added `deploy/cloudrun/post_deploy_smoke.ps1`. See [[gpu-vm-setup-gotchas]].
18 - WASB-C3 weights-license gate still open (unchanged; pre-existing).
19
20 ## (historical) release/rollout phase ? superseded by SHIPPED above
21 PR #507 squash-merged to master as **44a2f47** (branch deleted). Pre-merge gate reproduced locally: full suite 1764 passed / coverage 85.02% / ruff+mypy clean; CI 11/11 green; 2 optional review follow-ups landed (PyNvVideoCodec==2.1.0 pin in both deploy files; decode_backend persisted in native-runner detection_params). Real-L4 validation + golden gate PASS + latency/resource A/B all done (see PR #507 comment + below). **Remaining = the serving release (image rebuild + gated nvdec flip); decode_backend still defaults cpu so nothing in prod has changed yet.**
22
23 ### Serving release plan (live infra: Job `rally-worker` + service `rally-control-plane`, asia-southeast1; ONE image `rally-worker:latest` drives both ? gen_service_yaml.py:193)
24 - **Phase A (ship image, no behavior change):** `gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project gen-lang-client-0306026826 .` ? new `rally-worker:latest` (torch2.6+pynvc2.1.0+NVDEC libs). Push does NOT change prod (Job/service pin a digest). Cut over via `gcloud run jobs update rally-worker --region asia-southeast1 --image ?:latest` + redeploy service (`gen_service_yaml.py` ? `services replace`). decode_backend defaults cpu ? cpu path under torch2.6 (validated F1 0.576?0.582). Smoke 1 clip. Rollback: `jobs update --image <old digest>`.
25 - **Phase B (enable nvdec, gated + reversible):** flip via WORKER env `WASB_DECODE_BACKEND=nvdec_resident` (native runner reads env first ? no code/preset change, instant rollback by removing it). ? THE untested risk: NVDEC on the real Cloud Run L4 ? the image bakes libnvcuvid 580.159.03; Cloud Run's host-driver ABI is unverified there (only tested on a self-managed L4 VM). MUST smoke 1 clip on the Job before trusting it. Then canary; then durable default = add `decode_backend: nvdec_resident` to the cloud-serving preset (presets.py) + rebuild + Launch Report.
26 - Cost: build ~$0.3; each Job smoke ~$0.4?0.7 (L4). Rollback levers at every step.
27
28
29 ## You are here
30 The implementation is DONE and **PR #507 is open** (branch `feat/506-gpu-resident-nvdec`, 2026-07-07): default-OFF `indexing.wasb.decode_backend = "cpu" | "nvdec_resident"`, the validated NVDEC?affine-`grid_sample`?GPU-normalize path wired into `backend/pipeline/detectors/wasb_infer.py` (`NvdecResidentFrontend` + `run_video_nvdec_resident`, same resumable cache keyed by decode_backend), native-runner threading + healthcheck gates (CUDA + PyNvVideoCodec), and the env bump (wasb env ? py3.10 + torch 2.6.0+cu124 + PyNvVideoCodec; libnvcuvid/libnvidia-encode staged in `deploy/cloudrun/Dockerfile` + `deploy/digitalocean/gpu_setup.sh`). Verified locally: 1764 tests green, coverage 85% (floor 70), ruff/mypy/link/leak clean; a 14-agent adversarial review confirmed 7 minor findings, all fixed (notably `padding_mode="zeros"` = cv2 BORDER_CONSTANT for non-16:9 sources). Research evidence unchanged: [[gpu-resident-modernization]], issue #506

comments (F1 0.574 ? baseline 0.582 on GX010128 at SERVED, ~1.7×, GPU 47?79%).
31
32 ## Real-L4 validation ? DONE + golden gate PASS 2026-07-07 (this session, ~$3 of the $100 budget; VM torn down)
33 Validated end-to-end on an L4 (`rally-506`, torn down clean, no orphan disks) ? full evidence in the PR #507 comment. All GREEN: (1) `gpu_setup.sh` builds the env clean (py3.10 + torch 2.6.0+cu124 cuda True + PyNvVideoCodec 2.1.0 + NVDEC lib extraction); (2) on-box **parity gate PASS** (`pytest -k parity`, 480s ? needed test-name fix 9267255, the documented `-k parity` selected 0 tests); (3) crash/resume byte-identical; (4) cpu?nvdec cache isolation correct; (5) **Cloud Run image builds + `--gpus all` smoke PASS** (baked NVDEC libs decode; native healthcheck ok with nvdec_resident); throughput 43.7 fps / GPU 82% on GX010128.
34 **Golden gate @ SERVED = PASS** (10 baseline-comparable clips, 445 rallies): mean ?f1@0.5 **+0.020**, mean ?tolF1 +0.006, worst ?0.009 (**GX010128 must-pass: 0.574 vs 0.582 ?**), 8/10 improved-or-flat. The other 5 golden clips are 4K-sourced (baselines are 1080p) so a vs-baseline number would conflate resolution with decode ? the parity gate already covers nvdec?cpu on identical pixels, so they add nothing to the verdict.
35 **Latency + resource A/B** (2nd L4 `rally-506b`, torn down; GX010128, batch 8, telemetry in `gs://?nightly/dev/506/results/bench/`): nvdec **43.7 fps @ GPU 83% / CPU 30%** vs cpu-stream(no-prefetch) 15.0 fps @ GPU 28% / CPU 49%; GPU mem +170 MiB (of 24GB), RAM ~2?3 GB flat. The win is the **bottleneck flip (GPU?, CPU?)**; the 2.9× is vs the untuned streaming path ? vs prefetch-tuned cpu the realistic end-to-end is ~1.6?1.8× (#506 spike). All in the PR #507 comment.
36
37 ## Staged assets (durable)
38 - **Golden video working set**: all 14 originals in `gs://khelsutra-rally-corpus/videos/golden/<name>.mp4` + the 4 baseline-source 1080p proxies in `videos/golden_1080p/` (mahadevpura_2, Badminton_BXH_2, Boxhill_Doubles, GX020094). Frame-count-verified against each golden trajectory. (GOLDEN_VIDEOS.md could record this GCS location in a follow-up ? kept out of PR #507 to not muddy its scope.)
39 - Gate trajectories: `gs://khelsutra-rally-nightly/dev/506/results/golden/*_nvdec.csv`. Re-score with repo-root `score_golden_gate.py` (untracked dev tool, like `score_local.py`; labels from `?corpus/labels/`, baselines from `?corpus/trajectories/`).
40
41 ## Next steps
42 1. **Merge PR #507** ? real-L4-validated + golden-gate PASS; CI green locally.
43 2. **Flip the default**: owner Launch Report ? set `decode_backend=nvdec_resident` in the cloud-serving preset (`backend/config/presets.py`). The gate that gated this is now green.
44
45 ## Ramp-up kit
46 - PR #507 (full change description + verification record) and issue #506 (all comments ? measured evidence).
47 - New tests: `tests/test_wasb_nvdec_resident.py` (grid/cv2 parity incl. non-16:9, cache keying, CLI guards, the skippable on-box parity gate).
48 - Reference code still in `scratch/gpu_resident/` (`exp_t2_full.py`, `score_local.py`, `t2setup_env.sh`, `telemetry.sh`).
49
50 ## Key decisions (don't relitigate)
51 - GPU resize MUST be affine `grid_sample` matching `cv2.warpAffine` ? NOT `F.interpolate` (F1 0.58?0.07; only shows on hard clips). Locked by unit tests now.
52 - `padding_mode="zeros"` (cv2's BORDER_CONSTANT(0)), not "border" ? matters only for non-16:9 sources; identical to the validated run on 16:9.
53 - PyNvVideoCodec (MIT) for GPU decode; TorchCodec couldn't load on the DL image.
54 - Score with SERVED preset (cap6+R4); raw (0.02,2.0,2.0) is degenerate.
55 - Default-OFF; nvdec/cpu detection caches never cross-replay (manifest keyed by decode_backend).
56 - WASB weights/NYCU-dataset license (WASB-C3) is a separate launch gate, not a perf blocker.
57

6. user (2026-07-07T14:01:28.731Z)

```
---BRANCH---
master
---STATUS---
?? score_golden_gate.py
```

```
?? score_local.py
---LOG---
3be0935 docs(runlog): DEPLOY_REPORT addendum ? control-plane redeployed to the forwarding image (#510)
66756ad chore(serving): forward WASB decode/tuning knobs to worker + release hardening (#509)
143dbb9 feat(serving): default cloud-serving to nvdec_resident decode (#508)
```

7. assistant (2026-07-07T14:01:35.408Z)

Repo is on `master` @ `3be0935`, clean except untracked dev tools. Let me confirm master is current with origin and read the key files.

8. user (2026-07-07T14:01:38.088Z)

```
        ``?video=<md5>`` lands on grounded state after any restart/redeploy wipe. True ? the DB
        has the video's rows now (re-query). Non-md5 ids return False without touching the
        bucket; every failure degrades to False (the read just reports what the DB has)."""
        vid = (video_id or "").strip().lower()
        if not _MD5_HEX_RE.fullmatch(vid):
            return False
        if not probe_process_cache(db, cfg, vid):
            return False
        return cached_process_result(db, cfg, vid, "") is not None


    def _maybe_dispatch_remote(
        cfg,
        db,
        req: "ProcessRequest",
        video_id: str,
        seg_name: str,
        val_results,
        play_wm: int,
        cand_wm: int,
        notify,
    ) -> Optional[Tuple[Dict[str, Any], bool]]:
        """Cloud-serving (COMPUTE_DECOUPLED_SERVING M6 ? the API path): when ``deployment.compute_target
        == "cloud_run"``, run the heavy DETECT on a Cloud Run GPU Job and INGEST its rally intervals into
        THIS DB, so the rest of the flow (query ? compile/local-stitch) is unchanged.

        Returns ``(response_dict, ran)`` when the cloud path is taken ? ``ran`` is whether the cloud job
        actually EXECUTED (``False`` for a pre-dispatch config rejection, so the observability report
        doesn't claim a fake segmentation run) ? or ``None`` to fall through to the in-process segmenter
        (**DEFAULT-OFF**: ``get_compute_backend`` returns ``None`` for every non-``cloud_run`` target, so
        the in-process path stays byte-identical)."""
        from backend.compute import ComputeJobSpec, get_compute_backend

        def _failed(msg: str, ran: bool) -> Tuple[Dict[str, Any], bool]:
            # Shape-match the in-process segmenter-fail branch + drop any partial NEW rows (id > play_wm);
            # prior good rows (id <= play_wm) are preserved (destroy-AFTER-confirm).
            db.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)
            # Provenance even on failure, and HONEST about what ran: `path` is "cloud_run" only when the
            # job actually dispatched + executed remotely (ran=True, e.g. a non-zero worker rc / ingest
            # error); a pre-dispatch reject (ran=False) ran NOWHERE ? "not_run". `target` records the
```

```
            # intended backend, `requested` the picker choice, `dispatched` whether the remote job ran.
            return (
                {
                    "video_id": video_id,
                    "status": "failed",
                    "message": msg,
                    "results": [r.model_dump() for r in val_results],
                    "play_segments": [],
                    "compute": {
                        "path": "cloud_run" if ran else "not_run",
                        "requested": req.compute,
                        "target": "cloud_run",
                        "dispatched": ran,
                    },
                },
                ran,
            )

        # SPA compute-picker (item 3): a per-request choice overrides the server's configured default.
        # "local" forces the in-process segmenter; "cloud" forces a Cloud Run dispatch (guarded by
        # _cloud_available so we fail clearly rather than silently running local); None / unknown ? default.
        choice = (req.compute or "").strip().lower()
        if choice and choice not in ("local", "cloud"):
            logger.warning(
                "[API] ignoring unknown compute=%r (expected 'local'|'cloud'); using server default.",
                req.compute,
            )
            choice = ""
        if choice == "local":
            return None  # explicit "run on my machine" ? in-process regardless of the server's target
        if choice == "cloud":
            if not _cloud_available(cfg):
                return _failed(
                    "cloud-serving was requested but this server isn't configured for it "
                    "(no Cloud Run target). Pick 'run on my machine', or configure cloud serving.",
                    False,
                )
            compute = get_compute_backend(cfg, target_override="cloud_run")
        else:
            compute = get_compute_backend(
                cfg
            )  # server default (default-OFF unless cloud_run)
        if compute is None:
            return None  # default-OFF ? run the in-process segmenter below (unchanged)

        storage_cfg = getattr(cfg, "storage", None)

        # The Cloud Run job fetches the input FROM THE BUCKET ? it can't read a path local to this box.
        try:
            input_ref = storage.resolve_input_ref(req.video_path, storage_cfg)
        except ValueError as e:
            return _failed(
                f"cloud-serving: unresolvable input '{req.video_path}': {e}", False
            )
        if input_ref.backend == "local":
            # A browser upload lands on THIS box's local disk; the Cloud Run job (a different machine)
```

```
            # can't read it. Stage it UP to the input bucket first, then dispatch from there (#461). Only
            # possible when a CLOUD input_root is configured; without one we genuinely can't stage ? fail.
            # Classify the ROOT ITSELF (parse_uri), not storage.input_backend: the staged URI below is
            # built by joining under input_root, and storage.put routes by parse_uri(staged_uri).backend ?
            # so an explicit cloud root (gs://?, s3://?) stages fine with input_backend left at its
            # default, while a bare/local-looking root can't stage no matter what input_backend says
            # (resolve_input_uri only consults input_backend when input_root is EMPTY, and an empty root
            # gives us nothing to build the staged URI under).
            input_root = (getattr(storage_cfg, "input_root", "") or "").strip()
            try:
                root_backend = (
                    storage.parse_uri(input_root).backend if input_root else "local"
                )
            except ValueError:  # unsupported scheme in input_root ? nowhere real to stage to
                root_backend = "local"
            if root_backend == "local":
                return _failed(
                    "cloud-serving needs a cloud input (e.g. gs://?), or a cloud storage.input_root to "
                    "stage local uploads into; a path local to this machine can't be read by the Cloud "
                    "Run job.",
                    False,
                )
            local_src = input_ref.key  # resolved local path of the upload
            staged_uri = f"{input_root.rstrip('/')}/{video_id}/{os.path.basename(local_src)}"
            try:
                notify("staging upload to bucket")
                # Thread the active storage settings so backend-specific config survives (S3 endpoint_url/
                # region/addressing for R2·DO·MinIO; GCS project/client) ? not a fresh default client.
                storage.put(
                    local_src,
                    staged_uri,
                    config=(
                        storage_cfg.model_dump() if storage_cfg is not None else None
                    ),
                )
            except Exception as e:  # noqa: BLE001 ? a staging failure is a clean job failure, not a 500
                logger.error(
                    "[API] cloud-serving: failed to stage %s -> %s: %s",
                    local_src,
                    staged_uri,
                    e,
                    exc_info=True,
                )
                return _failed(
```

9. user (2026-07-07T14:01:38.218Z)

3be0935 docs(runlog): DEPLOY_REPORT addendum ? control-plane redeployed to the forwarding image (#510)

3be09357cf0783f1971e92774d6e8cc6830ff7c1

3be09357cf0783f1971e92774d6e8cc6830ff7c1

10. assistant (2026-07-07T14:01:43.170Z)

Now let me find the key supporting pieces: `_cloud_available`, the `[COMPUTE]` log line, and how the detector is selected.

11. user (2026-07-07T14:01:44.852Z)

```
backend\api\job_worker.py:58:    ``deployment.compute_target=cloud_run``, OR the per-request
``compute='cloud'`` override path is
backend\api\job_worker.py:60:    orchestration._cloud_available so drain concurrency tracks the
ACTUAL remote-dispatch capability,
backend\api\job_worker.py:63:    if getattr(dep, "compute_target", "") == "cloud_run":
backend\api\job_worker.py:360:        getattr(getattr(config, "deployment", None),
"compute_target", "") == "cloud_run"
backend\compute\base.py:8:The seam is **default-OFF**: only ``deployment.compute_target ==
"cloud_run"`` selects a
backend\config\loader.py:200:            ct = merged["deployment"].get("compute_target")
backend\config\loader.py:203:                # An active profile with no explicit compute_target
(e.g. config.json blanked it
backend\config\loader.py:205:                # active profile's compute_target (it stays
meaningful for the M2+ ComputeBackend).
backend\config\loader.py:206:                merged["deployment"]["compute_target"] = canonical
backend\config\loader.py:209:                    "[CONFIG] deployment profile %r normally uses
compute_target=%r but "
backend\config\loader.py:216:                "[CONFIG] deployment profile ACTIVE: %s
(compute_target=%s)",
backend\config\loader.py:218:                merged["deployment"].get("compute_target"),
backend\config\loader.py:254:# local storage and the compute_target, so this stays the handful
of knobs a fresh, non-technical
backend\config\presets.py:30:# Each profile maps 1:1 to its canonical compute_target. Used to
apply the preset and to warn
backend\config\presets.py:31:# when config.json explicitly overrides compute_target into a state
inconsistent with the profile.
backend\config\presets.py:39:# that EXIST today ? M1 is pure/additive: the new
``deployment.{profile,compute_target}`` plus
backend\config\presets.py:45:        "deployment": {"profile": "truly-local", "compute_target":
"local_gpu"},
backend\config\presets.py:52:        "deployment": {"profile": "cloud-serving",
"compute_target": "cloud_run"},
backend\config\presets.py:75:        "deployment": {"profile": "cpu-mock", "compute_target":
"cpu_mock"},
backend\compute\__init__.py:4:``ComputeBackend`` ONLY when a non-local
``deployment.compute_target`` is selected, and ``None``
backend\compute\__init__.py:44:    """Return the compute backend for
``config.deployment.compute_target``, or ``None`` to run
backend\compute\__init__.py:47:    Only ``compute_target == "cloud_run"`` returns a non-None
backend (a ``CloudRunCompute``). Its
backend\compute\__init__.py:54:    server's configured default. ``None`` falls back to
``config.deployment.compute_target``.
backend\compute\__init__.py:59:        else getattr(getattr(config, "deployment", None),
"compute_target", "") or ""
backend\config\models.py:471:    compute_target: Literal["", "local_gpu", "cloud_run",
"cpu_mock"] = ""
backend\config\models.py:473:    # bucket-poll CONCURRENTLY when compute_target=="cloud_run".
Each is an I/O-bound Cloud Run Job
backend\compute\cloud_run.py:17:The dispatched container is forced to run **INLINE**
(``compute_target=local_gpu`` + native detector)
backend\compute\cloud_run.py:47:# (compute_target=local_gpu disarms the cmd_infer dispatch
guard; native = the L4 WASB runner.)
backend\compute\cloud_run.py:49:    "deployment.compute_target=local_gpu",
backend\main.py:235:    _cloud_available,
backend\main.py:268:    # default (deployment.compute_target). Only honoured for these two
values; see _maybe_dispatch_remote.
```

```
backend\main.py:401:            "compute_target": getattr(deployment, "compute_target", "") or
"",
backend\main.py:404:            "cloud_available": _cloud_available(cfg),
backend\main.py:1581:                    segmenter_actual=segmenter_actual,
backend\orchestration.py:158:def _cloud_available(cfg) -> bool:
backend\orchestration.py:168:    if (getattr(deployment, "compute_target", "") or "") ==
"cloud_run":
backend\orchestration.py:391:def _maybe_dispatch_remote(
backend\orchestration.py:402:    """Cloud-serving (COMPUTE_DECOUPLED_SERVING M6 ? the API path):
when ``deployment.compute_target
backend\orchestration.py:440:    # _cloud_available so we fail clearly rather than silently
running local); None / unknown ? default.
backend\orchestration.py:451:        if not _cloud_available(cfg):
backend\orchestration.py:632:        "[COMPUTE] cloud_run dispatched: execution=%s job=%s
region=%s out=%s video_id=%s",
backend\orchestration.py:793:        # COMPUTE_DECOUPLED_SERVING M6 ? API: with
deployment.compute_target=cloud_run, run the heavy
backend\orchestration.py:863:            "[COMPUTE] video_id=%s requested=%s actual=%s
dispatched=%s status=%s",
backend\orchestration.py:872:        # proof lives on the job result + the [COMPUTE] log, not
the report ? obsreport's record_run
backend\orchestration.py:886:            segmenter_actual=segmenter_actual,
backend\reporting\__init__.py:69:            segmenter_actual=segmenter_actual,
backend\worker\__main__.py:177:    container runs this SAME cmd_infer INLINE (compute_target
forced local_gpu) ? never re-dispatch.
backend\worker\__main__.py:239:    # M6: a cloud compute_target DISPATCHES to a Cloud Run Job
instead of running in-process.
backend\worker\__main__.py:242:    # dispatcher forces compute_target=local_gpu), so it never
recursively re-dispatches.
backend\worker\__main__.py:363:            segmenter_actual=segmenter_actual,
```

12. assistant (2026-07-07T14:01:49.957Z)

Let me read `_cloud_available` and the caller around line 790-890, plus the `[COMPUTE]` log site.

13. user (2026-07-07T14:01:51.182Z)

```
150                             )  # OverflowError on 'inf'/'1e400'
151                         except (ValueError, OverflowError):
152                             meta["shot_count"] = sc
153                     db.add_play_segment(video_id, start, end, metadata=meta)
154                     n += 1
155         return n
156
157
158     def _cloud_available(cfg) -> bool:
159         """Whether THIS server can satisfy a per-request ``compute="cloud"`` dispatch (item
3).
160
161         True when the control plane is wired for Cloud Run: either the configured default
target is
162         already ``cloud_run`` (the operator chose cloud serving), or the Cloud Run env
coordinates
163         (``CLOUD_RUN_JOB`` + ``GOOGLE_CLOUD_PROJECT``) and an output bucket
(``storage.output_root``)
164         are present so a forced per-request override can actually dispatch + collect a
result.
165
166         Surfaced in ``/api/capabilities`` so the SPA only offers the cloud toggle when it
would work."""
167         deployment = getattr(cfg, "deployment", None)
168         if (getattr(deployment, "compute_target", "") or "") == "cloud_run":
169             return True
170         has_job = bool(
```

```
        os.environ.get("CLOUD_RUN_JOB") and os.environ.get("GOOGLE_CLOUD_PROJECT")
    )
    out_root = (getattr(getattr(cfg, "storage", None), "output_root", "") or "").strip()
    return bool(has_job and out_root)


def _wait_progress_message(seg_name: str, elapsed_sec: float) -> str:
    """The live job.progress line while a cloud GPU run is in flight (#482). Honest and
    minute-granular: the control plane knows ELAPSED time, not frames ? frame-level % comes
    with the worker-side heartbeat (the #482 follow-up). Under a minute reads as "starting"
    (queue/cold-start), so the very first tick already shows movement."""
    minutes = int(elapsed_sec) // 60
    if minutes < 1:
        return f"segmenting ({seg_name}) on cloud GPU ? starting up"
    return f"segmenting ({seg_name}) on cloud GPU ? {minutes} min elapsed"


def submit_time_video_id(input_ref, cfg) -> Optional[str]:
    """The video's md5 (the canonical ``video_id``) resolvable CHEAPLY at submit time ? gs://
```

14. user (2026-07-07T14:01:55.801Z)

```
                "status": "failed",
                "message": f"Segmenter '{seg_name}' not found. Available segmenters: {available}",
                "results": [r.model_dump() for r in val_results],
                "play_segments": [],
            }
            return response

        logger.info(f"[API] Using segmenter: {seg_name}")
        notify(f"segmenting ({seg_name})")
        segmenter_actual = seg_name
        # Destroy-AFTER-confirm: remember the pre-run rows, then keep new + drop old only
        # if the run yields segments; otherwise drop the new partial rows and keep the old.
        play_wm, cand_wm = db.segment_watermarks(video_id)
        # COMPUTE_DECOUPLED_SERVING M6 ? API: with deployment.compute_target=cloud_run, run the heavy
        # DETECT on a Cloud Run GPU Job + ingest its rallies HERE so query/compile are unchanged.
        # DEFAULT-OFF: returns None for every non-cloud target ? the in-process segmenter below runs
        # byte-identically. (player_pool/max_frames aren't threaded to the worker `infer` CLI yet, so
        # they're dropped on the cloud path in this slice ? a documented follow-up.)
        remote = _maybe_dispatch_remote(
            config, db, req, video_id, seg_name, val_results, play_wm, cand_wm, notify
        )
        if remote is not None:
            response, segmentation_ran = (
                remote  # `ran` = did the cloud job actually execute (report)
            )
            return response  # the cloud branch already stamped response["compute"] (path=cloud_run)
        # In-process from here on ? record the actual path for the provenance stamp in `finally`.
        compute_actual = "in_process"
        segmenter = segmenter_cls(db, config)
        try:
            # Serialize the CPU/GPU-heavy in-process detection (see
```

```
_LOCAL_COMPUTE_LANE). A remote
                # dispatch never reaches here (it returned above), so remote polls still overlap freely.
                with _LOCAL_COMPUTE_LANE:
                    result = segmenter.process_video(
                        video_id, local_video, req.player_pool, req.max_frames
                    )
            except Exception:
                # A raising segmenter must not leave half-written rows: restore the pre-run
                # state (same destroy-after-confirm contract as the Failure branch), then let
                # the job worker record the failure.
                db.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)
                raise
            segmentation_ran = True

            if isinstance(result, Failure):
                logger.error(f"[API] Segmenter failed: {result.message}")
                db.prune_segments(video_id, play_after=play_wm, cand_after=cand_wm)
                response = {
                    "video_id": video_id,
                    "status": "failed",
                    "message": f"Segmenter '{seg_name}' failed: {result.message}",
                    "results": [r.model_dump() for r in val_results],
                    "play_segments": [],
                }
                return response

            segments = result.value if hasattr(result, "value") else result
            notify("finalizing")
            _finalize_reingest(db, video_id, segments, play_wm, cand_wm)

            response = {
                "video_id": video_id,
                "status": "success",
                "message": f"Successfully indexed {len(segments)} play segments using '{seg_name}' segmenter.",  # type: ignore[arg-type]  # hasattr-unwrap above; Success arm has .value
                "results": [r.model_dump() for r in val_results],
                "play_segments": segments,
            }
            return response
        finally:
            # Compute-path PROVENANCE stamp (the validation method): the cloud branch already set
            # response["compute"] (path=cloud_run + the Cloud Run execution); for every other path stamp
            # the actual path here so the job result PROVES which backend ran (in_process / not_run) and
            # what the picker REQUESTED ? a silent fallback (requested cloud, ran local) is then visible.
            if isinstance(response, dict):
                response.setdefault(
                    "compute",
                    {
                        "path": compute_actual or "not_run",
                        "requested": getattr(req, "compute", None),
                    },
                )
            prov = response.get("compute", {}) if isinstance(response, dict) else {}
            logger.info(
                "[COMPUTE] video_id=%s requested=%s actual=%s dispatched=%s status=%s",
                video_id,
                getattr(req, "compute", None),
                prov.get("path"),
```

```
                prov.get("dispatched"),
                (response or {}).get("status") if isinstance(response, dict) else None,
            )
            # Always emit the per-video observability report (next to the video, or to
            # report.output_dir). Never raises. Surface the paths in the response. (The compute-path
            # proof lives on the job result + the [COMPUTE] log, not the report ? obsreport's record_run
            # doesn't surface arbitrary run signals.)
            paths = _main.emit_indexer_report(
                db=db,
                video_id=video_id,
                # Resolved LOCAL path, not the raw URI: obsreport writes the report next to the
                # video via os.path.abspath(video_path), and a remote URI (gs://, gcs://, local://)
                # collapses to a literal scheme dir at the repo root (see #397). The source URI is
                # still preserved in DB provenance via db.add_video.
                video_path=local_video,
                config=_report_config_for_input(config, req.video_path),
                val_results=val_results,
                val_context=val_context,
                segmenter_requested=req.segmenter,
                segmenter_actual=segmenter_actual,
                was_reingest=was_reingest,
                segmentation_ran=segmentation_ran,
            )
            if paths and isinstance(response, dict):
                response["reports"] = paths
            storage.cleanup_acquired_local_input(
                req.video_path, local_video, getattr(config, "storage", None)
            )


    def _remote_input_size_bytes(input_ref, storage_config) -> Optional[int]:
        """Best-effort size (bytes) of a remote storage source, or ``None`` if unknowable.

```

15. assistant (2026-07-07T14:02:03.984Z)

Now let me understand the detector/segmenter names that need weights and how WASB weights are located.

16. user (2026-07-07T14:02:05.433Z)

```
Found 20 files
backend\pipeline\segmenters\wasb_hybrid.py
backend\pipeline\detectors\wasb_infer.py
backend\pipeline\segmenters\fusion_hybrid.py
backend\pipeline\detectors\native_wasb_runner.py
backend\config\models.py
backend\worker\__main__.py
backend\pipeline\segmenters\trajectory_hybrid.py
backend\main.py
backend\eval\served_gate_a.py
backend\infrastructure\database_user_models.py
backend\pipeline\detectors\base.py
backend\pipeline\detectors\wasb_runner.py
backend\reporting\harvest.py
backend\pipeline\segmenters\tracknet_hybrid.py
backend\pipeline\detectors\stub_runner.py
backend\pipeline\personalization\serving.py
backend\eval\experiment.py
backend\eval\gemini_refine.py
```

```
backend\pipeline\segmenters\__init__.py
backend\pipeline\personalization\__init__.py
```

17. assistant (2026-07-07T14:02:10.889Z)

Let me find where the "WASB weights not found" / "environment not ready" message comes from, and how weights paths resolve.

18. user (2026-07-07T14:02:12.176Z)

```
backend\pipeline\detectors\tracknet_runner.py:43:    weights_path: str = ""  # WSL path to the
.h5/.keras weights (REQUIRED)
backend\pipeline\detectors\tracknet_runner.py:61:            weights_path=tn.weights_path,
backend\pipeline\detectors\tracknet_runner.py:142:        if not self.cfg.weights_path:
backend\pipeline\detectors\tracknet_runner.py:144:                "TrackNet weights_path is not
set. Set indexing.tracknet.weights_path "
backend\pipeline\detectors\tracknet_runner.py:197:            f"--model_weights
{wsl_tilde_quote(self.cfg.weights_path)} "
backend\pipeline\detectors\wasb_runner.py:45:    weights_path: str = (
backend\pipeline\detectors\wasb_runner.py:71:            weights_path=w.weights_path,
backend\pipeline\detectors\wasb_runner.py:183:            f"--weights
{wsl_tilde_quote(self.cfg.weights_path)} "
backend\pipeline\detectors\native_wasb_runner.py:87:    weights_path: str = ""  # REQUIRED (env
WASB_WEIGHTS or indexing.wasb.weights_path)
backend\pipeline\detectors\native_wasb_runner.py:120:
weights_path=env("WASB_WEIGHTS") or w.weights_path or "",
backend\pipeline\detectors\native_wasb_runner.py:202:        if not self.cfg.weights_path:
backend\pipeline\detectors\native_wasb_runner.py:204:                "WASB weights path is empty
? set WASB_WEIGHTS (or indexing.wasb.weights_path)."
backend\pipeline\detectors\native_wasb_runner.py:206:        weights =
os.path.expanduser(self.cfg.weights_path)
backend\pipeline\detectors\native_wasb_runner.py:208:            return False, f"WASB weights
not found at {weights}"
backend\pipeline\detectors\native_wasb_runner.py:298:        weights =
os.path.expanduser(self.cfg.weights_path)
backend\pipeline\segmenters\fusion_hybrid.py:77:                "weights_path":
cfg.weights_path,
backend\pipeline\segmenters\fusion_hybrid.py:83:            "weights_path": wcfg.weights_path,
backend\pipeline\segmenters\fusion_hybrid.py:105:            "weights_path": cfg.weights_path,
backend\pipeline\segmenters\tracknet_hybrid.py:43:            "weights_path": cfg.weights_path,
backend\pipeline\segmenters\wasb_hybrid.py:44:        return WasbRunner(cfg), {"weights_path":
cfg.weights_path}
backend\pipeline\segmenters\wasb_hybrid.py:58:            "weights_path": cfg.weights_path,
backend\pipeline\segmenters\trajectory_hybrid.py:16:                      detection_params
extras such as weights_path).
backend\pipeline\segmenters\trajectory_hybrid.py:285:            logger.error("%s detector
environment not ready: %s", label, msg)
backend\pipeline\segmenters\trajectory_hybrid.py:289:                message=f"{label} detector
environment not ready: {msg}",
```

19. user (2026-07-07T14:02:23.190Z)

```
80          defaults to ``cuda`` (the WSL parity path); ``stream_video`` defaults to True (the
perf win).
81          """
82
83          repo_dir: str = "~/models/WASB-SBDT"  # native clone of nttcom/WASB-SBDT
84          python_bin: str = (
85              "python"  # the `wasb` conda env python (invoked by path, no activate)
86          )
87          weights_path: str = ""  # REQUIRED (env WASB_WEIGHTS or indexing.wasb.weights_path)
88          sport: str = "badminton"
89          device: str = "cuda"  # auto | cuda | mps | cpu (auto = cuda-if-available-else-cpu)
90          scratch_dir: str = ""  # frame-extraction scratch (env); "" = next to the output dir
```

```
    timeout_sec: int = 1800  # 30 min; a hung GPU call is killed (0 = no timeout)
    keep_frames: bool = (
        False  # retain the decoded frame cache after success (debug only)
    )
    # NATIVE DEFAULT = STREAMING: skip the cv2 PNG extraction that dominates a long clip's
    # wall-clock (measured ~50s on a 20s clip) ? bit-identical to disk (verified on a real GPU
    # 2026-06-20). The GPU box reads from a local NVMe so streaming is the right default here.
    stream_video: bool = True
    # GPU-feed tuning (indexing.wasb.batch_size / prefetch_batches). 0 = omit the flag ?
    # wasb_infer.py's parity defaults (batch 8, no prefetch). The cloud-serving preset opts in.
    batch_size: int = 0
    prefetch_batches: int = 0
    # GPU-resident NVDEC decode (#506, indexing.wasb.decode_backend / WASB_DECODE_BACKEND).
    # "cpu" (default) = omit the flag ? wasb_infer.py's historical cv2 decode paths, argv
    # byte-identical. "nvdec_resident" = decode+resize+normalize on-GPU (torch-2.x env only).
    decode_backend: str = "cpu"

    @classmethod
    def from_indexing_cfg(cls, idx_cfg: "IndexingConfig") -> "NativeWasbConfig":
        if isinstance(idx_cfg, dict):
            idx_cfg = IndexingConfig(**idx_cfg)
        w = idx_cfg.wasb
        env = os.environ.get
        # stream_video is tri-state in config: None/absent => native default (stream); an
        # explicit True/False still wins (e.g. False for a container cv2 can't seek).
        sv = w.stream_video
        return cls(
            repo_dir=env("WASB_REPO_DIR") or w.repo_dir,
            python_bin=env("WASB_PYTHON") or w.python_bin,
            weights_path=env("WASB_WEIGHTS") or w.weights_path or "",
            sport=w.sport,
            device=env("WASB_DEVICE") or w.device,
            scratch_dir=env("WASB_SCRATCH")
            or env("RALLY_SCRATCH")
            or env("TMPDIR")
            or "",
            timeout_sec=w.timeout_sec,
            keep_frames=w.keep_frames,
            stream_video=(True if sv is None else bool(sv)),
            batch_size=int(w.batch_size or 0),
            prefetch_batches=int(w.prefetch_batches or 0),
            decode_backend=env("WASB_DECODE_BACKEND") or w.decode_backend,
        )


class NativeWasbRunner(DetectorRunner):
    """WASB runner that executes natively on a Linux GPU box ? no WSL. See module docstring."""

    def __init__(self, cfg: NativeWasbConfig):
        self.cfg = cfg
        # Cached CUDA-availability probe (used to resolve device="auto"). Set by healthcheck;
        # lazily probed by run_predict if healthcheck was skipped. None = not yet probed.
        self._cuda_available: Optional[bool] = None

```

```
    # --- low-level native exec (the single place tests patch) -------------------------
    def _run(
        self, argv: List[str], cwd: Optional[str] = None
    ) -> subprocess.CompletedProcess:
        logger.debug("native exec: %s (cwd=%s)", " ".join(argv), cwd)
        return subprocess.run(
            argv,
            cwd=cwd,
            capture_output=True,
            text=True,
            timeout=(self.cfg.timeout_sec or None),
        )

    # --- device resolution (M4 DeviceContext: device="auto" ? cuda-if-available-else-cpu) ------
    def _probe_cuda(self) -> bool:
        """Probe ``torch.cuda.is_available()`` in the wasb env's python (a subprocess). Any failure
        (missing python / timeout / import error) is treated as 'no CUDA'."""
        try:
            res = self._run([self.cfg.python_bin, "-c", _CUDA_PROBE_CODE])
        except (FileNotFoundError, subprocess.TimeoutExpired):
            return False
        return res.returncode == 0 and "cuda True" in (res.stdout or "")

    def _cuda_is_available(self) -> bool:
        """CUDA availability, cached across healthcheck + run_predict so we probe at most once."""
        if self._cuda_available is None:
            self._cuda_available = self._probe_cuda()
        return self._cuda_available

    def _effective_device(self) -> str:
        """The torch device to actually pass to ``wasb_infer.py``. Only ``auto`` needs the CUDA
        probe; an explicit cuda/mps/cpu resolves to itself (so ``device='cuda'`` is unchanged)."""
        if self.cfg.device == AUTO:
            return DeviceContext(AUTO, self._cuda_is_available()).effective
        return self.cfg.device

    def _repo_src(self) -> str:
        return os.path.join(os.path.expanduser(self.cfg.repo_dir), "src")

    def _copy_infer_script(self) -> bool:
        """Copy wasb_infer.py into the WASB repo src/ so it imports WASB modules + finds configs/."""
        repo_src = self._repo_src()
        try:
            os.makedirs(repo_src, exist_ok=True)
            shutil.copy(_INFER_PY, os.path.join(repo_src, "wasb_infer.py"))
            return True
        except OSError as e:
            logger.error("Failed to copy wasb_infer.py into %s: %s", repo_src, e)
            return False

    def _rm_rf(self, path: Optional[str]) -> None:
        """Best-effort recursive delete of a native path (post-success cleanup; never raises)."""
        if path:
            shutil.rmtree(path, ignore_errors=True)

    def healthcheck(self) -> Tuple[bool, str]:
        """Verify weights + repo present and the `wasb` python imports torch (and sees a
```

```
GPU on cuda)."""
        if not self.cfg.weights_path:
            return False, (
                "WASB weights path is empty ? set WASB_WEIGHTS (or indexing.wasb.weights_path)."
            )
        weights = os.path.expanduser(self.cfg.weights_path)
        if not os.path.exists(weights):
            return False, f"WASB weights not found at {weights}"
        repo_src = self._repo_src()
        if not os.path.isdir(repo_src):
            return False, (
                f"WASB-SBDT repo src/ not found at {repo_src} ? clone nttcom/WASB-SBDT "
                f"(set WASB_REPO_DIR / indexing.wasb.repo_dir)."
            )
        # Fail fast on a typo'd device (the device-policy seam) ? a clear error here beats a cryptic
        # argparse rc=2 from wasb_infer.py at run time.
        if self.cfg.device not in VALID_DEVICES:
            return False, (
                f"unknown device {self.cfg.device!r} ? valid: {', '.join(VALID_DEVICES)}."
```

20. user (2026-07-07T14:02:31.360Z)

```
        inactivity_min_density = idx_cfg.inactivity_min_density
        inactivity_temporal_density = idx_cfg.inactivity_temporal_density
        window_blob_fallback = idx_cfg.window_blob_fallback
        window_blob_factor = idx_cfg.window_blob_factor
        handoff_padding = idx_cfg.ai_handoff_padding
        ai_max_workers = idx_cfg.ai_max_workers
        log_candidates = idx_cfg.log_yolo_candidates
        store_candidates = idx_cfg.store_yolo_candidates

        detection_params = {
            "detector": self.name,
            "velocity_thresh": velocity_thresh,
            "merge_gap": merge_gap,
            "min_action_window_duration": min_window_duration,
            **runner_params,
        }

        # --- Phase 1: env healthcheck (fail fast with a clear message) ---
        ok, msg = runner.healthcheck()
        if not ok:
            logger.error("%s detector environment not ready: %s", label, msg)
            # A detector FAILURE, not a 0-rally video ? surface it so the caller marks the run
            # failed/retryable instead of writing an empty-but-"successful" result.
            return Failure(
                message=f"{label} detector environment not ready: {msg}",
                is_recoverable=True,
            )
        logger.info("%s detector env OK: %s", label, msg)

        # --- Phase 2: trajectory -> candidate rally windows ---
        # Trajectory detectors process the whole video in one pass (they carry
        # their own frame buffering), so unlike yolo_hybrid we don't pre-chunk.
        t_start = time.time()
        out_dir = os.path.join(output_base(self.config), self.family)
        csv_path = runner.run_predict(video_path, out_dir, video_id=video_id)
        if not csv_path:
            logger.error("%s produced no trajectory; aborting.", label)
            # The cold-start / timeout failure mode (e.g. WASB native inference timed out after
```

```
                # timeout_sec, or the detector env yielded nothing). This is a DETECTOR
failure ? NOT
                # a legitimately empty match ? so return a Failure the worker marks as
failed/retryable
```

21. assistant (2026-07-07T14:02:36.757Z)

Let me look at the segmenters registry to understand names and which need weights.

22. user (2026-07-07T14:02:37.446Z)

```
from backend.pipeline.segmenters.base import BasePlaySegmenter, segmenter_registry
from backend.pipeline.segmenters.fusion_hybrid import FusionSegmenter
from backend.pipeline.segmenters.gemini import GeminiPlaySegmenter
from backend.pipeline.segmenters.motion import MotionPlaySegmenter
from backend.pipeline.segmenters.tracknet_hybrid import TrackNetHybridSegmenter
from backend.pipeline.segmenters.wasb_hybrid import WasbHybridSegmenter
from backend.pipeline.segmenters.yolo_hybrid import HybridYoloSegmenter

__all__ = [
    "segmenter_registry",
    "BasePlaySegmenter",
    "MotionPlaySegmenter",
    "GeminiPlaySegmenter",
    "HybridYoloSegmenter",
    "TrackNetHybridSegmenter",
    "WasbHybridSegmenter",
    "FusionSegmenter",
]

```

23. user (2026-07-07T14:02:44.245Z)

```
backend/pipeline/segmenters/base.py:70:            name = temp_inst.name
backend/pipeline/segmenters/fusion_hybrid.py:243:segmenter_registry.register(FusionSegmenter)
backend/pipeline/segmenters/gemini.py:95:        sport_name = self.config.sport
backend/pipeline/segmenters/gemini.py:103:        provider_name = self.config.ai_provider
backend/pipeline/segmenters/gemini.py:574:        self, video_id: str, validated: List[Dict[str,
Any]], model_name: str
backend/pipeline/segmenters/gemini.py:633:        model_name = self.config.ai_model
backend/pipeline/segmenters/gemini.py:707:segmenter_registry.register(GeminiPlaySegmenter)
backend/pipeline/segmenters/motion.py:243:segmenter_registry.register(MotionPlaySegmenter)
backend/pipeline/segmenters/tracknet_hybrid.py:48:segmenter_registry.register(TrackNetHybridSegm
enter)
backend/pipeline/segmenters/trajectory_hybrid.py:190:        sport_name = self.config.sport
backend/pipeline/segmenters/trajectory_hybrid.py:200:        provider_name =
self.config.ai_provider
backend/pipeline/segmenters/trajectory_hybrid.py:202:            model_name = "local-noai"
backend/pipeline/segmenters/trajectory_hybrid.py:210:            model_name =
self.config.ai_model
backend/pipeline/segmenters/wasb_hybrid.py:68:segmenter_registry.register(WasbHybridSegmenter)
backend/pipeline/segmenters/yolo_hybrid.py:75:        sport_name = self.config.sport
backend/pipeline/segmenters/yolo_hybrid.py:83:        provider_name = self.config.ai_provider
backend/pipeline/segmenters/yolo_hybrid.py:84:        model_name = self.config.ai_model
backend/pipeline/segmenters/yolo_hybrid.py:324:        provider_name: str,
backend/pipeline/segmenters/yolo_hybrid.py:434:        self, video_id: str, gemini_segments:
List[dict], model_name: str
backend/pipeline/segmenters/yolo_hybrid.py:507:        provider_name = self.config.ai_provider
backend/pipeline/segmenters/yolo_hybrid.py:508:        model_name = self.config.ai_model
backend/pipeline/segmenters/yolo_hybrid.py:596:segmenter_registry.register(HybridYoloSegmenter)
```

24. user (2026-07-07T14:02:49.192Z)

```
"""WASB-based hybrid play segmenter.
```

```
Phase 2 uses **WASB** (nttcom/WASB-SBDT, MIT) shuttle-trajectory detection via
WSL2 (backend/pipeline/detectors/wasb_runner.py). WASB ships pretrained
badminton weights and runs today, so this is the live shuttle-motion substrate.

For the per-video *calibration* goal we mainly use Phase 2 (trajectory ->
candidate windows); the windowing thresholds (velocity_thresh / merge_gap /
min_window_duration) are what get tuned against golden rally labels.

The pipeline engine (healthcheck -> trajectory -> windows -> AI handoff -> DB)
lives in trajectory_hybrid.TrajectoryHybridSegmenter; this class only wires the
WASB runner and identity.
"""

from typing import Any, Dict, Tuple

from backend.config.models import IndexingConfig
from backend.pipeline.detectors.base import DetectorRunner
from backend.pipeline.detectors.wasb_runner import WasbConfig, WasbRunner
from backend.pipeline.segmenters.base import segmenter_registry
from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter


class WasbHybridSegmenter(TrajectoryHybridSegmenter):
    family = "wasb"
    display_label = "WASB"

    @property
    def name(self) -> str:
        return "wasb_hybrid"

    @property
    def description(self) -> str:
        return (
            "Two-stage pipeline: WASB shuttle-trajectory detection (WSL2, pretrained badminton "
            "weights) for precise candidate rallies + AI provider for semantic boundaries."
        )

    def _build_runner(
        self, idx_cfg: IndexingConfig
    ) -> Tuple[DetectorRunner, Dict[str, Any]]:
        cfg = WasbConfig.from_indexing_cfg(idx_cfg)
        return WasbRunner(cfg), {"weights_path": cfg.weights_path}

    def _build_native_runner(
        self, idx_cfg: IndexingConfig
    ) -> Tuple[DetectorRunner, Dict[str, Any]]:
        # Lazy import: the native runner pulls no heavy deps, but keep parity with the stub
        # seam (imported only when detector_impl='native' is actually selected).
        from backend.pipeline.detectors.native_wasb_runner import (
            NativeWasbConfig,
            NativeWasbRunner,
        )

        cfg = NativeWasbConfig.from_indexing_cfg(idx_cfg)
        return NativeWasbRunner(cfg), {
            "weights_path": cfg.weights_path,
            "detector_impl": "native",
            "device": cfg.device,
            # Record the decode path so persisted candidates are auditable: the cpu and
            # nvdec_resident caches are non-interchangeable (#506), so downstream evals /
```

```
                # dataset exports must be able to tell which decode produced a candidate.
                "decode_backend": cfg.decode_backend,
            }


    segmenter_registry.register(WasbHybridSegmenter)

```

25. assistant (2026-07-07T14:03:54.866Z)

Let me look at the config models for the weights paths and the ProcessRequest/compute field.

26. user (2026-07-07T14:03:56.910Z)

```
46:class TrackNetConfig(BaseModel):
51:    weights_path: str = ""
80:    weights_path: str = (
168:class FusionConfig(BaseModel):
202:class IndexingConfig(BaseModel):
293:    tracknet: TrackNetConfig = Field(default_factory=TrackNetConfig)
294:    wasb: WasbIndexConfig = Field(default_factory=WasbIndexConfig)
295:    fusion: FusionConfig = Field(default_factory=FusionConfig)
```

27. user (2026-07-07T14:04:03.550Z)

```
    class TrackNetConfig(BaseModel):
        wsl_distro: str = "Ubuntu"
        repo_dir: str = "~/models/TrackNetV4"
        conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
        conda_env: str = "TrackNetV4"
        weights_path: str = ""
        queue_length: int = 5
        stage_in_wsl: bool = True
        wsl_stage_dir: str = "~/clips"
        timeout_sec: int = 1800  # 30 min; kills a hung WSL/GPU call (0 = no timeout)
        velocity_threshold: float = 0.02
        # Cache hygiene: after a SUCCESSFUL run the staged video copy (and, for WASB, the
        # decoded frame cache) are deleted automatically ? they are pure intermediates,
        # regenerable from the source, and the dominant disk consumer (~GBs/video). Set
        # true only for debugging. On FAILURE they are always kept so a re-run can resume.
        keep_frames: bool = False
        # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of the
        # ~15×-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS, not warns).
        # 0 = no extra floor (the headroom guard still applies).
        min_free_gb: float = 0.0


    class WasbIndexConfig(BaseModel):
        """Typed config for the live WASB shuttle substrate (indexing.wasb).

        Mirrors backend/pipeline/detectors/wasb_runner.py::WasbConfig.from_indexing_cfg so
        these knobs actually take effect ? previously this whole block was silently dropped
        because nested config sections defaulted to extra='ignore'.
        """

        wsl_distro: str = "Ubuntu"
        repo_dir: str = "~/models/WASB-SBDT"
        conda_sh: str = "~/miniconda3/etc/profile.d/conda.sh"
        conda_env: str = "wasb"
        weights_path: str = (
            "~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar"
        )
```

```
    sport: str = "badminton"
    wsl_stage_dir: str = "~/clips"
    timeout_sec: int = 1800  # 30 min; kills a hung WSL/GPU call (0 = no timeout)
    velocity_threshold: float = 0.02
    # --- Linux-native runner only (detector_impl='native'); ignored by the WSL runner. ---
    # Env vars override these (WASB_DEVICE / WASB_PYTHON) for a config-free GPU box.
    # torch device for the native runner. "auto" = cuda-if-available-else-cpu (the M4 CPU fallback
    # for a no-GPU box); "cuda" stays strict (hard-fails without a GPU ? no silent slow-CPU downgrade).
    device: str = "cuda"  # auto | cuda | mps | cpu
    python_bin: str = (
        "python"  # the `wasb` conda env python (invoked by path, no `conda activate`)
    )
    # Cache hygiene: after a SUCCESSFUL run the staged video copy + decoded frame cache
    # (the ~GBs/video disk hog) are deleted automatically ? pure intermediates, regenerable
    # from the source. Set true only for debugging. On FAILURE they are kept so a re-run resumes.
    keep_frames: bool = False
    # Absolute FLOOR (GB) on WSL free space (at wsl_stage_dir) before a run, on top of the
    # ~15×-source headroom the frame-stage guard always enforces (D11 ? it now BLOCKS, not warns).
    # 0 = no extra floor (the headroom guard still applies).
    min_free_gb: float = 0.0
    # FAST PATH ? decode the video on the fly and feed frames straight to the detector,
    # skipping the slow per-frame PNG extraction that dominates a long clip (~46k PNGs for a
    # 12-min 1080p60 video, which overran the 30-min timeout). Output is bit-identical to the
    # PNG path (PNG is lossless) ? VERIFIED on a real GPU 2026-06-20 (native stream == WSL disk,
    # 1194/1195 frames byte-identical; native run-to-run byte-identical / deterministic).
    # Tri-state:
    #   None       = per-runner default ? the WSL runner keeps DISK extraction (unchanged Windows
    #                behaviour); the Linux-native runner (GPU box) defaults to STREAMING (the perf win).
    #   True/False = force on/off for BOTH runners (e.g. set False for a container cv2 can't seek).
    stream_video: Optional[bool] = None
    # GPU-feed tuning for the native runner (measured on the first real L4 serving run,
    # 2026-07-03: the default batch 8 with single-process loading left the GPU <1% utilized ?
    # the pipeline was CPU-feed-bound). None = wasb_infer.py's parity defaults (batch 8, no
    # prefetch ? byte-identical to the historical path). The cloud-serving preset opts into
    # the measured fast values; local/WSL behaviour is unchanged unless set explicitly.
    batch_size: Optional[int] = None
    # Depth of the bounded producer-thread queue that overlaps CPU-side batch assembly
    # (decode + PIL + ToTensor) with GPU inference in streaming mode. None/0 = off.
    prefetch_batches: Optional[int] = None
    # GPU-RESIDENT DECODE (#506) ? DEFAULT OFF. "nvdec_resident" moves decode+resize+normalize
    # onto the GPU with no host round-trip: PyNvVideoCodec NVDEC decode ? affine `grid_sample`
    # resize replicating WASB's cv2.warpAffine EXACTLY (the #506 lesson: a naive F.interpolate
    # silently over-detects ? F1 0.58?0.07 ? and only shows on hard clips) ? GPU ImageNet
    # normalize ? HRNet. Validated on a real L4 (GX010128, SERVED preset, 2026-07-06):
```

```
~1.7×
    # fps (35.8?60.2), GPU util 47?79% / CPU 62?35%, F1 0.574 ? baseline 0.582. Native (Linux)
    # runner only ? needs the torch-2.x + PyNvVideoCodec WASB env (deploy/cloudrun/Dockerfile)
    # with libnvcuvid/libnvidia-encode present, and device=cuda; the WSL runner ignores it.
    # Flip gate: the full 15-clip golden set at the SERVED preset (GX010128 must pass).
    decode_backend: Literal["cpu", "nvdec_resident"] = "cpu"

    @model_validator(mode="after")
    def _nvdec_resident_requires_cuda(self) -> "WasbIndexConfig":
        # NVDEC decode is CUDA-only; reject the contradictory combo at config load instead of
        # failing deep inside a paid GPU run. "auto" is allowed here ? the native runner's
        # healthcheck hard-fails if auto resolves to cpu with this backend on.
        if self.decode_backend == "nvdec_resident" and self.device not in ("cuda", "auto"):
            raise ValueError(
                f"indexing.wasb.decode_backend='nvdec_resident' requires device 'cuda' or "
                f"'auto' (got {self.device!r}) ? NVDEC decode runs on the GPU."
            )
        return self

    @field_validator("batch_size")
    @classmethod
    def _batch_size_positive(cls, v: Optional[int]) -> Optional[int]:
        # A DataLoader batch must be >=1; reject at config time so a typo can't reach
        # `wasb_infer.py --batch-size` as an invalid value (the runner only passes the flag
        # when truthy, so 0 stays "off/parity-default").
        if v is not None and v < 1:
            raise ValueError(f"indexing.wasb.batch_size must be >= 1 when set (got {v})")
        return v

    @field_validator("prefetch_batches")
```

28. assistant (2026-07-07T14:04:09.186Z)

Now let me read the existing test file to match conventions.

29. user (2026-07-07T14:04:10.661Z)

```
"""API cloud-serving dispatch (COMPUTE_DECOUPLED_SERVING M6 ? the API path).

Proves the /api/process job path can offload DETECT to a Cloud Run GPU Job and ingest the rally
intervals back into the API DB, so /api/query + /api/compile (local stitch) are unchanged ? all
GPU-free + cloud-free via a FAKE CloudRunJobClient that writes report.json + intervals.csv + the
done-marker into a local "bucket". Cardinal invariant: DEFAULT-OFF ? only compute_target=cloud_run
dispatches; every other target runs the in-process segmenter byte-identically.
"""

import csv
import json
import os

import pytest

```

```python
pytest.importorskip("httpx")
from fastapi.testclient import TestClient

import backend.compute as compute_pkg
import backend.main as m
from backend.compute.cloud_run import CloudRunJobClient
from backend.config.models import AppConfig
from backend.infrastructure.database import Database
from backend.infrastructure.execution_policy import ExecutionPolicy
from backend.pipeline.validators.base import ValidationResult

_FAST = ExecutionPolicy(poll_every_n_sec=0.0, op_timeout_sec=5.0, max_wait_sec=5.0)


def _pass():
    return [ValidationResult(validator_name="x", passed=True, message="ok", details={})]


class _InlineExecutor:
    def submit(self, fn, *a, **k):
        fn(*a, **k)


class _FakeCloudClient(CloudRunJobClient):
    """Simulates the Cloud Run worker: on dispatch, write report.json + intervals.csv + the done-marker
    into the local bucket exactly where the real worker would, so the bucket-poll + ingest resolve."""

    def __init__(self, *, video_id="vidCloud", returncode=0, rows=None):
        self.calls: list[list[str]] = []
        self.video_id, self.returncode = video_id, returncode
        self.rows = (
            rows
            if rows is not None
            else [(2.74, 6.91, 5, "winner"), (8.74, 11.38, 3, "unforced_error")]
        )

    def run_job(self, job_name, argv, *, region, project=None):
        self.calls.append(list(argv))
        out_uri = argv[argv.index("--out-uri") + 1].rstrip("/")
        done_uri = argv[argv.index("--done-uri") + 1]
        if self.returncode == 0:
            outputs = os.path.join(out_uri, "outputs", self.video_id)
            os.makedirs(outputs, exist_ok=True)
            with open(os.path.join(outputs, "report.json"), "w", encoding="utf-8") as f:
                json.dump(
                    {
                        "video_id": self.video_id,
                        "segment_count": len(self.rows),
                        "status": "success",
                    },
                    f,
                )
            with open(
                os.path.join(outputs, "intervals.csv"),
                "w",
                newline="",
                encoding="utf-8",
            ) as f:
                w = csv.writer(f)
                w.writerow(["start", "end", "shot_count", "ending_reason"])
                for s, e, sc, er in self.rows:
                    w.writerow([f"{s:.3f}", f"{e:.3f}", sc, er])
        os.makedirs(os.path.dirname(done_uri), exist_ok=True)
```

```python
            with open(done_uri, "w", encoding="utf-8") as f:
                json.dump(
                    {
                        "video_id": self.video_id,
                        "returncode": self.returncode,
                        "segment_count": len(self.rows),
                        "status": "success",
                    },
                    f,
                )
            return "exec-1"

        def cancel_execution(self, execution):
            raise AssertionError(
                "cancel must never fire on the happy path"
            )  # pragma: no cover


    @pytest.fixture
    def cloud_env(tmp_path, monkeypatch):
        """API wired for cloud_run with a LOCAL bucket (the local storage backend makes the bucket read an
        identity file read ? no GCS). A gs:// input is 'fetched' to a local fake file for hash+validate."""
        db = Database(str(tmp_path / "t.db"))  # noqa: F841
        monkeypatch.setattr(m, "db_instance", db)
        monkeypatch.setattr(m, "db_instance", db)
        bucket = tmp_path / "bucket"
        bucket.mkdir()
        cfg = AppConfig.model_validate(
            {  # noqa: F841
                "deployment": {"compute_target": "cloud_run"},
                "storage": {"backend": "local", "output_root": str(bucket)},
            }
        )
        monkeypatch.setattr(m, "global_config", cfg)
        monkeypatch.setattr(m, "global_config", cfg)
        monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
        # A real gs:// input would be fetched to hash+validate; return a local fake so that path is offline.
        fake_local = tmp_path / "fetched.mp4"
        fake_local.write_bytes(b"FAKEVIDEOBYTES")
        monkeypatch.setattr(
            m.storage, "acquire_local_input", lambda path, scfg: str(fake_local)
        )
        monkeypatch.setattr(
            m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
        )
        return db, bucket, monkeypatch


    def _inject_fake(monkeypatch, fake):
        """Make backend.compute.get_compute_backend (re-imported inside _maybe_dispatch_remote) build a
        real CloudRunCompute around the FAKE client + the fast no-wait policy + a fixed token. ``**kw``
        forwards the per-request ``target_override`` (item 3) so the cloud-picker path resolves too."""
        real = compute_pkg.get_compute_backend
        monkeypatch.setattr(
            compute_pkg,
            "get_compute_backend",
            lambda config, **kw: real(
                config, run_client=fake, policy=_FAST, token="tok", **kw
            ),
```

```python
    )


def _meta(row):
    md = row.get("metadata")
    return json.loads(md) if isinstance(md, str) else (md or {})


def test_cloud_dispatch_ingests_segments(cloud_env):
    db, _bucket, monkeypatch = cloud_env
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)

    out = m._execute_processing(
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
    )

    assert out["status"] == "success"
    assert len(out["play_segments"]) == 2
    rows = db.query_play_segments(out["video_id"], {})
    assert len(rows) == 2
    metas = [_meta(r) for r in rows]
    assert all(md["source"] == "cloud_run" for md in metas)
    assert {md.get("shot_count") for md in metas} == {5, 3}
    assert {md.get("ending_reason") for md in metas} == {"winner", "unforced_error"}


def test_cloud_dispatch_builds_correct_spec(cloud_env):
    _db, bucket, monkeypatch = cloud_env
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)

    m._execute_processing(
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
    )

    assert len(fake.calls) == 1  # exactly one Cloud Run Job execution
    argv = fake.calls[0]
    assert argv[argv.index("--video-uri") + 1] == "gs://b/clip.mp4"
    assert argv[argv.index("--out-uri") + 1] == str(bucket)
    assert argv[argv.index("--segmenter") + 1] == "wasb_hybrid"
    joined = " ".join(argv)
    assert (
        "deployment.compute_target=local_gpu" in joined
    )  # container runs INLINE (never re-dispatch)
    assert "indexing.detector_impl=native" in joined
    assert (
        "indexing.skip_ai_handoff=false" in joined
    )  # the engine's AI-handoff stance is forwarded


def test_cloud_dispatch_forwards_skip_ai_handoff_true(cloud_env):
    """When the engine runs no-AI (skip_ai_handoff=True), the cloud job must too ? else it tries the
    Gemini handoff (which the cloud job has no key for) and yields 0 rallies. The flag is forwarded."""
    _db, _bucket, monkeypatch = cloud_env
    m.global_config.indexing.skip_ai_handoff = True
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    m._execute_processing(
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
    )
    assert "indexing.skip_ai_handoff=true" in " ".join(fake.calls[0])
```

```
def test_cloud_dispatch_forwards_wasb_serving_knobs(cloud_env):
    """The worker Job has no DEPLOYMENT_PROFILE, so it never applies the cloud-serving preset ?
    the control-plane MUST forward its resolved WASB serving knobs (decode_backend + GPU-feed
    tuning) or the worker silently reverts to the model defaults (cpu decode, batch 8), making the
    preset flip (#506/#507) a no-op on the real dispatch path. Regression guard for that wiring."""
    _db, _bucket, monkeypatch = cloud_env
    m.global_config.indexing.wasb.decode_backend = "nvdec_resident"
    m.global_config.indexing.wasb.batch_size = 64
    m.global_config.indexing.wasb.prefetch_batches = 4
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    m._execute_processing(
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
    )
    joined = " ".join(fake.calls[0])
    assert "indexing.wasb.decode_backend=nvdec_resident" in joined
    assert "indexing.wasb.batch_size=64" in joined
    assert "indexing.wasb.prefetch_batches=4" in joined


def test_cloud_dispatch_omits_unset_wasb_tuning(cloud_env):
    """decode_backend defaults to cpu (always forwarded ? harmless, keeps worker == control-plane),
    but batch_size/prefetch_batches default to None and must NOT be forwarded (None = the worker's
    parity default; forwarding an empty value would reach argparse as a bad token)."""
    _db, _bucket, monkeypatch = cloud_env
    # defaults: decode_backend="cpu", batch_size=None, prefetch_batches=None
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    m._execute_processing(
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
    )
    joined = " ".join(fake.calls[0])
    assert "indexing.wasb.decode_backend=cpu" in joined
    assert "indexing.wasb.batch_size" not in joined
    assert "indexing.wasb.prefetch_batches" not in joined


def test_cloud_failure_marks_failed_no_rows(cloud_env):
    db, _bucket, monkeypatch = cloud_env
    # pre-seed a prior good row ? must be preserved on a cloud failure (destroy-AFTER-confirm).
    db.add_video("vidLocalPrior", "gs://b/clip.mp4", "processed", [])
    fake = _FakeCloudClient(returncode=1)
    _inject_fake(monkeypatch, fake)

    out = m._execute_processing(
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
    )
    assert out["status"] == "failed"
    assert "returncode 1" in out["message"]
    assert out["play_segments"] == []
    assert db.query_play_segments(out["video_id"], {}) == []
    # It DID dispatch + run remotely (then failed rc=1), so the path is honestly cloud_run + dispatched.
    assert (
        out["compute"]["path"] == "cloud_run" and out["compute"]["dispatched"] is True
    )
```

```

def test_cloud_requires_a_bucket_input(cloud_env, tmp_path):
    _db, _bucket, monkeypatch = cloud_env
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    local_path = str(tmp_path / "local_only.mp4")

    out = m._execute_processing(
        m.ProcessRequest(video_path=local_path, segmenter="wasb_hybrid")
    )
    assert out["status"] == "failed"
    assert "cloud input" in out["message"]
    assert fake.calls == []  # never dispatched a local-only path


def test_cloud_stages_local_upload_to_bucket(cloud_env, tmp_path, monkeypatch):
    """A browser upload lands on THIS box's local disk; under cloud-serving with a CLOUD input_root it
    is STAGED to the bucket and the L4 job dispatches from the staged gs:// URI (not hard-failed) ? #461.
    The DB video record is repointed at the staged source so a later /api/compile can re-fetch it."""
    db, _bucket, _mp = cloud_env
    m.global_config.storage.input_backend = "gcs"
    m.global_config.storage.input_root = "gs://in-bucket/videos"
    puts: list = []

    def _fake_put(src, uri, **kw):
        puts.append((str(src), uri, kw.get("config")))
        return uri

    monkeypatch.setattr(m.storage, "put", _fake_put)
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    local_path = str(tmp_path / "upload.mp4")

    out = m._execute_processing(
        m.ProcessRequest(video_path=local_path, segmenter="wasb_hybrid")
    )

    assert out["status"] == "success"
    # 1) the local upload was staged UP to the input bucket, namespaced by video_id
    assert len(puts) == 1
    src, dest, put_cfg = puts[0]
    assert src == local_path
    assert dest.startswith("gs://in-bucket/videos/") and dest.endswith("/upload.mp4")
    # the active storage settings are threaded into put() (so a non-default S3 endpoint / GCS project /
    # etc. survives) ? not a fresh default client. The model_dump carries input_root/input_backend.
    assert put_cfg is not None and put_cfg.get("input_root") == "gs://in-bucket/videos"
    assert put_cfg.get("input_backend") == "gcs"
    # 2) the L4 job dispatched from the STAGED gs:// URI, not the unreadable local path
    assert len(fake.calls) == 1
    argv = fake.calls[0]
    assert argv[argv.index("--video-uri") + 1] == dest
    # 3) the DB video record now points at the staged source (so compile re-fetches it)
    row = db.get_video(out["video_id"])
    assert row is not None and row["filepath"] == dest


def test_cloud_stages_when_input_root_is_cloud_and_input_backend_default(
    cloud_env, tmp_path, monkeypatch
):
    """input_root='gs://?' with input_backend left at its DEFAULT ('local') is a valid
```

```
    cloud config ?
    the storage layer classifies an explicit root by its scheme (resolve_input_uri joins under
    input_root before ever consulting input_backend), so the staging gate must too (PR #467 review).
    Regression: the gate previously required input_backend != 'local' and hard-failed this shape."""
    _db, _bucket, _mp = cloud_env
    m.global_config.storage.input_root = "gs://in-bucket/videos"
    # input_backend deliberately NOT set ? stays at the model default 'local'
    assert m.global_config.storage.input_backend == "local"
    puts: list = []

    def _fake_put(src, uri, **kw):
        puts.append((str(src), uri))
        return uri

    monkeypatch.setattr(m.storage, "put", _fake_put)
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    local_path = str(tmp_path / "upload.mp4")

    out = m._execute_processing(
        m.ProcessRequest(video_path=local_path, segmenter="wasb_hybrid")
    )

    assert out["status"] == "success"  # NOT the 'needs a cloud input' pre-dispatch failure
    assert len(puts) == 1
    src, dest = puts[0]
    assert src == local_path
    assert dest.startswith("gs://in-bucket/videos/") and dest.endswith("/upload.mp4")
    # dispatch continued from the STAGED gs:// URI
    assert len(fake.calls) == 1
    argv = fake.calls[0]
    assert argv[argv.index("--video-uri") + 1] == dest


def test_cloud_local_upload_without_input_root_still_fails(cloud_env, tmp_path, monkeypatch):
    """No cloud input_root configured ? we genuinely can't stage a local upload ? hard-fail (no
    dispatch), rather than dispatching a path the Cloud Run job can't read."""
    _db, _bucket, _mp = cloud_env  # fixture leaves input_root unset (input_backend=local)
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    called = []
    monkeypatch.setattr(m.storage, "put", lambda *a, **k: called.append(a))

    out = m._execute_processing(
        m.ProcessRequest(video_path=str(tmp_path / "up.mp4"), segmenter="wasb_hybrid")
    )
    assert out["status"] == "failed"
    assert "cloud input" in out["message"]
    assert fake.calls == [] and called == []  # neither staged nor dispatched


def test_default_off_runs_in_process(tmp_path, monkeypatch):
    """compute_target='' ? get_compute_backend returns None ? the in-process segmenter runs
    byte-identically (the gate is inert). No Cloud Run dispatch."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    monkeypatch.setattr(m, "db_instance", db)
```

```
    cfg = AppConfig()  # compute_target defaults to ""  # noqa: F841
    monkeypatch.setattr(m, "global_config", cfg)
    monkeypatch.setattr(m, "global_config", cfg)
    monkeypatch.setattr(m, "global_config", cfg)
    monkeypatch.setattr(
        m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
    )
    video = tmp_path / "v.mp4"
    video.write_bytes(b"hashable-bytes")

    class _InProcSeg:
        def __init__(self, db_, cfg_):
            self.db = db_

        def process_video(self, vid, path, *a, **k):
            sid = self.db.add_play_segment(vid, 0.0, 5.0)
            return [
                {
                    "id": sid,
                    "start_time": 0.0,
                    "end_time": 5.0,
                    "duration": 5.0,
                    "metadata": {},
                }
            ]

    monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _InProcSeg)
    out = m._execute_processing(
        m.ProcessRequest(video_path=str(video), segmenter="motion")
    )
    assert out["status"] == "success"
    assert len(out["play_segments"]) == 1
    assert "cloud_run" not in out["message"]


def test_cloud_ingest_failure_marks_failed_no_leak(cloud_env):
    """A rc=0 marker but a MISSING intervals.csv (a partial push) must fail cleanly + leak no rows
    (destroy-AFTER-confirm), not 500 or hang."""
    db, _bucket, monkeypatch = cloud_env

    class _NoCsvClient(_FakeCloudClient):
        def run_job(self, job_name, argv, *, region, project=None):
            self.calls.append(list(argv))
            out_uri = argv[argv.index("--out-uri") + 1].rstrip("/")
            done_uri = argv[argv.index("--done-uri") + 1]
            outputs = os.path.join(out_uri, "outputs", self.video_id)
            os.makedirs(outputs, exist_ok=True)
            with open(os.path.join(outputs, "report.json"), "w", encoding="utf-8") as f:
                json.dump({"video_id": self.video_id, "status": "success"}, f)
            # deliberately DO NOT write intervals.csv
            os.makedirs(os.path.dirname(done_uri), exist_ok=True)
            with open(done_uri, "w", encoding="utf-8") as f:
                json.dump({"video_id": self.video_id, "returncode": 0}, f)
            return "exec"

    _inject_fake(monkeypatch, _NoCsvClient())
    out = m._execute_processing(
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
    )
    assert out["status"] == "failed"
    assert "could not read the result" in out["message"]
    assert db.query_play_segments(out["video_id"], {}) == []
```

```python
def test_cloud_video_id_divergence_is_tolerated_with_warning(cloud_env, caplog):
    """If the worker re-hashes different bytes (result.video_id != local id), ingest still proceeds
    under the LOCAL id but logs a WARNING so the orphaned-report condition is diagnosable."""
    db, _bucket, monkeypatch = cloud_env
    _inject_fake(monkeypatch, _FakeCloudClient(video_id="DIVERGED"))
    with caplog.at_level("WARNING", logger="backend.main"):
        out = m._execute_processing(
            m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="wasb_hybrid")
        )
    assert out["status"] == "success"
    assert len(db.query_play_segments(out["video_id"], {})) == 2
    assert any(
        "DIVERGED" in r.getMessage() and "!= local" in r.getMessage()
        for r in caplog.records
    )


class _InProcSeg:
    """Minimal in-process segmenter (writes one play row) ? proves the local path actually ran."""

    def __init__(self, db_, cfg_):
        self.db = db_

    def process_video(self, vid, path, *a, **k):
        sid = self.db.add_play_segment(vid, 0.0, 5.0)
        return [
            {
                "id": sid,
                "start_time": 0.0,
                "end_time": 5.0,
                "duration": 5.0,
                "metadata": {},
            }
        ]


# --- item 3: SPA compute-picker (per-request compute override + cloud_available capability) ---


def test_per_request_local_forces_in_process(cloud_env):
    """Server configured for cloud (compute_target=cloud_run), but a per-request compute='local'
    forces the in-process segmenter and NEVER dispatches to Cloud Run."""
    db, _bucket, monkeypatch = cloud_env
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _InProcSeg)

    out = m._execute_processing(
        m.global_config,
        m.db_instance,
        m.ProcessRequest(
            video_path="gs://b/clip.mp4", segmenter="motion", compute="local"
        ),
    )
    assert out["status"] == "success"
    assert len(out["play_segments"]) == 1
    assert "cloud_run" not in out["message"]
    assert (
        fake.calls == []
    )  # explicit "run on my machine" ? no dispatch despite a cloud server
```

```python


def test_per_request_cloud_forces_dispatch_on_local_server(tmp_path, monkeypatch):
    """Server default is LOCAL (compute_target=''), but a per-request compute='cloud' forces a Cloud
    Run dispatch when the control plane is wired (CLOUD_RUN_JOB + GOOGLE_CLOUD_PROJECT + a bucket)."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    bucket = tmp_path / "bucket"
    bucket.mkdir()
    cfg = AppConfig.model_validate(
        {  # noqa: F841
            "deployment": {"compute_target": ""},  # server default = in-process
            "storage": {"backend": "local", "output_root": str(bucket)},
        }
    )
    monkeypatch.setattr(m, "global_config", cfg)
    monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
    fake_local = tmp_path / "fetched.mp4"
    fake_local.write_bytes(b"FAKEVIDEOBYTES")
    monkeypatch.setattr(
        m.storage, "acquire_local_input", lambda path, scfg: str(fake_local)
    )
    monkeypatch.setattr(
        m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
    )
    monkeypatch.setenv("CLOUD_RUN_JOB", "rally-worker")
    monkeypatch.setenv("GOOGLE_CLOUD_PROJECT", "proj")
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)

    out = m._execute_processing(
        m.global_config,
        m.db_instance,
        m.ProcessRequest(
            video_path="gs://b/clip.mp4", segmenter="wasb_hybrid", compute="cloud"
        ),
    )
    assert out["status"] == "success"
    assert len(out["play_segments"]) == 2
    assert (
        len(fake.calls) == 1
    )  # per-request override dispatched despite the LOCAL server default


def test_per_request_cloud_unconfigured_fails_clearly(tmp_path, monkeypatch):
    """compute='cloud' on a server NOT wired for it fails clearly (no silent local fallback, no
    dispatch) so the picker can surface 'cloud isn't available here'."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    cfg = AppConfig.model_validate(
        {"deployment": {"compute_target": ""}}
    )  # no bucket, no env  # noqa: F841
    monkeypatch.setattr(m, "global_config", cfg)
    monkeypatch.setattr(
        m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
    )
    monkeypatch.delenv("CLOUD_RUN_JOB", raising=False)
    monkeypatch.delenv("GOOGLE_CLOUD_PROJECT", raising=False)
    video = tmp_path / "v.mp4"
    video.write_bytes(b"hashable-bytes")

    out = m._execute_processing(
```

```
        m.global_config,
        m.db_instance,
        m.ProcessRequest(
            video_path=str(video), segmenter="wasb_hybrid", compute="cloud"
        ),
    )
    assert out["status"] == "failed"
    assert "isn't configured" in out["message"]


def test_unknown_compute_value_falls_back_to_server_default(cloud_env, caplog):
    """An unrecognised compute value is ignored (logged) and the server default applies ? here a
    cloud_run server still dispatches to the cloud."""
    db, _bucket, monkeypatch = cloud_env
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    with caplog.at_level("WARNING", logger="backend.main"):
        out = m._execute_processing(
            m.global_config,
            m.db_instance,
            m.ProcessRequest(
                video_path="gs://b/clip.mp4", segmenter="wasb_hybrid", compute="banana"
            ),
        )
    assert out["status"] == "success"
    assert len(fake.calls) == 1  # fell back to the configured cloud_run default
    assert any("unknown compute" in r.getMessage() for r in caplog.records)


def test_capabilities_reports_cloud_available(cloud_env):
    """A cloud_run-configured server advertises deployment.cloud_available=True (the SPA shows the
    cloud toggle only then)."""
    _db, _bucket, _monkeypatch = cloud_env
    app = m.create_app(m.global_config, m.db_instance)
    client = TestClient(app)
    resp = client.get("/api/capabilities")
    caps = resp.json()
    assert caps["deployment"]["cloud_available"] is True
    assert caps["deployment"]["compute_target"] == "cloud_run"


def test_capabilities_cloud_unavailable_by_default(monkeypatch, tmp_path):
    """A default (local) server with no Cloud Run env advertises cloud_available=False."""
    cfg = AppConfig()
    db = Database(str(tmp_path / "t.db"))
    app = m.create_app(cfg, db)
    monkeypatch.delenv("CLOUD_RUN_JOB", raising=False)
    monkeypatch.delenv("GOOGLE_CLOUD_PROJECT", raising=False)
    client = TestClient(app)
    resp = client.get("/api/capabilities")
    caps = resp.json()
    assert caps["deployment"]["cloud_available"] is False


def test_full_api_process_to_query_via_cloud(cloud_env):
    """End-to-end through the public API: POST /api/process (gs:// input) ? the inline worker dispatches
    to (fake) Cloud Run ? job done ? POST /api/query returns the ingested segments."""
    db, _bucket, monkeypatch = cloud_env
    fake = _FakeCloudClient(video_id="vidCloud2")
    _inject_fake(monkeypatch, fake)
    app = m.create_app(m.global_config, db)
```

```python
    client = TestClient(app)

    r = client.post(
        "/api/process",
        json={"video_path": "gs://b/clip.mp4", "segmenter": "wasb_hybrid"},
    )
    assert r.status_code == 202
    job_id = r.json()["job_id"]

    job = client.get(f"/api/jobs/{job_id}").json()
    assert job["status"] == "done"
    video_id = job["video_id"]
    assert video_id

    q = client.post("/api/query", json={"video_id": video_id})
    assert q.status_code == 200
    segs = q.json()["play_segments"]
    assert len(segs) == 2
    assert all(_meta(s)["source"] == "cloud_run" for s in segs)


def test_local_override_survives_api_boundary(cloud_env):
    """REGRESSION: the picker choice must survive the async submit?worker?reconstruct boundary. On a
    cloud_run server, POST /api/process with compute='local' must persist (jobs.compute) + reconstruct
    (_job_to_request) the choice and run IN-PROCESS ? not dispatch to Cloud Run. (An earlier cut
    dropped compute in create_job/_job_to_request, silently nullifying the picker through /api/process.)"""
    db, _bucket, monkeypatch = cloud_env
    fake = _FakeCloudClient()
    _inject_fake(monkeypatch, fake)
    monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _InProcSeg)
    app = m.create_app(m.global_config, db)
    client = TestClient(app)

    r = client.post(
        "/api/process",
        json={
            "video_path": "gs://b/clip.mp4",
            "segmenter": "motion",
            "compute": "local",
        },
    )
    assert r.status_code == 202
    job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
    assert job["status"] == "done"
    assert (
        fake.calls == []
    )  # 'local' honoured across the boundary ? never dispatched to Cloud Run
    segs = db.query_play_segments(job["video_id"], {})
    assert len(segs) == 1
    assert all(
        _meta(s).get("source") != "cloud_run" for s in segs
    )  # in-process rows, not cloud


def test_cloud_override_survives_api_boundary_on_local_server(tmp_path, monkeypatch):
    """REGRESSION (mirror): on a LOCAL-default server wired for cloud (env + bucket), POST /api/process
    with compute='cloud' must survive the boundary and dispatch to Cloud Run."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    bucket = tmp_path / "bucket"
```

```python
    bucket.mkdir()
    cfg = AppConfig.model_validate(
        {  # noqa: F841
            "deployment": {"compute_target": ""},  # server default = in-process
            "storage": {"backend": "local", "output_root": str(bucket)},
        }
    )
    monkeypatch.setattr(m, "global_config", cfg)
    monkeypatch.setattr(m, "_get_executor", lambda: _InlineExecutor())
    fake_local = tmp_path / "fetched.mp4"
    fake_local.write_bytes(b"FAKEVIDEOBYTES")
    monkeypatch.setattr(
        m.storage, "acquire_local_input", lambda path, scfg: str(fake_local)
    )
    monkeypatch.setattr(
        m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
    )
    monkeypatch.setenv("CLOUD_RUN_JOB", "rally-worker")
    monkeypatch.setenv("GOOGLE_CLOUD_PROJECT", "proj")
    fake = _FakeCloudClient(video_id="vidBoundary")
    _inject_fake(monkeypatch, fake)
    app = m.create_app(m.global_config, db)
    client = TestClient(app)

    r = client.post(
        "/api/process",
        json={
            "video_path": "gs://b/clip.mp4",
            "segmenter": "wasb_hybrid",
            "compute": "cloud",
        },
    )
    assert r.status_code == 202
    job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
    assert job["status"] == "done"
    assert (
        len(fake.calls) == 1
    )  # 'cloud' honoured across the boundary ? dispatched despite local default
    segs = db.query_play_segments(job["video_id"], {})
    assert len(segs) == 2
    assert all(_meta(s)["source"] == "cloud_run" for s in segs)


# --- compute-path PROVENANCE (the validation method: prove which backend ran) ------------------


def test_cloud_dispatch_stamps_compute_provenance(cloud_env):
    """A cloud run stamps result['compute'] with path=cloud_run + the REAL Cloud Run execution name
    (the GCP-verifiable proof) + the requested choice + job/region/outputs ? so the intended path is
    provable from the job result alone (no guessing from segment metadata)."""
    _db, _bucket, monkeypatch = cloud_env
    _inject_fake(monkeypatch, _FakeCloudClient())

    out = m._execute_processing(
        m.ProcessRequest(
            video_path="gs://b/clip.mp4", segmenter="wasb_hybrid", compute="cloud"
        )
    )
    prov = out["compute"]
    assert prov["path"] == "cloud_run"
    assert prov["requested"] == "cloud"
    assert (
```

```
        prov["execution"] == "exec-1"
    )  # the fake client's execution name (real one in prod)
    assert prov["outputs_uri"].endswith("/outputs/vidCloud/")
    assert prov["job"] and prov["region"]  # the dispatch coordinates are recorded


def test_local_override_stamps_in_process_provenance(cloud_env):
    """On a cloud server, compute='local' runs in-process AND the result proves it: path=in_process,
    requested=local ? so a 'ran local despite a cloud server' is auditable (no silent ambiguity)."""
    _db, _bucket, monkeypatch = cloud_env
    _inject_fake(monkeypatch, _FakeCloudClient())
    monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _InProcSeg)

    out = m._execute_processing(
        m.ProcessRequest(
            video_path="gs://b/clip.mp4", segmenter="motion", compute="local"
        )
    )
    assert out["status"] == "success"
    assert out["compute"] == {"path": "in_process", "requested": "local"}


def test_default_in_process_stamps_provenance(tmp_path, monkeypatch):
    """A default-OFF (no cloud) run stamps path=in_process, requested=None ? the byte-identical local
    path is now self-describing."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    monkeypatch.setattr(m, "db_instance", db)
    monkeypatch.setattr(m, "global_config", AppConfig())
    m.app.state.config = AppConfig()
    monkeypatch.setattr(
        m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
    )
    monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _InProcSeg)
    video = tmp_path / "v.mp4"
    video.write_bytes(b"hashable-bytes")

    out = m._execute_processing(
        m.ProcessRequest(video_path=str(video), segmenter="motion")
    )
    assert out["compute"] == {"path": "in_process", "requested": None}


def test_cloud_requested_unconfigured_stamps_not_dispatched(tmp_path, monkeypatch):
    """compute='cloud' on a server that can't dispatch fails clearly AND records it: target cloud_run,
    requested=cloud, dispatched=False ? proving the cloud job never ran (no silent local fallback)."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    monkeypatch.setattr(m, "db_instance", db)
    monkeypatch.setattr(
        m,
        "global_config",
        AppConfig.model_validate({"deployment": {"compute_target": ""}}),
    )
    m.app.state.config = AppConfig.model_validate(
        {"deployment": {"compute_target": ""}}
    )
    monkeypatch.setattr(
        m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
    )
```

```
    monkeypatch.delenv("CLOUD_RUN_JOB", raising=False)
    monkeypatch.delenv("GOOGLE_CLOUD_PROJECT", raising=False)
    video = tmp_path / "v.mp4"
    video.write_bytes(b"hashable-bytes")

    out = m._execute_processing(
        m.ProcessRequest(
            video_path=str(video), segmenter="wasb_hybrid", compute="cloud"
        )
    )
    assert out["status"] == "failed"
    # Honest: it ran NOWHERE (pre-dispatch reject) ? path=not_run, but the intent is recorded.
    assert out["compute"] == {
        "path": "not_run",
        "requested": "cloud",
        "target": "cloud_run",
        "dispatched": False,
    }


def test_validation_failure_stamps_not_run_provenance(tmp_path, monkeypatch):
    """A validation failure never reaches a segmenter ? path=not_run (nothing executed)."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    monkeypatch.setattr(m, "global_config", AppConfig())
    m.app.state.config = AppConfig()
    fail = [
        ValidationResult(validator_name="x", passed=False, message="bad", details={})
    ]
    monkeypatch.setattr(m.registry, "run_all_with_context", lambda *a, **k: (fail, {}))
    video = tmp_path / "v.mp4"
    video.write_bytes(b"hashable-bytes")

    out = m._execute_processing(m.ProcessRequest(video_path=str(video)))
    assert out["status"] == "failed"
    assert out["compute"]["path"] == "not_run"


def test_full_api_process_job_result_carries_cloud_provenance(cloud_env):
    """End-to-end: POST /api/process (compute=cloud) ? the job result the SPA polls carries the
    cloud_run provenance + execution, so the UI/operator can PROVE the cloud path was taken."""
    _db, _bucket, monkeypatch = cloud_env
    _inject_fake(monkeypatch, _FakeCloudClient(video_id="vidProv"))
    app = m.create_app(m.global_config, _db)
    client = TestClient(app)

    r = client.post(
        "/api/process",
        json={
            "video_path": "gs://b/clip.mp4",
            "segmenter": "wasb_hybrid",
            "compute": "cloud",
        },
    )
    assert r.status_code == 202
    job = client.get(f"/api/jobs/{r.json()['job_id']}").json()
    assert job["status"] == "done"
    assert job["result"]["compute"]["path"] == "cloud_run"
    assert job["result"]["compute"]["execution"] == "exec-1"
```

```
# --- #468: local compute lane ? CPU/GPU-heavy in-process work serializes, remote polls don't --------


def test_in_process_detection_runs_under_the_local_lane(cloud_env):
    """The in-process segmenter holds orchestration._LOCAL_COMPUTE_LANE while it runs, so raising the
    drain worker count (to overlap I/O-bound remote dispatch) never runs two local detections at once.
    A remote dispatch returns before this path, so remote polls never hold the lane (they overlap)."""
    import backend.orchestration as orch

    _db, _bucket, monkeypatch = cloud_env
    held = {}

    class _LaneCheckSeg:
        def __init__(self, db_, cfg_):
            self.db = db_

        def process_video(self, vid, path, *a, **k):
            # the wrapping `with _LOCAL_COMPUTE_LANE` holds the size-1 lane ? a non-blocking re-acquire
            # must FAIL while detection runs (proving concurrent local detections are excluded).
            got = orch._LOCAL_COMPUTE_LANE.acquire(blocking=False)
            held["busy"] = not got
            if got:
                orch._LOCAL_COMPUTE_LANE.release()
            sid = self.db.add_play_segment(vid, 0.0, 5.0)
            return [
                {"id": sid, "start_time": 0.0, "end_time": 5.0, "duration": 5.0, "metadata": {}}
            ]

    monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _LaneCheckSeg)
    out = m._execute_processing(
        m.global_config,
        m.db_instance,
        m.ProcessRequest(video_path="gs://b/clip.mp4", segmenter="motion", compute="local"),
    )
    assert out["status"] == "success"
    assert held.get("busy") is True  # detection ran while the local lane was held


def test_wait_progress_message_is_honest_and_minute_granular():
    """#482: the CP knows ELAPSED time, not frames ? the heartbeat says so plainly, shows
    movement from the first tick ("starting up" during queue/cold-start), then counts minutes."""
    from backend.orchestration import _wait_progress_message

    early = _wait_progress_message("wasb_hybrid", 12.0)
    assert "wasb_hybrid" in early and "starting up" in early
    later = _wait_progress_message("wasb_hybrid", 125.0)
    assert later.endswith("2 min elapsed")

```

30. assistant (2026-07-07T14:04:53.633Z)

Let me read fusion_hybrid.py to confirm which weights it needs, and check the SPA compute-picker (item 2).

31. user (2026-07-07T14:04:55.251Z)

```python
"""Multi-signal fusion segmenter (MULTI_SIGNAL_FUSION_PLAN P2).

Shuttle stays PRIMARY: this reuses an existing trajectory substrate (WASB by default,
TrackNet optional) to seed candidate windows exactly as `wasb_hybrid`/`tracknet_hybrid`
do ? it subclasses the same `TrajectoryHybridSegmenter` engine. It then ENRICHES each
shuttle-seeded window via two overridable seams the engine exposes:

- ``_enrich_candidate_stats`` ? always adds the **net-crossing count** (Gate A signal,
  GPU-free, computed from the trajectory we already have); and, only when the person cue
  is explicitly enabled (``indexing.fusion.cue_flags.person``), the person-cue features
  (`fusion_features`), persisting them into ``candidate_segments.stats`` for the learned
  fusion (P3). The person path lazily builds ONE `PersonDetector` and only then loads the
  torchvision/supervision model ? so the default (person-OFF) path never loads a detector
  model nor touches the GPU. (torch/cv2 themselves are imported by the registry regardless,
  via the pre-existing yolo_hybrid ? person_detector chain; the lazy part is the model.)
- ``_select_for_handoff`` ? optional served Gate A/B: when ``indexing.fusion.min_crossings``
  (and, with the person cue, ``min_players``) are raised above 0, sub-threshold windows are
  dropped from the AI handoff (the held-shuttle / one-sided-feed false-positive fix).
  **Persisted candidates remain the full superset** regardless, so the offline experiment
  harness can re-score any variant.

Defaults reproduce the substrate exactly: net_crossings/person features are persisted but
nothing is gated (thresholds 0, person OFF), so `FusionSegmenter` with default config seeds
and hands off identically to `wasb_hybrid`. Registered as ``fusion_hybrid``, NOT the default
segmenter ? it is opt-in. Promotion to default happens only on measured LOVO lift through the
experiment harness (EXPERIMENT_HARNESS.md), never silently.
"""

from typing import Any, Dict, List, Optional, Tuple

from backend.config.models import FusionConfig, IndexingConfig
from backend.pipeline.detectors import fusion_features as ff
from backend.pipeline.detectors import rally_gate
from backend.pipeline.detectors.base import DetectorRunner
from backend.pipeline.detectors.tracknet_runner import TrackNetConfig, TrackNetRunner
from backend.pipeline.detectors.wasb_runner import WasbConfig, WasbRunner
from backend.pipeline.segmenters.base import segmenter_registry
from backend.pipeline.segmenters.trajectory_hybrid import TrajectoryHybridSegmenter


class FusionSegmenter(TrajectoryHybridSegmenter):
    """Shuttle trajectory (primary) + net-crossing Gate A + optional person cue (Gate B)."""

    family = "fusion"
    display_label = "Fusion"

    def __init__(self, db: Any, config: Any):
        super().__init__(db, config)
        # Built lazily only if the person cue is enabled (no detector model loaded otherwise).
        self._person: Optional[Any] = None
        # Cache the play-axis/net estimate per trajectory so rally_gate doesn't re-estimate it
        # over the whole trajectory once per candidate window (it depends only on
```

```
`points`).
        self._axis_cache_key: Optional[int] = None
        self._axis_cache: Optional[tuple] = None

    @property
    def name(self) -> str:
        return "fusion_hybrid"

    @property
    def description(self) -> str:
        return (
            "Multi-signal fusion: shuttle trajectory (WASB/TrackNet) seeds candidate rallies, "
            "net-crossing Gate A + optional person-occupancy Gate B adjudicate them, and the "
            "AI provider sets semantic boundaries."
        )

    def _build_runner(
        self, idx_cfg: IndexingConfig
    ) -> Tuple[DetectorRunner, Dict[str, Any]]:
        if isinstance(idx_cfg, dict):
            idx_cfg = IndexingConfig(**idx_cfg)
        substrate = idx_cfg.fusion.substrate
        if substrate == "tracknet":
            cfg = TrackNetConfig.from_indexing_cfg(idx_cfg)
            return TrackNetRunner(cfg), {
                "weights_path": cfg.weights_path,
                "queue_length": cfg.queue_length,
                "fusion_substrate": "tracknet",
            }
        wcfg = WasbConfig.from_indexing_cfg(idx_cfg)
        return WasbRunner(wcfg), {
            "weights_path": wcfg.weights_path,
            "fusion_substrate": "wasb",
        }

    def _build_native_runner(
        self, idx_cfg: IndexingConfig
    ) -> Tuple[DetectorRunner, Dict[str, Any]]:
        # Native (no-WSL) path supports the WASB substrate only ? there is no native TrackNet
        # runner yet. Fusion's default substrate is WASB, so this covers the common case.
        substrate = idx_cfg.fusion.substrate
        if substrate != "wasb":
            raise NotImplementedError(
                "detector_impl='native' supports the WASB substrate only ? set "
                "indexing.fusion.substrate='wasb' (no native TrackNet runner yet)."
            )
        from backend.pipeline.detectors.native_wasb_runner import (
            NativeWasbConfig,
            NativeWasbRunner,
        )

        cfg = NativeWasbConfig.from_indexing_cfg(idx_cfg)
        return NativeWasbRunner(cfg), {
            "weights_path": cfg.weights_path,
            "detector_impl": "native",
            "fusion_substrate": "wasb",
            "device": cfg.device,
            # Audit the decode path alongside the CPU/nvdec-isolated cache (#506).
            "decode_backend": cfg.decode_backend,
        }

```

```
    def _enrich_candidate_stats(
        self,
        cw: Dict[str, Any],
```

32. assistant (2026-07-07T14:05:34.545Z)

Let me see how `seg_name` is derived and the `_execute_processing` signature/start.

33. user (2026-07-07T14:05:36.504Z)

```
    local_video = ""
    try:
        local_video = storage.acquire_local_input(
            req.video_path, getattr(config, "storage", None)
        )
        video_id = _main.compute_video_id(local_video)
    except Exception:
        storage.cleanup_acquired_local_input(
            req.video_path, local_video, getattr(config, "storage", None)
        )
        raise
    try:
        existing_video = db.get_video(video_id)
    except Exception:
        storage.cleanup_acquired_local_input(
            req.video_path, local_video, getattr(config, "storage", None)
        )
        raise
    was_reingest = existing_video is not None

    # typed AppConfig: attribute access for the default segmenter (with safe fallback)
    default_seg = getattr(
        getattr(config, "indexing", None), "default_segmenter", "motion"
    )
    seg_name = req.segmenter or default_seg
    segmenter_actual = None
    segmentation_ran = False
    val_results: List[Any] = []
    val_context: Dict[str, Any] = {}
    # Compute-path PROVENANCE (the validation method): which backend actually ran the DETECT ?
    # "in_process" once the local segmenter is reached, set on the response as "cloud_run" by
    # _maybe_dispatch_remote when it dispatches. None ? nothing ran (validation/config fail).
    compute_actual: Optional[str] = None
    response: Optional[Dict[str, Any]] = None
    try:
        if (
            existing_video
            and existing_video.get("status") == "processed"
            and not req.force
        ):
            cached_segments = db.query_play_segments(video_id, {})
            if cached_segments:
                logger.info(
                    "[API] Reusing cached segmentation for video_id=%s; force=true bypasses cache.",
                    video_id,
                )
                response = _cached_process_payload(
                    video_id,
                    existing_video.get("validation_results", []),
                    cached_segments,
                    getattr(req, "compute", None),
```

```
                )
                return response

        # 1. Run pluggable validators (with-context variant surfaces ffprobe_meta for the report)
        notify("validating")
        logger.info(f"Running validations for {req.video_path}...")
        val_results, val_context = registry.run_all_with_context(local_video, config)
        failures = [r for r in val_results if not r.passed]

        if failures:
            # add_video is an UPSERT (no cascade) ? a failed re-validation only updates
            # status; it no longer destroys the video's prior good segments.
            db.add_video(
                video_id,
                req.video_path,
                "failed",
                [r.model_dump() for r in val_results],
            )
            response = {
                "video_id": video_id,
                "status": "failed",
                "message": _validation_failure_message(failures),
                "results": [r.model_dump() for r in val_results],
            }
            return response

        # 2. Run segmenter & indexer
        logger.info("[API] Video passed checks. Segmenting and indexing...")
        db.add_video(
            video_id, req.video_path, "processed", [r.model_dump() for r in val_results]
        )

        segmenter_cls = segmenter_registry.get(seg_name)
        if not segmenter_cls:
            # Submit-time validation already 400s unknown names; this is the defensive
            # in-worker path (e.g. config default pointing at an unregistered segmenter).
            available = list(segmenter_registry.list_available().keys())
            response = {
                "video_id": video_id,
                "status": "failed",
                "message": f"Segmenter '{seg_name}' not found. Available segmenters: {available}",
                "results": [r.model_dump() for r in val_results],
                "play_segments": [],
            }
            return response

        logger.info(f"[API] Using segmenter: {seg_name}")
        notify(f"segmenting ({seg_name})")
        segmenter_actual = seg_name
```

34. assistant (2026-07-07T14:05:42.023Z)

Now let me look at the SPA compute-picker (item 2). Let me find the frontend code.

35. user (2026-07-07T14:05:43.715Z)

```
assets
index.html
THIRD_PARTY_NOTICES.md
ui_version.json
---
backend/ui/assets/index-5O3c-FCB.js
```

36. assistant (2026-07-07T14:05:48.938Z)

The `backend/ui` is a built bundle. Let me find the SPA source.

37. user (2026-07-07T14:05:52.367Z)

```
__pycache__/
artifacts/
backend/
deploy/
docs/
eval_baselines/
output/
scratch/
scripts/
tests/
tools/
training/
---FRONTEND DIRS---
---COMPUTE PICKER SRC---
./artifacts/promo/capture.js
```

38. user (2026-07-07T14:06:02.596Z)

```
compute:
compute:
compute:
compute:
compute:
compute:
compute:
compute:
compute:
compute:
compute:
compute:
compute:o
compute:i.compute?String(i.compute):void 0
compute:C
---FIND SPA SRC OUTSIDE REPO---
Annotation Setup
AUDIT_REPORT_khelsutra-guru_2026-07-02.md
badminton-highlight-indexer
bash.exe.stackdump
khelsutra
khelsutra-screencast
marketing
pr-158-screenshots
rally-annotator
sports-data-collector
sports-obsreport
../khelsutra/site/package.json
../khelsutra/web/package.json
../khelsutra-screencast/package.json
../rally-annotator/web/package.json
```

39. user (2026-07-07T14:06:12.218Z)

```
---UI BUILD REFERENCES---
backend/pipeline/validators/pts_continuity.py
backend/ui/THIRD_PARTY_NOTICES.md
backend/ui/index.html
backend/ui/ui_version.json
backend/utils/freespace.py
```

backend/utils/workspace.py
docs/PER_VIDEO_WORKSPACE.md
docs/UI_ALPHA_EXECUTION_PLAN.md
docs/UI_PROPOSAL.md
docs/mockups/ui_alpha_mockup.html
requirements-trackers.txt
requirements.txt
scripts/bundle_ui.sh
tests/test_api_cloud_dispatch.py
tests/test_freespace.py
tests/test_trajectory_hybrid_equivalence.py
tests/test_ui_bundle.py
tests/test_workspace.py
training/requirements-train.txt

40. user (2026-07-07T14:06:16.403Z)

```
#!/usr/bin/env bash
# Bundle the KhelSutra web SPA into the engine so a downloaded engine serves the real UI
# single-origin (PACKAGING_PLAN.md P1b).
#
# OPTION 1 (now, owner-chosen): build the SPA from a khelsutra checkout and vendor the built `dist/`
# into `backend/ui/`, which is COMMITTED ? so the wheel + CI build fully offline/reproducibly. Refresh
# the snapshot by re-running this on an SPA release; commit the diff.
#
# OPTION 3 (designed-for, future): point --from at a published dist tarball/URL (a GitHub Release
# artifact) to vendor a pre-built SPA without a local khelsutra checkout or Node. Already wired below.
#
# Usage:
#   scripts/bundle_ui.sh [SOURCE]
#     SOURCE = a khelsutra checkout dir (default: $KHELSUTRA_DIR or ../khelsutra) [builds it]
#            | a path/URL ending .tgz/.tar.gz [unpacks it]
#   BUILD=0 scripts/bundle_ui.sh <khelsutra-dir>   # skip npm; reuse an existing web/dist
set -euo pipefail

ENGINE_ROOT="$(cd "$(dirname "${BASH_SOURCE[0]}")/.." && pwd)"
UI_DEST="$ENGINE_ROOT/backend/ui"
SOURCE="${1:-${KHELSUTRA_DIR:-$ENGINE_ROOT/../khelsutra}}"

stage_dist() {  # populates $1 (a dist dir) into a clean $UI_DEST
  local dist="$1"
  [ -f "$dist/index.html" ] || { echo "ERROR: no index.html in $dist ? not a built SPA dist" >&2; exit 1; }
  rm -rf "$UI_DEST"
  mkdir -p "$UI_DEST"
  cp -r "$dist"/. "$UI_DEST"/
}

SPA_REPO="unknown"
SPA_COMMIT="unknown"

case "$SOURCE" in
  *.tgz|*.tar.gz|http://*|https://*)
    # OPTION 3: a published dist artifact.
    tmp="$(mktemp -d)"; trap 'rm -rf "$tmp"' EXIT
    if [[ "$SOURCE" == http*://* ]]; then
      echo "[bundle_ui] fetching dist artifact: $SOURCE"
      curl -fsSL "$SOURCE" -o "$tmp/dist.tgz"; SRC_TGZ="$tmp/dist.tgz";
```

```
SPA_REPO="$SOURCE"
      else
        SRC_TGZ="$SOURCE"; SPA_REPO="$SOURCE"
      fi
      tar -xzf "$SRC_TGZ" -C "$tmp"
      # accept either dist/* at the root or a top-level dir containing index.html
      if [ -f "$tmp/index.html" ]; then DIST="$tmp"; else DIST="$(dirname "$(find "$tmp" -name index.html | head -1)")"; fi
      stage_dist "$DIST"
      ;;
    *)
      # OPTION 1: a khelsutra checkout ? build it.
      WEB="$SOURCE/web"
      [ -d "$WEB" ] || { echo "ERROR: $WEB not found (pass a khelsutra checkout dir or a dist tarball)" >&2; exit 1; }
      SPA_REPO="$(git -C "$SOURCE" config --get remote.origin.url 2>/dev/null || echo "$SOURCE")"
      SPA_COMMIT="$(git -C "$SOURCE" rev-parse HEAD 2>/dev/null || echo unknown)"
      if [ "${BUILD:-1}" = "1" ]; then
        echo "[bundle_ui] building SPA in $WEB (npm ci && npm run build)"
        ( cd "$WEB" && npm ci && npm run build )
      else
        echo "[bundle_ui] BUILD=0 ? reusing existing $WEB/dist"
      fi
      stage_dist "$WEB/dist"
      ;;
  esac

  # Third-party notices for the vendored bundle (the python license-scan lane can't see JS/font blobs).
  cat > "$UI_DEST/THIRD_PARTY_NOTICES.md" <<NOTICE
  # Third-party notices ? vendored KhelSutra SPA (\`backend/ui/\`)

  This directory is the BUILT KhelSutra web SPA (MIT), vendored from $SPA_REPO @ $SPA_COMMIT by
  \`scripts/bundle_ui.sh\` (PACKAGING_PLAN.md P1b). The minified \`assets/*.js\` bundle includes
  third-party runtime dependencies (React and the SPA's other libraries ? MIT/BSD), and the
  \`assets/*.woff2\` are Noto Sans Kannada/Devanagari under the SIL Open Font License 1.1 (decision D7).
  Per the project's MIT/Apache/BSD-only guardrail, the SPA's dependencies are vetted in the SPA repo;
  see its package manifest/lockfile for the authoritative per-package licenses.
  NOTICE

  # Provenance + the SPA<->engine contract version (the engine's API_VERSION at bundle time). The
  # content hash is over the SPA build output only (relative paths ? reproducible across checkouts),
  # excluding the two generated metadata files.
  # ? Normalize sha256sum's mode marker to the POSIX text form ("<hash>  <path>", two spaces):
  # MSYS/git-bash sha256sum emits "<hash> *<path>" (binary-mode asterisk), which silently changed
  # the OUTER hash on Windows even over byte-identical payloads (#495 review). The sed keeps the
  # recipe's value identical on every platform; tests/test_ui_bundle.py mirrors it in Python.
  ENGINE_API_VERSION="$(grep -E '^API_VERSION\b[^=]*=' "$ENGINE_ROOT/backend/main.py" | grep -oE '[0-9]+' | head -1)"
  CONTENT_HASH="$( (cd "$UI_DEST" && find . -type f ! -name ui_version.json ! -name THIRD_PARTY_NOTICES.md | LC_ALL=C sort | xargs sha256sum | sed 's/^\([0-9a-f]*\) \*/\1  /') | sha256sum | cut -d' ' -f1)"
  BUILT_AT="$(date -u +%Y-%m-%dT%H:%M:%SZ 2>/dev/null || echo unknown)"
```

```
    cat > "$UI_DEST/ui_version.json" <<JSON
    {
      "spa_repo": "$SPA_REPO",
      "spa_commit": "$SPA_COMMIT",
      "engine_api_version": ${ENGINE_API_VERSION:-null},
      "content_hash": "$CONTENT_HASH",
      "built_at": "$BUILT_AT"
    }
    JSON

    echo "[bundle_ui] vendored SPA -> $UI_DEST (spa_commit=$SPA_COMMIT, engine_api_version=$ENGINE_API_VERSION)"
    echo "[bundle_ui] files:"; find "$UI_DEST" -type f | sed "s|$UI_DEST/||" | sort
```

41. assistant (2026-07-07T14:06:21.318Z)

```
The SPA source lives in a separate repo (`../khelsutra/web`). Let me inspect it for the compute-
picker.
```

42. user (2026-07-07T14:06:24.372Z)

```
/usr/bin/bash: line 1: cd: /c/Users/avidu/Projects/khelsutra: No such file or directory
---BRANCH/STATUS---
master
?? score_golden_gate.py
?? score_local.py
---COMPUTE REFS---
```

43. user (2026-07-07T14:06:35.436Z)

```
PWD=/c/Users/avidu/Projects/khelsutra-guru/khelsutra
---REMOTE---
https://github.com/avidullu/khelsutra.git
---BRANCH/STATUS---
master
---COMPUTE REFS---
web/src/api/client.ts
web/src/api/types.ts
web/src/App.tsx
web/src/components/IngestPanel.test.tsx
web/src/components/IngestPanel.tsx
web/src/components/SetupWizard.test.tsx
web/src/components/SetupWizard.tsx
web/src/components/upload.test.tsx
web/src/hooks/useIngestJob.test.ts
web/src/hooks/useIngestJob.ts
```

44. assistant (2026-07-07T14:06:43.783Z)

```
The SPA lives in a separate repo (`github.com/avidullu/khelsutra`). Let me pull it and inspect
the compute-picker code.
```

45. user (2026-07-07T14:06:48.487Z)

```
From https://github.com/avidullu/khelsutra
   9bf4104..c7265a4  master     -> origin/master
 * [new branch]      fix/161-video-name-on-retry -> origin/fix/161-video-name-on-retry
Updating 9bf4104..c7265a4
Fast-forward
---HEAD vs origin---
c7265a4c71678f9d4d3f3d201e8251db63a74897
c7265a4c71678f9d4d3f3d201e8251db63a74897
```

46. user (2026-07-07T14:06:54.415Z)

```
web\src\api\client.ts-91-  cid?: string,
web\src\api\client.ts-92-  locale?: string,
web\src\api\client.ts:93:  compute?: string,
web\src\api\client.ts-94-): Promise<ProcessResponse> {
web\src\api\client.ts:95:  // `locale` (the UI language) is recorded for PMF analytics (which
markets use the tool). `compute`
web\src\api\client.ts:96:  // is the compute-picker choice ("local" | "cloud") ? where DETECT
runs; omitted ? the server
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\client.ts-97-  // default. Both
optional + omitted when absent (JSON.stringify drops undefined).
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\client.ts-98-  return
request<ProcessResponse>("/process", {
web\src\api\client.ts-99-    method: "POST",
web\src\api\client.ts:100:    body: JSON.stringify({ video_path: videoPath, locale, compute }),
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\client.ts-101-    cid,
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\client.ts-102-  });
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\App.tsx-363-        )}
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\App.tsx-364-
web\src\App.tsx:365:        {/* P2c first-run wizard: before anything is loaded, walk a new user
through AI-key / compute /
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\App.tsx-366-            videos (from
/api/capabilities). Shows once per browser; then the ingest panel takes over. */}
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\App.tsx-367-        {!videoId &&
!wizardDone && <SetupWizard onClose={() => setWizardDone(true)} />}
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\types.ts-105-   *  skipped and the
engine's server-side fail-fast still applies. */
web\src\api\types.ts-106-  max_upload_mb?: number;
web\src\api\types.ts:107:  /** `cloud_available` = can THIS server satisfy a per-request
compute="cloud" dispatch (a Cloud Run
web\src\api\types.ts:108:   *  GPU Job)? The ingest compute-picker shows the "in the cloud"
option only when true. */
web\src\api\types.ts:109:  deployment?: { profile?: string; compute_target?: string;
cloud_available?: boolean };
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\types.ts-110-  /** Which storage
media THIS box can store INTO (P3). The medium picker renders only `usable`
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\types.ts-111-   *  backends;
`free_gb` is `null` for a cloud bucket (no readable capacity). Absent on older engines
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\types.ts-133-/** POST /api/assets ?
201 (P4). The video was stored into the chosen medium and registered as a
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\types.ts-134- *  library asset
keyed by the whole-file MD5 (`video_id`). `stored_uri` is the durable location to
web\src\api\types.ts:135: *  feed into POST /api/process (a bucket/Drive URI, or a local path
for "this computer"). */
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\api\types.ts-136-export interface
AssetResponse {
web\src\api\types.ts-137-  video_id: string;
--
web\src\hooks\useIngestJob.ts-29-  startedAt: number;
web\src\hooks\useIngestJob.ts-30-  locale?: string;
web\src\hooks\useIngestJob.ts:31:  compute?: string;
web\src\hooks\useIngestJob.ts-32-  displayName?: string;
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-33-}
--
web\src\hooks\useIngestJob.ts-67-      startedAt: Number(parsed.startedAt),
web\src\hooks\useIngestJob.ts-68-      locale: parsed.locale ? String(parsed.locale) :
undefined,
web\src\hooks\useIngestJob.ts:69:      compute: parsed.compute ? String(parsed.compute) :
undefined,
web\src\hooks\useIngestJob.ts-70-      displayName: parsed.displayName ?
String(parsed.displayName) : undefined,
```

```
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-71-    };
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-236-
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-237-  const
submit = useCallback(
web\src\hooks\useIngestJob.ts:238:    (rawUri: string, locale?: string, compute?: string,
displayName?: string) => {
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-239-      const
uri = rawUri.trim();
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-240-      if
(!uri) {
--
web\src\hooks\useIngestJob.ts-263-        let jobId: string;
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-264-        try {
web\src\hooks\useIngestJob.ts:265:          const resp = await processVideo(uri, cid, locale,
compute);
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-266-
jobId = resp.job_id;
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-267-        }
catch (err) {
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-270-        }
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-271-        if
(run.cancelled) return;
web\src\hooks\useIngestJob.ts:272:        const active: ActiveIngestJob = { jobId, uri, cid,
startedAt, locale, compute, displayName };
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-273-
writeActiveJob(active);
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.ts-274-
replaceJobInUrl(jobId);
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.test.ts-106-
expect(window.location.search).toContain("job=jPersist");
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.test.ts-107-
const saved = JSON.parse(window.localStorage.getItem(ACTIVE_JOB_KEY) ?? "{}") as Record<string,
unknown>;
web\src\hooks\useIngestJob.test.ts:108:    expect(saved).toMatchObject({ jobId: "jPersist", uri:
"gs://b/x.mp4", locale: "en", compute: "local" });
web\src\hooks\useIngestJob.test.ts-109-    expect(result.current.phase).toMatchObject({ status:
"polling", jobId: "jPersist", resumed: false });
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\hooks\useIngestJob.test.ts-110-  });
--
web\src\components\IngestPanel.test.tsx-20-  return vi.fn(async (url: string | URL, init?:
RequestInit) => {
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-21-
const u = String(url);
web\src\components\IngestPanel.test.tsx:22:    // The panel probes /api/capabilities on mount
(compute-picker) ? answer it benignly so the
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-23-
// ingest-flow assertions don't see a stray "unexpected fetch". No cloud ? no picker.
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-24-
if (u === "/api/capabilities") return httpOk(200, {});
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-34-
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-35-/**
Like fetchSeq but lets a test set the capabilities payload + capture the /api/process body ?
used
web\src\components\IngestPanel.test.tsx:36: *  to drive and assert the compute-picker. */
web\src\components\IngestPanel.test.tsx-37-function fetchAll(opts: {
web\src\components\IngestPanel.test.tsx-38-  caps?: unknown;
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-158-
});
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-159-
web\src\components\IngestPanel.test.tsx:160:  // --- compute-picker (item 3)
```

```
--------------------------------------------------------------------
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-161-
web\src\components\IngestPanel.test.tsx:162:  it("hides the compute picker when the server can't
dispatch to the cloud", async () => {
web\src\components\IngestPanel.test.tsx-163-    vi.stubGlobal("fetch", fetchAll({ caps: {
deployment: { cloud_available: false } } }));
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-164-
render(<IngestPanel onLoaded={vi.fn()} />);
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-165-
await act(async () => {}); // let the capabilities probe resolve
web\src\components\IngestPanel.test.tsx:166:    expect(screen.queryByTestId("compute-
picker")).toBeNull();
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-167-
});
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-168-
web\src\components\IngestPanel.test.tsx:169:  it("defaults to local and sends compute=local when
the cloud is available", async () => {
web\src\components\IngestPanel.test.tsx-170-    const bodies: Record<string, unknown>[] = [];
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-171-
vi.stubGlobal(
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-178-
);
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-179-
render(<IngestPanel onLoaded={vi.fn()} />);
web\src\components\IngestPanel.test.tsx:180:    await screen.findByTestId("compute-picker");
web\src\components\IngestPanel.test.tsx-181-    typeUri("gs://bucket/match.mp4");
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-182-
clickProcess();
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-183-
await act(async () => {});
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-184-
expect(bodies).toHaveLength(1);
web\src\components\IngestPanel.test.tsx:185:    expect(bodies[0].compute).toBe("local");
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-186-
});
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-187-
--
web\src\components\IngestPanel.test.tsx-234-      default_medium: "local",
web\src\components\IngestPanel.test.tsx-235-      backends: [
web\src\components\IngestPanel.test.tsx:236:        { id: "local", kind: "local", label: "This
computer", usable: true, free_gb: 100 },
web\src\components\IngestPanel.test.tsx-237-        { id: "gcs", kind: "gcs", label: "GCS",
usable: true, free_gb: null },
web\src\components\IngestPanel.test.tsx-238-        { id: "s3", kind: "s3", label: "S3", usable:
false, free_gb: null },
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-361-
);
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-362-
render(<IngestPanel onLoaded={vi.fn()} />);
web\src\components\IngestPanel.test.tsx:363:    await screen.findByTestId("compute-picker");
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-364-
fireEvent.click(screen.getByTestId("medium-bucket"));
web\src\components\IngestPanel.test.tsx-365-    fireEvent.change(screen.getByLabelText(\storage
destination\i), { target: { value: "gcs://b/videos" } });
--
C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx-373-
expect(screen.getByTestId("selected-file").textContent).toContain("2 MB selected");

[Showing results with pagination = limit: 120]
```

47. assistant (2026-07-07T14:07:01.274Z)

Now let me read the `IngestPanel.tsx` compute-picker logic.

48. user (2026-07-07T14:07:03.401Z)

```
1-// The front-half ingest screen (appliance S1/S2; tracker B-P2 + T4). Two ways to point the
tool at a
2:// video: UPLOAD a local file (primary ? alpha users have a file, not a URI; chunked to the
engine's
3-// /api/upload/* endpoints) or paste a `gs://` bucket URI / host path (secondary). Both
converge on
4-// the SAME back-half: POST /api/process ? poll GET /api/jobs/{id} ? hand the video_id to App.
Honest
--
17-import { StatusChip, Spinner } from "./StatusChip";
18-
19:type Compute = "local" | "cloud";
20-// The three storage-medium choices the picker offers. "bucket" covers s3 OR gcs (the user
pastes a
21-// full scheme'd URI); "drive" is a mounted Google Drive folder (STUDIO_MEDIA_LIFECYCLE.md
P5, D14).
22:type Medium = "local" | "bucket" | "drive";
23-
24-interface Props {
--
77-  const [selectedFile, setSelectedFile] = useState<File | null>(null);
78-  const { phase, submit, cancel: cancelIngest, reset: resetIngest, busy: ingestBusy } =
useIngestJob(onLoaded);
79:  // The UI locale recorded with the submission (PMF analytics + the localized reel
watermark).
80:  const locale = i18n.resolvedLanguage ?? i18n.language;
81-
82:  // Compute-picker (PACKAGING compute-picker, item 3): if this server can dispatch to a
cloud GPU
83:  // (capabilities.deployment.cloud_available), let the user choose where DETECT runs.
Default LOCAL
84-  // (free, no surprise cost); CLOUD requires an explicit cost acknowledgement before it can
run.
85-  const [cloudAvailable, setCloudAvailable] = useState(false);
86:  const [compute, setCompute] = useState<Compute>("local");
87-  const [cloudAck, setCloudAck] = useState(false);
88-  // Storage-medium picker (P5): which media THIS box can store into
(capabilities.storage.backends,
89:  // usable-only). Absent/local-only ? no picker (today's flow unchanged).
90-  const [backends, setBackends] = useState<StorageBackendInfo[]>([]);
91:  const [medium, setMedium] = useState<Medium>("local");
92-  const [mediumUri, setMediumUri] = useState("");
93-  const [storing, setStoring] = useState(false);
--
101-      .then((c) => {
102-        if (!alive) return;
103:        setCloudAvailable(c.deployment?.cloud_available === true);
104-        setBackends(c.storage?.backends ?? []);
105-        setMaxUploadBytes(
--
113-  }, []);
114-  // Only send a choice when the picker is actually shown; otherwise omit it (? server
default).
115:  const sendCompute = cloudAvailable ? compute : undefined;
116:  // CLOUD selected but the cost note not yet accepted ? block submit (don't silently run
local, and
117-  // don't dispatch a paid job without consent).
118:  const cloudBlocked = cloudAvailable && compute === "cloud" && !cloudAck;
119-
120:  // Which medium segments to offer: local always; "bucket" if s3 OR gcs is usable; "drive"
if a Drive
121:  // mount is usable. The picker only appears when there's a REAL choice (a non-local usable
backend).
```

```
122-  const canUse = (id: string) => backends.some((b) => b.id === id && b.usable);
123-  const canBucket = canUse("s3") || canUse("gcs");
124-  const canDrive = canUse("gdrive");
125:  const mediumOptions: Medium[] = ["local", ...(canBucket ? ["bucket" as const] : []), ...(canDrive ? ["drive" as const] : [])];
126-  const showMediumPicker = mediumOptions.length > 1;
127:  const needsMediumUri = medium !== "local";
128:  // A non-local medium needs a destination URI before anything can be stored/submitted.
129-  const mediumBlocked = needsMediumUri && mediumUri.trim() === "";
130-
131:  // Store the obtained source (an uploaded local path, or a pasted at-rest URI) INTO the chosen medium,
132-  // then process the durable copy. A failed store is loud (surfaced like every other ingest error).
133-  const storeThenProcess = async (src: { uploadedPath?: string; sourceUri?: string }, displayName?: string) => {
--
138-      const asset = await createAsset({ ...src, mediumUri: mediumUri.trim() }, cid);
139-      log("info", "asset.stored", { cid, video_id: asset.video_id, medium: asset.medium, copied: asset.copied });
140:      submit(asset.stored_uri, locale, sendCompute, displayName ?? sourceLabel(asset.stored_uri) ?? undefined); // index the stored copy
141-    } catch (err) {
142-      setStoreError(classifyError(err));
--
147-  };
148-
149:  // On upload completion: local medium ? process the upload directly (today's flow); a non-local
150-  // medium ? store a durable copy into it first, then process. The arrow is recreated each render
151-  // (useUpload reads onCompleteRef.current), so `medium`/`mediumUri`/`sendCompute` are always fresh.
152-  const upload = useUpload((path, filename) => {
153:    if (medium === "local") submit(path, locale, sendCompute, filename);
154-    else void storeThenProcess({ uploadedPath: path }, filename);
155-  }, maxUploadBytes);
--
161-    };
162-  }, [busy, onBusyChange]);
163:  // Uploading a FILE into a non-local medium needs a destination first. A pasted cloud/host link is
164-  // already an at-rest SOURCE and should process directly regardless of the storage picker state.
165-  const fileInputsDisabled = busy || cloudBlocked || mediumBlocked;
--
170-    if (uriInputsDisabled) return;
171-    // The URI form is always a SOURCE (a bucket URL / host path to index directly). The medium picker
172:    // only controls where a locally picked file is stored before indexing.
173-    const displayName = sourceLabel(uri);
174-    setSelectedFile(null);
175-    setActiveSource(displayName);
176:    submit(uri, locale, sendCompute, displayName ?? undefined);
177-  };
178-  const onFile = (e: ChangeEvent<HTMLInputElement>) => {
--
217-  // Shared mapping with the build flow (lib/errorText): an `engine` failure carries the engine's own
218-  // message (a 400 scheme/path/extension detail, a worker error); every other code maps to a friendly
219:  // localized line. Surface whichever half failed (upload or engine) ? they're mutually exclusive.
220-  const errorState =
221-    upload.phase.status === "error"
```

```
--
253-        {/* UPLOAD-MOMENT DISCLOSURE (khelsutra #154): on a cloud-capable server (the hosted
alpha) the
254-            footage leaves the device, so testers get an honest, plain-English disclosure of
where it's
255:            stored + a link to the full policy. Suppressed on a local appliance
(cloudAvailable=false),
256-            where nothing is uploaded. Copy: ALPHA_LAUNCH_REVIEW_2026-07-04.md §4.4. */}
257-        {cloudAvailable && (
--
276-            LOCAL; CLOUD reveals a cost note + a required acknowledgement before any submit
can run. */}
277-        {cloudAvailable && (
278:        <fieldset className="mt-4 rounded-xl border border-black/10 p-3" data-
testid="compute-picker">
279-          <legend className="px-1 text-xs font-semibold uppercase tracking-wide text-
ink/40">
280:            {t("ingest.compute.legend")}
281-          </legend>
282-          <div className="mt-1 flex flex-wrap gap-2">
283:            {(["local", "cloud"] as const).map((opt) => (
284-              <button
285-                key={opt}
286-                type="button"
287:                aria-pressed={compute === opt}
288-                disabled={busy}
289-                onClick={() => setCompute(opt)}
290-                className={
291-                  "min-h-11 rounded-xl border px-4 py-2 text-sm font-semibold
disabled:opacity-40 " +
292:                  (compute === opt
293-                    ? "border-green bg-green/10 text-ink"
294-                    : "border-black/15 text-ink/70 hover:border-green")
295-                }
296-              >
297:                {t(`ingest.compute.${opt}`)}
298-              </button>
299-            ))}
300-          </div>
301:          {compute === "cloud" && (
302:            <div className="mt-3" data-testid="compute-cost">
303:              <p className="text-xs text-ink/60">{t("ingest.compute.cloudCost")}</p>
304-              <label className="mt-2 flex items-center gap-2 text-sm text-ink/80">
305-                <input
--
310-                  className="h-4 w-4"
311-                />
312:                {t("ingest.compute.ack")}
313-              </label>
314-              {cloudBlocked && (
315:                <p className="mt-1 text-xs text-amber-700" data-testid="compute-ack-
required">
316:                  {t("ingest.compute.ackRequired")}
317-                </p>
318-              )}
--
322-        )}
323-
324:        {/* MEDIUM PICKER (P5, D14): how the user wants to add this video. A non-local
medium still
325-            stores a durable MD5-keyed copy before indexing, but the copy uses user-facing
source
326-            language because the operator is choosing the route into Studio, not thinking in
backend
--
```

```
376-        )}
377-
378:        {/* PRIMARY: upload a local file (T4). A native file input, visually a button
(?44px, A4). */}
379-        <div className="mt-4">
380-        <label
--
568-                  </li>
569-                  <li>
570:                    <Trans i18nKey="errors.fileTooLargeHelp.computer" components={{ b:
<strong /> }} />
571-                  </li>
572-                  <li>
```

49. assistant (2026-07-07T14:07:59.105Z)

This is a two-repo change. Let me read the full capabilities effect in `IngestPanel.tsx` and the main.py capabilities payload to confirm `compute_target` is available.

50. user (2026-07-07T14:08:00.355Z)

```
94        const [storeError, setStoreError] = useState<IngestError | null>(null);
95        // The engine's max upload size (BYTES), so an oversized file fails INSTANTLY client-
side instead of
96        // uploading into a 413. Absent ? undefined ? no guard (the engine still fail-fasts).
97        const [maxUploadBytes, setMaxUploadBytes] = useState<number | undefined>(undefined);
98        useEffect(() => {
99          let alive = true;
100         getCapabilities()
101           .then((c) => {
102             if (!alive) return;
103             setCloudAvailable(c.deployment?.cloud_available === true);
104             setBackends(c.storage?.backends ?? []);
105             setMaxUploadBytes(
106               typeof c.max_upload_mb === "number" && c.max_upload_mb > 0 ? c.max_upload_mb *
1024 * 1024 : undefined,
107             );
108           })
109           .catch((err) => log("warn", "capabilities.unavailable", { error: String(err) }));
110         return () => {
111           alive = false;
112         };
113       }, []);
114       // Only send a choice when the picker is actually shown; otherwise omit it (? server
default).
115       const sendCompute = cloudAvailable ? compute : undefined;
116       // CLOUD selected but the cost note not yet accepted ? block submit (don't silently
run local, and
117       // don't dispatch a paid job without consent).
118       const cloudBlocked = cloudAvailable && compute === "cloud" && !cloudAck;
```

51. assistant (2026-07-07T14:08:08.232Z)

Let me confirm the capabilities payload exposes `compute_target` and check the API_VERSION for the bundle test.

52. user (2026-07-07T14:08:08.943Z)

```
392                 "ffmpeg_path": ffmpeg,
393                 "ffprobe_path": ffprobe,
394             },
395             "gpu": {"nvidia_smi_present": bool(nvidia_smi)},
396             "gemini_key_present": bool(os.environ.get("GEMINI_API_KEY")),
397             "wasb_weights_present": bool(weights_path) and os.path.exists(weights_path),
```

```
398             **uploads_api.upload_caps(cfg),  # #480 item 5: the SPA pre-flight reads max_upload_mb
399             "deployment": {
400                 "profile": getattr(deployment, "profile", "") or "",
401                 "compute_target": getattr(deployment, "compute_target", "") or "",
402                 # item 3: can this server satisfy a per-request compute="cloud" dispatch? The SPA shows
403                 # the "run in the cloud" toggle only when true (else it's local-only).
404                 "cloud_available": _cloud_available(cfg),
405             },
406             # P3: which storage media THIS box can store INTO (local always; s3/gcs if SDK+config;
407             # gdrive if a mount resolves). The P5 medium picker renders only usable backends. No secrets.
408             "storage": storage_capabilities(
409                 getattr(cfg, "storage", None), local_free_path=output_base(cfg)
```

53. assistant (2026-07-07T14:08:15.532Z)

Line 397 already computes `wasb_weights_present`. Let me read how `weights_path` is resolved in the capabilities handler.

54. user (2026-07-07T14:08:16.508Z)

```
360             "ui": _bundled_ui_version(),
361         }
362
363
364     @app.get("/api/capabilities")
365     def capabilities(request: Request):
366         """Honest probe of what THIS box can do ? the first-run wizard renders the compute step from
367         it (PACKAGING_PLAN.md P1). No secrets: reports only whether a Gemini key is PRESENT, never its
368         value. Best-effort + never raises (the wizard degrades gracefully)."""
369         cfg = request.app.state.config
370         indexing = getattr(cfg, "indexing", None)
371         deployment = getattr(cfg, "deployment", None)
372         ffmpeg = shutil.which(ffmpeg_bin())  # P2a: honor RALLY_FFMPEG / RALLY_FFPROBE
373         ffprobe = shutil.which(ffprobe_bin())
374         # WASB weights are best-effort: env first (matches native_wasb_runner), then config.
375         wasb = getattr(indexing, "wasb", None)
376         weights_path = (
377             os.environ.get("WASB_WEIGHTS") or getattr(wasb, "weights_path", "") or ""
378         )
379         # Cheap GPU heuristic: nvidia-smi on PATH (matches scripts/doctor.py). Deliberately NOT
380         # torch.cuda.is_available() ? torch is absent in the LIGHT tier (so it would false-negative on a
381         # real GPU box) and importing it would add seconds to this probe. The detector healthcheck is the
382         # authority on whether GPU inference actually works; this is just the wizard's first-order signal.
383         nvidia_smi = shutil.which("nvidia-smi")
384         return {
385             "api_version": API_VERSION,
386             "engine_version": _engine_version(),
387             "segmenters": sorted(segmenter_registry.list_available().keys()),
388             "default_segmenter": getattr(indexing, "default_segmenter", None),
389             "ffmpeg": {
390                 "ffmpeg": bool(ffmpeg),
391                 "ffprobe": bool(ffprobe),
392                 "ffmpeg_path": ffmpeg,
393                 "ffprobe_path": ffprobe,
394             },
```

```
            "gpu": {"nvidia_smi_present": bool(nvidia_smi)},
            "gemini_key_present": bool(os.environ.get("GEMINI_API_KEY")),
            "wasb_weights_present": bool(weights_path) and os.path.exists(weights_path),
            **uploads_api.upload_caps(cfg),  # #480 item 5: the SPA pre-flight reads max_upload_mb
            "deployment": {
```

55. assistant (2026-07-07T14:09:50.906Z)

The capabilities payload exposes `compute_target` ? clean signal for the SPA. Let me read the SPA test helpers to match conventions.

56. user (2026-07-07T14:09:52.558Z)

```
 *  to drive and assert the compute-picker. */
function fetchAll(opts: {
  caps?: unknown;
  process?: unknown;
  jobs?: JobStatus[];
  onProcessBody?: (body: Record<string, unknown>) => void;
}): FetchFn {
  let i = 0;
  const jobs = opts.jobs ?? [];
  return vi.fn(async (url: string | URL, init?: RequestInit) => {
    const u = String(url);
    if (u === "/api/capabilities") return httpOk(200, opts.caps ?? {});
    if (u === "/api/process" && (init?.method ?? "GET") === "POST") {
      opts.onProcessBody?.(JSON.parse(String(init?.body ?? "{}")) as Record<string, unknown>);
      return opts.process ?? httpOk(202, { job_id: "j", status: "queued", poll: "/api/jobs/j" });
    }
    if (u.startsWith("/api/jobs/")) {
      const body = jobs[Math.min(i, jobs.length - 1)];
      i += 1;
      return httpOk(200, body);
    }
    throw new Error(`unexpected fetch ${u}`);
  }) as unknown as FetchFn;
}

beforeEach(async () => {
  window.localStorage.clear();
  window.history.replaceState(null, "", "/");
  await i18n.changeLanguage("en"); // guarantees init completed + a known locale for the assertions
});

afterEach(() => {
  cleanup();
  vi.restoreAllMocks();
  vi.unstubAllGlobals();
  vi.useRealTimers();
});

const typeUri = (value: string) =>
  fireEvent.change(screen.getByLabelText(/video location/i), { target: { value } });
const clickProcess = () => fireEvent.click(screen.getByRole("button", { name: /process video/i }));
const clickStartUpload = () => fireEvent.click(screen.getByRole("button", { name: /start upload and find rallies/i }));

describe("IngestPanel", () => {
  it("submits a URI, polls, and hands the video_id to onLoaded", async () => {
    vi.useFakeTimers();
```

```
    vi.stubGlobal(
      "fetch",
      fetchSeq(httpOk(202, { job_id: "j", status: "queued", poll: "/api/jobs/j" }), [
        { job_id: "j", status: "running", progress: "segmenting" },
        { job_id: "j", status: "done", video_id: "vidZ", result: { status: "success", video_id: "vidZ" } },
      ]),
    );
    const onLoaded = vi.fn();
    render(<IngestPanel onLoaded={onLoaded} />);
    typeUri("gs://bucket/match.mp4");
    clickProcess();

    await act(async () => {
      await vi.advanceTimersByTimeAsync(0); // flush submit ? start polling
    });
    await act(async () => {
      await vi.advanceTimersByTimeAsync(1000); // running tick
    });
    await act(async () => {
      await vi.advanceTimersByTimeAsync(1500); // done tick
    });
    expect(onLoaded).toHaveBeenCalledWith("vidZ", "match.mp4");
  });

  it("shows the engine's own 400 detail (e.g. an unsupported scheme)", async () => {
    vi.stubGlobal("fetch", fetchSeq(httpOk(400, { detail: "unsupported storage URI scheme: ftp://" }), []));
    render(<IngestPanel onLoaded={vi.fn()} />);
    typeUri("ftp://nope");
    clickProcess();
    expect(await screen.findByText(/unsupported storage URI scheme/i)).toBeTruthy();
    fireEvent.click(screen.getByRole("button", { name: /try again/i }));
    expect(screen.queryByTestId("ingest-error")).toBeNull();
  });

  it("shows a failed pipeline verdict as the engine's reason, not as no rallies", async () => {
    vi.useFakeTimers();
    vi.stubGlobal(
      "fetch",
      fetchSeq(httpOk(202, { job_id: "jFail", status: "queued", poll: "/api/jobs/jFail" }), [
        { job_id: "jFail", status: "done", video_id: "vidFail", result: { status: "failed", message: "Video failed validation: blur: blurry" } },
      ]),
    );
    render(<IngestPanel onLoaded={vi.fn()} />);
    typeUri("gs://bucket/blurry.mp4");
    clickProcess();
    await act(async () => {
      await vi.advanceTimersByTimeAsync(0);
    });
    await act(async () => {
      await vi.advanceTimersByTimeAsync(1000);
    });
    expect(screen.getByTestId("ingest-error").textContent).toMatch(/Video failed validation: blur: blurry/i);
    expect(screen.getByTestId("ingest-error").textContent).not.toMatch(/found no rallies/i);
  });

  it("shows a friendly, localized message when the engine is unreachable", async () => {
    vi.stubGlobal("fetch", vi.fn(async () => {
      throw new Error("offline");
```

```
    }) as unknown as FetchFn);
    render(<IngestPanel onLoaded={vi.fn()} />);
    typeUri("gs://bucket/match.mp4");
    clickProcess();
    expect(await screen.findByText(/can't reach khelsutra/i)).toBeTruthy();
  });

  it("rejects an empty URI with a hint and never submits to the engine", () => {
    // The mount-time capabilities probe may fire; assert specifically that /api/process never does.
    const f = fetchAll({ caps: {} });
    vi.stubGlobal("fetch", f);
    render(<IngestPanel onLoaded={vi.fn()} />);
    clickProcess();
    expect(screen.getByTestId("ingest-error")).toBeTruthy();
    const submitted = (f as unknown as { mock: { calls: unknown[][] } }).mock.calls.some(
      ([u]) => String(u) === "/api/process",
    );
    expect(submitted).toBe(false);
  });

  // --- compute-picker (item 3) -------------------------------------------------------------------

  it("hides the compute picker when the server can't dispatch to the cloud", async () => {
    vi.stubGlobal("fetch", fetchAll({ caps: { deployment: { cloud_available: false } } }));
    render(<IngestPanel onLoaded={vi.fn()} />);
    await act(async () => {}); // let the capabilities probe resolve
```

57. user (2026-07-07T14:09:57.000Z)

```
  it("defaults to local and sends compute=local when the cloud is available", async () => {
    const bodies: Record<string, unknown>[] = [];
    vi.stubGlobal(
      "fetch",
      fetchAll({
        caps: { deployment: { cloud_available: true } },
        onProcessBody: (b) => bodies.push(b),
        jobs: [{ job_id: "j", status: "done", video_id: "vidL", result: { status: "success", video_id: "vidL" } }],
      }),
    );
    render(<IngestPanel onLoaded={vi.fn()} />);
    await screen.findByTestId("compute-picker");
    typeUri("gs://bucket/match.mp4");
    clickProcess();
    await act(async () => {});
    expect(bodies).toHaveLength(1);
    expect(bodies[0].compute).toBe("local");
  });

  it("shows a pasted host-path basename while processing and clears it on cancel", async () => {
    vi.useFakeTimers();
    vi.stubGlobal(
      "fetch",
      fetchAll({
        caps: {},
        jobs: [{ job_id: "j", status: "running", progress: "hashing" }],
      }),
    );
```

```
    render(<IngestPanel onLoaded={vi.fn()} />);

    typeUri("C:\\videos\\Sunday Match.mov");
    clickProcess();
    await act(async () => {
      await vi.advanceTimersByTimeAsync(0);
    });

    expect(screen.getByTestId("ingest-source").textContent).toContain("Sunday Match.mov");
    fireEvent.click(screen.getByRole("button", { name: /cancel/i }));
    expect(screen.queryByTestId("ingest-source")).toBeNull();
  });

  it("shows a decoded URL basename while processing", async () => {
    vi.useFakeTimers();
    vi.stubGlobal(
      "fetch",
      fetchAll({
        caps: {},
        jobs: [{ job_id: "j", status: "running", progress: "hashing" }],
      }),
    );
    render(<IngestPanel onLoaded={vi.fn()} />);

    typeUri("https://cdn.example/video/Sunday%20Final.mp4?download=1");
    clickProcess();
    await act(async () => {
      await vi.advanceTimersByTimeAsync(0);
    });

    expect(screen.getByTestId("ingest-source").textContent).toContain("Sunday Final.mp4");
  });

  // --- medium picker (P5, store-into-medium)
```
----

58. user (2026-07-07T14:10:11.893Z)

```
          </div>
          {compute === "cloud" && (
            <div className="mt-3" data-testid="compute-cost">
              <p className="text-xs text-ink/60">{t("ingest.compute.cloudCost")}</p>
              <label className="mt-2 flex items-center gap-2 text-sm text-ink/80">
                <input
                  type="checkbox"
                  checked={cloudAck}
                  disabled={busy}
                  onChange={(e) => setCloudAck(e.target.checked)}
                  className="h-4 w-4"
                />
                {t("ingest.compute.ack")}
              </label>
              {cloudBlocked && (
                <p className="mt-1 text-xs text-amber-700" data-testid="compute-ack-required">
                  {t("ingest.compute.ackRequired")}
                </p>
              )}
            </div>
          )}
        </fieldset>
```

59. assistant (2026-07-07T14:10:16.567Z)

Let me check for existing cloud-selection test patterns (clicking the cloud button + ack).

60. user (2026-07-07T14:10:18.923Z)

```
22:    // The panel probes /api/capabilities on mount (compute-picker) ? answer it benignly so
the
36: *  to drive and assert the compute-picker. */
160:  // --- compute-picker (item 3)
------------------------------------------------------------------
162:  it("hides the compute picker when the server can't dispatch to the cloud", async () => {
166:    expect(screen.queryByTestId("compute-picker")).toBeNull();
169:  it("defaults to local and sends compute=local when the cloud is available", async () => {
180:    await screen.findByTestId("compute-picker");
363:    await screen.findByTestId("compute-picker");
375:    expect(screen.getByTestId("compute-picker")).toBeTruthy();
384:    expect(screen.queryByTestId("compute-picker")).toBeNull();
466:  it("gates cloud on a cost ack, then sends compute=cloud", async () => {
477:    await screen.findByTestId("compute-picker");
482:    expect(screen.getByTestId("compute-cost")).toBeTruthy();
489:    fireEvent.click(screen.getByRole("checkbox"));
494:    expect(bodies[0].compute).toBe("cloud");
```

61. user (2026-07-07T14:10:23.427Z)

```
466       it("gates cloud on a cost ack, then sends compute=cloud", async () => {
467         const bodies: Record<string, unknown>[] = [];
468         vi.stubGlobal(
469           "fetch",
470           fetchAll({
471             caps: { deployment: { cloud_available: true } },
472             onProcessBody: (b) => bodies.push(b),
473             jobs: [{ job_id: "j", status: "done", video_id: "vidC", result: { status:
"success", video_id: "vidC" } }],
474           }),
475         );
476         render(<IngestPanel onLoaded={vi.fn()} />);
477         await screen.findByTestId("compute-picker");
478         // Type the URI while LOCAL (the input is enabled), then switch to CLOUD.
479         typeUri("gs://bucket/match.mp4");
480         fireEvent.click(screen.getByRole("button", { name: /in the cloud/i }));
481         // Cost note + ack appear; submit is gated until the cost is accepted.
482         expect(screen.getByTestId("compute-cost")).toBeTruthy();
483         expect(screen.getByTestId("compute-ack-required")).toBeTruthy();
484         const submitBtn = screen.getByRole("button", { name: /process video/i }) as
HTMLButtonElement;
485         expect(submitBtn.disabled).toBe(true);
486         clickProcess(); // a disabled button does nothing ? still no submit
487         expect(bodies).toHaveLength(0);
488         // Accept the cost ? the gate lifts ? run on the cloud.
489         fireEvent.click(screen.getByRole("checkbox"));
490         expect(submitBtn.disabled).toBe(false);
491         clickProcess();
492         await act(async () => {});
493         expect(bodies).toHaveLength(1);
494         expect(bodies[0].compute).toBe("cloud");
495       });
496
497       it("shows the upload-moment privacy disclosure + a link to the full policy on a cloud
server (#154)", async () => {
498         vi.stubGlobal("fetch", fetchAll({ caps: { deployment: { cloud_available: true } }
}));
499         render(<IngestPanel onLoaded={vi.fn()} />);
500         const box = await screen.findByTestId("upload-disclosure");
501         // Honest about WHERE footage goes (the blocker this closes), and no "never leaves
your device" lie.
```

```
        expect(box.textContent).toMatch(/Google Cloud bucket in Singapore/);
        expect(box.textContent).not.toMatch(/never leave/i);
        const link = screen.getByRole("link", { name: /privacy policy/i }) as HTMLAnchorElement;
        expect(link.getAttribute("href")).toBe("https://khelsutra.guru/privacy");
```

62. assistant (2026-07-07T14:10:53.009Z)

I have enough understanding of both repos. Let me verify the `FusionConfig.substrate` default before implementing.

63. user (2026-07-07T14:10:54.094Z)

```
class FusionConfig(BaseModel):
    """Config for the multi-signal `fusion_hybrid` segmenter (MULTI_SIGNAL_FUSION_PLAN P2).
    Every default reproduces control behaviour: the fusion segmenter persists the
    shuttle net-crossing count (Gate A signal) and ? only when ``cue_flags['person']`` is
    true ? the person-cue features, but it does NOT gate the served handoff unless a
    threshold is raised (``min_crossings``/``min_players`` default 0 = OFF). The court mask
    is optional: ``court_polygon=None`` ? Gate B counts every detection (player-count-only);
    supplied ? off-court persons (spectators / adjacent courts) are excluded.
    """
    # Which trajectory substrate seeds the windows (shuttle stays primary).
    substrate: Literal["wasb", "tracknet"] = "wasb"
    # Windowing velocity threshold for the fusion family (the engine reads
    # indexing.<family>.velocity_threshold). Default matches the substrates' 0.02; set
    # equal to indexing.wasb.velocity_threshold to A/B fusion against a tuned wasb run.
    velocity_threshold: float = 0.02
    # Per-cue enable flags, e.g. {"person": true}. Person cue is OFF unless explicitly
    # enabled ? so the default path never imports torch / touches the GPU.
    cue_flags: dict[str, bool] = Field(default_factory=dict)
    # Served Gate A: drop windows whose shuttle net-crossings < this from the AI handoff.
    # 0 = OFF (no gating; persisted candidates are always the full superset for offline re-scoring).
    min_crossings: int = Field(default=0, ge=0)
    # Served Gate B: when the person cue is on, drop windows with median in-court players
```

64. assistant (2026-07-07T14:11:01.171Z)

I have full understanding of both repos. Let me set up tracking and start implementing. First, the backend branch (from `origin/master`).

65. user (2026-07-07T14:11:04.856Z)

```
branch 'fix/hosted-local-detection-guardrail' set up to track 'origin/master'.
Switched to a new branch 'fix/hosted-local-detection-guardrail'
---VERIFY---
fix/hosted-local-detection-guardrail
3be0935 docs(runlog): DEPLOY_REPORT addendum ? control-plane redeployed to the forwarding image (#510)
---origin/master..HEAD (should be empty)---
```

66. assistant (2026-07-07T14:11:08.766Z)

Now let me implement the backend guardrail. First the helper near `_cloud_available`:

67. user (2026-07-07T14:11:19.180Z)

The file C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\backend\orchestration.py has been updated successfully. (file state is current in your context ? no need to Read it back)

68. assistant (2026-07-07T14:11:23.049Z)

Now the guardrail in the `choice == "local"` branch:

69. user (2026-07-07T14:11:29.078Z)

The file C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\backend\orchestration.py has been updated successfully. (file state is current in your context ? no need to Read it back)

70. assistant (2026-07-07T14:11:52.058Z)

Now let me add the tests. I'll insert a dedicated section after the API-boundary tests (before the provenance section).

71. user (2026-07-07T14:12:14.270Z)

The file C:\Users\avidu\Projects\khelsutra-guru\badminton-highlight-indexer\tests\test_api_cloud_dispatch.py has been updated successfully. (file state is current in your context ? no need to Read it back)

72. assistant (2026-07-07T14:12:40.969Z)

Now let me verify the changed backend files pass ruff/mypy and run the targeted tests. Let me check the Python environment first.

73. user (2026-07-07T14:12:44.353Z)

```
/c/Users/avidu/AppData/Local/Programs/Python/Python313/python
Python 3.13.13
---ruff/mypy present?---
ruff 0.15.16
mypy 2.1.0 (compiled: yes)
```

74. user (2026-07-07T14:12:58.842Z)

```
===RUFF===
All checks passed!
===MYPY===
backend\annotations\base.py:88: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\sports\__init__.py:9: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs  [annotation-unchecked]
backend\pipeline\validators\base.py:44: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 1 source file
```

75. assistant (2026-07-07T14:13:04.843Z)

Ruff and mypy clean. Now the targeted tests:

76. user (2026-07-07T14:13:42.765Z)

```
..................................                                             [100%]
============================== warnings summary ===============================
```

```
tests/test_api_cloud_dispatch.py::test_full_api_process_to_query_via_cloud
tests/test_api_cloud_dispatch.py::test_full_api_process_to_query_via_cloud
tests/test_api_cloud_dispatch.py::test_local_override_survives_api_boundary
tests/test_api_cloud_dispatch.py::test_local_override_survives_api_boundary
tests/test_api_cloud_dispatch.py::test_cloud_override_survives_api_boundary_on_local_server
tests/test_api_cloud_dispatch.py::test_cloud_override_survives_api_boundary_on_local_server
tests/test_api_cloud_dispatch.py::test_full_api_process_job_result_carries_cloud_provenance
tests/test_api_cloud_dispatch.py::test_full_api_process_job_result_carries_cloud_provenance
  C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-
packages\google\auth\_default.py:113: UserWarning: Your application has authenticated using end
user credentials from Google Cloud SDK without a quota project. You might receive a "quota
exceeded" or "API not enabled" error. See the following page for troubleshooting:
https://cloud.google.com/docs/authentication/adc-troubleshooting/user-creds.
    warnings.warn(_CLOUD_SDK_CREDENTIALS_WARNING)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
34 passed, 8 warnings in 34.52s
```

77. assistant (2026-07-07T14:13:51.059Z)

All 34 tests pass (30 existing + 4 new). Now the SPA side. Let me create a branch in the khelsutra repo and make the compute-picker change.

78. user (2026-07-07T14:13:53.487Z)

```
branch 'fix/hosted-compute-picker-defaults-cloud' set up to track 'origin/master'.
Switched to a new branch 'fix/hosted-compute-picker-defaults-cloud'
fix/hosted-compute-picker-defaults-cloud
```

79. user (2026-07-07T14:14:02.849Z)

The file C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

80. user (2026-07-07T14:14:11.547Z)

The file C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

81. assistant (2026-07-07T14:14:17.810Z)

Now the SPA test. Let me add it in the compute-picker section:

82. user (2026-07-07T14:14:25.363Z)

The file C:\Users\avidu\Projects\khelsutra-guru\khelsutra\web\src\components\IngestPanel.test.tsx has been updated successfully. (file state is current in your context ? no need to Read it back)

83. assistant (2026-07-07T14:14:30.932Z)

Now let me run the SPA gate. Let me check the package.json scripts.

84. user (2026-07-07T14:14:33.873Z)


```
{
  "dev": "vite",
  "build": "vite build",
  "preview": "vite preview",
  "typecheck": "tsc --noEmit",
  "test": "vitest run",
  "test:coverage": "vitest run --coverage",
  "e2e": "playwright test -c e2e/playwright.config.mjs",
```

```
  "e2e:ci": "vite build && node e2e/contract-smoke.mjs && playwright test -c
e2e/playwright.config.mjs",
  "screencast": "vite build && playwright test -c e2e/playwright.config.mjs",
  "screencast:mp4": "node e2e/to-mp4.mjs"
}
```

85. user (2026-07-07T14:14:39.540Z)

```
node_modules present
v26.1.0
11.13.0
```

86. user (2026-07-07T14:14:53.838Z)

```
stderr | src/components/IngestPanel.test.tsx > IngestPanel >
names the video on the retry screen when a URI submission fails (#161)
{"ts":"2026-07-
07T14:14:53.522Z","level":"error","event":"api.http_error","cid":"2547d0ed-802c-478f-8a5e-
fb996ebc3e29","method":"POST","url":"/api/process","status":400,"body":"{\"detail\":\"unsupporte
d storage URI scheme: ftp://\"}"}

stderr | src/components/IngestPanel.test.tsx > IngestPanel >
names the video on the retry screen when a URI submission fails (#161)
{"ts":"2026-07-
07T14:14:53.523Z","level":"error","event":"ingest.submit_error","cid":"2547d0ed-802c-478f-8a5e-
fb996ebc3e29","uri":"gs://bucket/SemiFinal.mp4","code":"engine","detail":"unsupported storage
URI scheme: ftp://"}

stdout | src/components/IngestPanel.test.tsx > IngestPanel >
still names the video on the retry screen after a reload, once the resumed job fails
(#161)
{"ts":"2026-07-
07T14:14:53.536Z","level":"info","event":"ingest.resume","cid":"c1","uri":"gs://b/reload.mp4","j
ob_id":"jR"}
{"ts":"2026-07-07T14:14:53.536Z","level":"info","event":"api.request","cid":"8e7c5c5a-6652-45ab-
b0a0-f2dfb5bd12fa","method":"GET","url":"/api/capabilities"}

stdout | src/components/IngestPanel.test.tsx > IngestPanel >
still names the video on the retry screen after a reload, once the resumed job fails
(#161)
{"ts":"2026-07-
07T14:14:53.539Z","level":"info","event":"api.response","cid":"8e7c5c5a-6652-45ab-b0a0-
f2dfb5bd12fa","method":"GET","url":"/api/capabilities","status":200}

stdout | src/components/IngestPanel.test.tsx > IngestPanel >
still names the video on the retry screen after a reload, once the resumed job fails
(#161)
{"ts":"2026-07-
07T14:14:53.540Z","level":"info","event":"api.request","cid":"c1","method":"GET","url":"/api/job
s/jR"}

stdout | src/components/IngestPanel.test.tsx > IngestPanel >
still names the video on the retry screen after a reload, once the resumed job fails
(#161)
{"ts":"2026-07-
07T14:14:53.540Z","level":"info","event":"api.response","cid":"c1","method":"GET","url":"/api/jo
bs/jR","status":200}

stderr | src/components/IngestPanel.test.tsx > IngestPanel >
still names the video on the retry screen after a reload, once the resumed job fails
(#161)
{"ts":"2026-07-
07T14:14:53.540Z","level":"error","event":"ingest.job_failed","cid":"c1","uri":"gs://b/reload.mp
4","job_id":"jR","code":"engine","detail":"worker crashed"}
```

```
 ? src/components/IngestPanel.test.tsx (23 tests)
1028ms

 Test Files  1 passed (1)
      Tests  23 passed (23)
   Start at  19:44:48
   Duration  5.25s (transform 340ms, setup 33ms, collect 1.05s, tests 1.03s,
environment 2.53s, prepare 247ms)
```

87. assistant (2026-07-07T14:14:59.190Z)

```
All 23 IngestPanel tests pass (22 existing + 1 new). Now the full SPA gate ? typecheck, lint,
full test suite, and build:
```

88. user (2026-07-07T14:15:06.383Z)

```
===TYPECHECK===

> khelsutra-web@0.0.0 typecheck
> tsc --noEmit

===LINT===
npm error Missing script: "lint"
npm error
npm error Did you mean this?
npm error   npm link # Symlink a package folder
npm error
npm error To see a list of scripts, run:
npm error   npm run
npm error A complete log of this run can be found in: C:\Users\avidu\AppData\Local\npm-
cache\_logs\2026-07-07T14_15_05_491Z-debug-0.log
```

89. assistant (2026-07-07T14:15:12.143Z)

```
Typecheck passes; no `lint` script. Let me check the SPA CI to run the same gates, then the full
test suite + build.
```

90. user (2026-07-07T14:15:15.483Z)

```
---CI CONTENT (web-related)---
../.github/workflows/test.yml
---eslint config?---
          cache: "npm"
      # script in a DOM); the Cloudflare Worker + VM-host RUNTIMES stay dependency-free. `npm
test`
      - run: npm ci
      - run: npm test
  web:
    # Operator-console SPA (web/): typecheck + unit tests (with a coverage floor) + production
build
        working-directory: web
          cache: "npm"
          cache-dependency-path: web/package-lock.json
      - run: npm ci
      - run: npm run typecheck
      - run: npm run test:coverage
        run: npm run e2e:ci
          name: web-playwright-artifacts
          path: web/e2e-results/_raw
          cache: "npm"
      - run: npm ci
      - run: npm run test:layout
      - run: npm run test:render
      - run: npm run test:a11y
```

91. assistant (2026-07-07T14:15:21.027Z)

CI runs typecheck + `test:coverage` + `e2e:ci`. Let me run the full unit suite with coverage and the production build:

92. user (2026-07-07T14:15:34.646Z)

```
  ...tatePanel.tsx |     100 |   88.88 |    92.3 |     100 | 39,52,55-56
  ...ileResult.tsx |     100 |   95.45 |     100 |     100 | 31,56
  FfmpegBanner.tsx |     100 |    90.9 |     100 |     100 | 24
  IngestPanel.tsx  |   96.76 |   91.44 |   94.44 |   96.76 | ...06-211,509-516
  ...eSwitcher.tsx |     100 |   83.33 |     100 |     100 | 27
  QueryBar.tsx     |     100 |   88.88 |     100 |     100 | 30
  RallyGrid.tsx    |   98.11 |   73.33 |      50 |   98.11 | 42
  ...ionFooter.tsx |     100 |     100 |     100 |     100 |
  SetupWizard.tsx  |   97.94 |   91.83 |     100 |   97.94 | 18-19,116
  StatusChip.tsx   |     100 |     100 |     100 |     100 |
  TapFilters.tsx   |     100 |     100 |     100 |     100 |
  VideoLibrary.tsx |     100 |   89.47 |     100 |     100 | 33,82
 src/hooks         |   97.58 |   81.25 |     100 |   97.58 |
  useIngestJob.ts  |   96.61 |    82.4 |     100 |   96.61 | ...99,114,199-201
  ...ySelection.ts |     100 |   83.33 |     100 |     100 | 9
  useUpload.ts     |     100 |   75.86 |     100 |     100 | 81-91,97,102
 src/i18n          |     100 |      25 |     100 |     100 |
  index.ts         |     100 |      25 |     100 |     100 | 72-76
 src/lib           |   97.68 |   94.44 |     100 |   97.68 |
  errorText.ts     |     100 |     100 |     100 |     100 |
  errors.ts        |     100 |   93.33 |     100 |     100 | 8,15
  floor.ts         |     100 |     100 |     100 |     100 |
  format.ts        |     100 |     100 |     100 |     100 |
  log.ts           |   81.81 |   88.88 |     100 |   81.81 | 13-14
  query.ts         |   97.07 |   93.45 |     100 |   97.07 | ...95,284,286,288
  selection.ts     |     100 |    92.3 |     100 |     100 | 30
  video.ts         |     100 |   93.33 |     100 |     100 | 21
-------------------|---------|----------|---------|---------|-------------------

=============================== Coverage summary ===============================
Statements   : 94.87% ( 2460/2593 )
Branches     : 88.82% ( 795/895 )
Functions    : 89.47% ( 136/152 )
Lines        : 94.87% ( 2460/2593 )
================================================================================
```

93. assistant (2026-07-07T14:15:39.562Z)

Let me confirm the test pass count and coverage floor:

94. user (2026-07-07T14:15:53.479Z)

{"ts":"2026-07-07T14:15:47.700Z","level":"warn","event":"upload.aborting_partial","cid":"31457963-b02c-4c50-875c-0e323c6a6911","upload_id":"up1","error":"AbortError: Upload aborted"}
stdout | src/hooks/useUpload.test.ts > useUpload > surfaces a classified error when a part is rejected
stdout | src/hooks/useUpload.test.ts > useUpload > surfaces a classified error when a part is rejected
stdout | src/hooks/useUpload.test.ts > useUpload > surfaces a classified error when a part is rejected
stderr | src/hooks/useUpload.test.ts > useUpload > surfaces a classified error when a part is rejected
{"ts":"2026-07-07T14:15:47.886Z","level":"error","event":"api.http_error","cid":"3617582b-f08e-4fdf-9ec9-67b4eec20784","method":"PUT","url":"/api/upload/up1/part/0","status":413,"body":"{\"detail\":\"Part exceeds the per-part limit\"}"}

```
stdout | src/hooks/useUpload.test.ts > useUpload > surfaces a
classified error when a part is rejected
stderr | src/hooks/useUpload.test.ts > useUpload > surfaces a
classified error when a part is rejected
{"ts":"2026-07-
07T14:15:47.886Z","level":"warn","event":"upload.aborting_partial","cid":"3617582b-f08e-4fdf-
9ec9-67b4eec20784","upload_id":"up1","error":"ApiError: PUT /api/upload/up1/part/0 failed: 413"}
stdout | src/hooks/useUpload.test.ts > useUpload > surfaces a
classified error when a part is rejected
stderr | src/hooks/useUpload.test.ts > useUpload > surfaces a
classified error when a part is rejected
{"ts":"2026-07-
07T14:15:47.886Z","level":"error","event":"upload.failed","cid":"3617582b-f08e-4fdf-9ec9-
67b4eec20784","filename":"match.mp4","error":"ApiError: PUT /api/upload/up1/part/0 failed: 413"}
{"ts":"2026-07-
07T14:15:48.152Z","level":"warn","event":"capabilities.unavailable","error":"TypeError: Cannot
read properties of undefined (reading 'ok')"}
stdout | src/components/VideoLibrary.test.tsx > VideoLibrary >
degrades to a soft error (never crashes, never hides ingest) when the list can't load
stderr | src/components/VideoLibrary.test.tsx > VideoLibrary >
degrades to a soft error (never crashes, never hides ingest) when the list can't load
{"ts":"2026-07-
07T14:15:48.196Z","level":"error","event":"api.network_error","cid":"09da3394-876a-49b4-8bcb-
084ab1b3affd","method":"GET","url":"/api/videos","error":"TypeError: Failed to fetch"}
stderr | src/components/VideoLibrary.test.tsx > VideoLibrary >
degrades to a soft error (never crashes, never hides ingest) when the list can't load
{"ts":"2026-07-
07T14:15:48.197Z","level":"warn","event":"library.load_failed","cid":"09da3394-876a-49b4-8bcb-
084ab1b3affd","error":"ApiError: Network error calling GET /api/videos"}
{"ts":"2026-07-
07T14:15:48.374Z","level":"error","event":"api.network_error","cid":"9b3a9c09-0853-417a-8bf6-
81476e5b1e9c","method":"GET","url":"/api/capabilities","error":"TypeError: Failed to fetch"}
{"ts":"2026-07-
07T14:15:48.375Z","level":"warn","event":"wizard.capabilities_unavailable","cid":"9b3a9c09-0853-
417a-8bf6-81476e5b1e9c","error":"ApiError: Network error calling GET /api/capabilities"}
---floor config---
vite.config.ts:    // Coverage floor (CI runs `npm run test:coverage`): a merge can't DROP
coverage below these
vite.config.ts:    // thresholds, so untested new code is caught. `all` counts every source file
(not only the ones a
vite.config.ts:    coverage: {
vite.config.ts:      // Floor set just below current (stmts/lines 93.3, branches 88.1, funcs
85.9) ? a real ratchet
vite.config.ts:      // coverage grows. (functions stays a touch looser: App.tsx is the engine-
orchestrating shell,
vite.config.ts:      thresholds: {
vite.config.ts:        statements: 92,
vite.config.ts:        branches: 87,
vite.config.ts:        functions: 84,
vite.config.ts:        lines: 92,
```

95. assistant (2026-07-07T14:15:58.612Z)

Coverage (stmts 94.87 / branches 88.82 / funcs 89.47 / lines 94.87) is above the floor (92/87/84/92). Let me confirm a clean exit and test totals, then build:

96. user (2026-07-07T14:16:14.256Z)

```
 ? src/components/TapFilters.test.tsx (4 tests)
94ms
 ? src/components/CompileResult.test.tsx (4 tests)
79ms
 Test Files  32 passed (32)
      Tests  310 passed (310)
   Duration  8.23s (transform 2.16s, setup 598ms, collect 15.64s, tests 13.10s,
```

```
environment 51.85s, prepare 7.84s)
EXIT=0
===BUILD===
dist/assets/noto-sans-devanagari-latin-wght-normal-Bxfr4UJH.woff2
 25.38 kB
dist/assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
 27.58 kB
dist/assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2
 90.80 kB
dist/assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
121.19 kB
dist/assets/index-BjemzRyy.css
 28.70 kB ? gzip:   6.03 kB
dist/assets/index-kJhH4dVE.js
701.65 kB ? gzip: 209.26 kB

(!) Some chunks are larger than 500 kB after minification. Consider:
- Using dynamic import() to code-split the application
- Use build.rollupOptions.output.manualChunks to improve chunking:
https://rollupjs.org/configuration-options/#output-manualchunks
- Adjust chunk size limit for this warning via build.chunkSizeWarningLimit.
? built in 1.90s
```

97. assistant (2026-07-07T14:16:24.600Z)

```
SPA gate green: typecheck ?, **310 tests pass** ?, coverage above floor ?, production build ?.
(Playwright `e2e:ci` needs browser binaries + a live engine ? not run locally, same as it only
runs in CI.)

Now the backend full pre-merge gate. Let me run ruff + mypy (repo CI scope) first:
```

98. user (2026-07-07T14:16:30.876Z)

```
===RUFF (full repo)===
    |

E702 Multiple statements on one line (semicolon)
   --> score_local.py:181:102
    |
179 |             rows.append(score_one(label, path, gts, mods, args.fps, args.frame_width))
180 |         except Exception as e:
181 |             rows.append({"label": label, "ok": False, "error": "{}:
{}".format(type(e).__name__, e)}); traceback.print_exc()
    |
^
182 |     print_table(rows, len(gts), args.name, mods["tm"])
183 |     return 0
    |

Found 18 errors.
[*] 4 fixable with the `--fix` option.
===MYPY (backend training)===
backend\annotations\base.py:88: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\sports\__init__.py:9: note: By default the bodies of untyped functions are not checked,
consider using --check-untyped-defs  [annotation-unchecked]
backend\pipeline\validators\base.py:44: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not
checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 148 source files
```

99. assistant (2026-07-07T14:16:39.501Z)

The 18 ruff errors are in the untracked root dev scripts (`score_local.py`, `score_golden_gate.py`), not my change. Let me confirm ruff is clean on tracked files:

100. user (2026-07-07T14:16:42.858Z)

```
===files with ruff errors===
===are they tracked?===
(empty above = untracked)
===ruff on tracked python only===
All checks passed!
```

101. assistant (2026-07-07T14:16:49.004Z)

Confirmed ? those ruff errors are in untracked local dev tools; all **tracked** files pass ruff, and mypy is clean. Now the full suite with coverage. Let me check how `run_all_tests.py` invokes it:

102. user (2026-07-07T14:16:50.896Z)

```
run_all_tests.py:8:    python run_all_tests.py -v          # verbose (passthrough to pytest)
run_all_tests.py:10:    python run_all_tests.py --cov=backend --cov-report=term-missing   #
coverage
run_all_tests.py:12:Anything after the script name is passed straight through to pytest.
run_all_tests.py:18:  package isn't installed (they use ``pytest.importorskip``). To exercise
them too:
run_all_tests.py:20:- Exit code is pytest's own, so this is CI / pre-PR friendly (0 = all
green).
run_all_tests.py:42:    cmd = [sys.executable, "-m", "pytest", *args]
run_all_tests.py:60:        + ("PASS - all tests green" if rc == 0 else f"FAIL - pytest exit
{rc}")
pytest.ini:1:[pytest]
pytest.ini:2:# When `pytest` is invoked without an explicit path, only collect from tests/.
pyproject.toml:50:    "pytest",
pyproject.toml:51:    "pytest-cov",
pyproject.toml:65:# tests skip via pytest.importorskip. For local dev against a sibling
checkout:
```

103. user (2026-07-07T14:16:56.019Z)

```
1       [pytest]
2       # When `pytest` is invoked without an explicit path, only collect from tests/.
3       # scratch/ holds debugging scripts that hit real APIs (Gemini upload, etc.)
4       # and should never run as part of the regular test suite.
5       testpaths = tests
6
```

104. assistant (2026-07-07T14:17:01.742Z)

Let me check the backend CI to run the exact same coverage command:

105. user (2026-07-07T14:17:03.158Z)

```
.github/workflows/ci.yml:16:  # Lint gate (ruff.toml at repo root): pyflakes/pycodestyle +
bugbear.
.github/workflows/ci.yml:17:  # Also the repo-wide syntax guard ? ruff reports invalid-syntax on
every
.github/workflows/ci.yml:29:      - name: Install ruff
.github/workflows/ci.yml:30:        # Pinned so an upstream ruff release can't redden every PR
overnight; bump deliberately.
.github/workflows/ci.yml:31:        run: python -m pip install ruff==0.15.16
.github/workflows/ci.yml:33:        run: ruff check . --output-format github
.github/workflows/ci.yml:66:  # Type gate (mypy.ini at repo root): lenient green baseline with
an explicit
.github/workflows/ci.yml:68:  # up. Typed core deps are installed so mypy sees their real
```

signatures
.github/workflows/ci.yml:79:   - name: Install mypy + typed core deps
.github/workflows/ci.yml:80:   # mypy pinned alongside ruff so a type-checker release can't redden CI overnight.
.github/workflows/ci.yml:81:   run: python -m pip install mypy==2.1.0 pydantic fastapi uvicorn python-dotenv google-genai numpy
.github/workflows/ci.yml:83:   run: mypy backend training
.github/workflows/ci.yml:99:   run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai
.github/workflows/ci.yml:101:   run: python -m pytest tests/test_import_smoke.py -v
.github/workflows/ci.yml:132:   # actually run in CI instead of skipping via pytest.importorskip.
.github/workflows/ci.yml:136:   # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
.github/workflows/ci.yml:139:   # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
.github/workflows/ci.yml:146:   run: python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70
.github/workflows/ci.yml:168:   run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai numpy
.github/workflows/ci.yml:175:   python -m pytest tests/test_golden_fixtures.py tests/test_golden_real_fixtures.py -q
.github/workflows/nightly.yml:43: # Lint gate (ruff.toml at repo root) ? mirrors ci.yml::lint.
.github/workflows/nightly.yml:53:   - name: Install ruff
.github/workflows/nightly.yml:54:   # Pinned so an upstream ruff release can't redden the nightly overnight; bump deliberately.
.github/workflows/nightly.yml:55:   run: python -m pip install ruff==0.15.16
.github/workflows/nightly.yml:57:   run: ruff check . --output-format github

106. assistant (2026-07-07T14:17:12.628Z)

The CI command is `python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70`. Let me run the full suite in the background and monitor:

107. user (2026-07-07T14:17:14.436Z)

Command running in background with ID: bg9kpzy5f. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\bg9kpzy5f.output. You will be notified when it completes. To check interim output, use Read on that file path.

108. assistant (2026-07-07T14:17:19.188Z)

While the full suite runs, let me review my final diffs and update the session handoff. First, the backend diff:

109. user (2026-07-07T14:17:20.359Z)

```
diff --git a/backend/orchestration.py b/backend/orchestration.py
index 5e9f510..bfe39f5 100644
--- a/backend/orchestration.py
+++ b/backend/orchestration.py
@@ -155,6 +155,38 @@ def _ingest_remote_segments(
     return n


+# Trajectory segmenters whose IN-PROCESS detect needs model weights present on THIS box. A hosted
+# cloud-serving control-plane bakes NO detector weights into its image (only the worker Cloud Run Job
+# GCSFuse-mounts them), so forcing compute="local" for one of these there can never succeed ? the
+# guardrail in _maybe_dispatch_remote fails such a request FAST instead of the cryptic deep
+# "WASB weights not found". (Names are the registered segmenter identities; see
```

```
backend/pipeline/segmenters.)
+_WEIGHTS_BACKED_SEGMENTERS = frozenset({"wasb_hybrid", "tracknet_hybrid", "fusion_hybrid"})
+
+
+def _local_detector_weights_present(cfg, seg_name: str) -> bool:
+    """Whether the trajectory weights the IN-PROCESS ``seg_name`` detector needs are present +
readable
+    on THIS box. Mirrors the native/WSL runners' resolution (env override ?
``indexing.<det>.weights_path``)
+    and the ``/api/capabilities`` probe, so the guardrail agrees with the healthcheck that
would
+    otherwise fail deep in the run. Fusion follows its configured substrate (WASB default /
TrackNet).
+    A non-weights segmenter returns True (nothing local to check)."""
+    name = (seg_name or "").strip().lower()
+    if name not in _WEIGHTS_BACKED_SEGMENTERS:
+        return True
+    idx = getattr(cfg, "indexing", None)
+    substrate = (
+        getattr(getattr(idx, "fusion", None), "substrate", "") or "wasb"
+    ).strip().lower()
+    if name == "tracknet_hybrid" or (name == "fusion_hybrid" and substrate == "tracknet"):
+        path = getattr(getattr(idx, "tracknet", None), "weights_path", "") or ""
+    else:  # wasb_hybrid, or fusion on the (default) WASB substrate
+        path = (
+            os.environ.get("WASB_WEIGHTS")
+            or getattr(getattr(idx, "wasb", None), "weights_path", "")
+            or ""
+        )
+    return bool(path) and os.path.exists(os.path.expanduser(path))
+
+
 def _cloud_available(cfg) -> bool:
     """Whether THIS server can satisfy a per-request ``compute="cloud"`` dispatch (item 3).

@@ -446,6 +478,23 @@ def _maybe_dispatch_remote(
         )
         choice = ""
     if choice == "local":
+        # A hosted cloud-serving control-plane bakes NO detector weights into its image (only
the
+        # worker Cloud Run Job GCSFuse-mounts them), so forcing a weights-backed trajectory
detector
+        # to run IN-PROCESS here can never succeed ? it dies deep in the segmenter with a
cryptic
+        # "WASB weights not found" (observed live: job f29d41d0?, 2026-07-07). Fail FAST with
an
+        # actionable message instead of that silent trap. A truly-local box (not cloud-
capable), or a
+        # cloud-wired box that ACTUALLY has the weights on disk, is unaffected ? it runs in-
process
+        # below exactly as before (the weights-present check is the escape hatch).
+        if (
+            (seg_name or "").strip().lower() in _WEIGHTS_BACKED_SEGMENTERS
+            and _cloud_available(cfg)
+            and not _local_detector_weights_present(cfg, seg_name)
+        ):
+            return _failed(
+                f"this hosted server can't run detection locally ? the '{seg_name}' model
weights "
+                "live only on the cloud worker, not on the control-plane. Choose 'Run in
cloud'.",
+                False,
+            )
         return None  # explicit "run on my machine" ? in-process regardless of the server's
```

```
target
     if choice == "cloud":
         if not _cloud_available(cfg):
```

110. assistant (2026-07-07T14:17:27.704Z)

Now the SPA diff:

111. user (2026-07-07T14:17:30.044Z)

```
diff --git a/web/src/components/IngestPanel.tsx b/web/src/components/IngestPanel.tsx
index 38ca3a9..edfd5d9 100644
--- a/web/src/components/IngestPanel.tsx
+++ b/web/src/components/IngestPanel.tsx
@@ -81,7 +81,9 @@ export function IngestPanel({ onLoaded, onBusyChange }: Props) {

   // Compute-picker (PACKAGING compute-picker, item 3): if this server can dispatch to a cloud
GPU
   // (capabilities.deployment.cloud_available), let the user choose where DETECT runs. Default
LOCAL
-  // (free, no surprise cost); CLOUD requires an explicit cost acknowledgement before it can
run.
+  // (free, no surprise cost) EXCEPT on a hosted cloud-serving control-plane
(compute_target=cloud_run):
+  // there the control-plane bakes no detector weights, so LOCAL can never run ? default CLOUD
instead
+  // (set from capabilities below), which requires an explicit cost acknowledgement before it
can run.
   const [cloudAvailable, setCloudAvailable] = useState(false);
   const [compute, setCompute] = useState<Compute>("local");
   const [cloudAck, setCloudAck] = useState(false);
@@ -101,6 +103,13 @@ export function IngestPanel({ onLoaded, onBusyChange }: Props) {
       .then((c) => {
         if (!alive) return;
         setCloudAvailable(c.deployment?.cloud_available === true);
+        // Hosted cloud-serving control-plane (compute_target=cloud_run) bakes NO detector
weights, so
+        // in-process LOCAL detection there always fails ("weights not found") ? default the
picker to
+        // CLOUD so the SPA never sends compute=local by default on the hosted build. A local
appliance
+        // that merely also offers cloud (compute_target?cloud_run) keeps the free LOCAL
default.
+        if (c.deployment?.cloud_available === true && c.deployment?.compute_target ===
"cloud_run") {
+          setCompute("cloud");
+        }
         setBackends(c.storage?.backends ?? []);
         setMaxUploadBytes(
           typeof c.max_upload_mb === "number" && c.max_upload_mb > 0 ? c.max_upload_mb * 1024 *
1024 : undefined,
```

112. assistant (2026-07-07T14:17:42.542Z)

Both diffs are clean. Let me check the full-suite progress:

113. user (2026-07-07T14:17:44.719Z)

```
========================================================================
 Full test suite:  C:\Users\avidu\AppData\Local\Programs\Python\Python313\python.exe -m pytest
--cov=backend --cov=training --cov-report=term --cov-fail-under=70 -q
 obsreport (optional reporting dep): NOT installed ? reporting tests will SKIP
========================================================================
................
```

114. assistant (2026-07-07T14:17:54.587Z)

The suite is running (reporting tests will skip without `obsreport`). Let me check how CI installs that optional dep so I can match it:

115. user (2026-07-07T14:17:55.696Z)

```
            sudo apt-get install -y libgl1 libglib2.0-0t64 || sudo apt-get install -y libgl1 libglib2.0-0
        - name: Install CPU-only torch (no CUDA wheels on a CPU runner)
          # torch/torchvision are NOT in pyproject deps ? their CPU/CUDA wheels
          # need an out-of-band index that PEP 621 can't express. Keep installing
          # them separately here, before the editable install pulls the rest.
          run: python -m pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
        - name: Install trackers (git-only dep ? PyPI wheel is broken)
          # D13/P12: ByteTrack moved out of `supervision` (removed in 0.30.0) to the
          # standalone `trackers` package. Its PyPI wheel ships metadata only (no
          # source ? roboflow/trackers#474), so it MUST come from git. PEP 621 can't
          # express a git URL, so it's not in pyproject.toml (same pattern as torch).
          # Installed here so the two D13/P12 tests in tests/test_person_detector.py
          # actually run in CI instead of skipping via pytest.importorskip.
          run: python -m pip install -r requirements-trackers.txt
        - name: Install package (editable) + dev tooling
          # Editable PEP 621 install: runtime deps + the `dev` group
          # (pytest, pytest-cov, httpx for the fastapi TestClient, ruff, mypy).
          run: python -m pip install -e .[dev]
        # obsreport (private, soft dep) is absent here: reporting tests skip via
        # pytest.importorskip ? run_all_tests.py prints which. The coverage floor
        # is therefore calibrated against THIS lane's number (local runs with
        # obsreport installed measure a little higher).
        # Floor calibrated 2026-06-11 against this lane's measured 72% (local
        # with obsreport: 73%). Ratchet it UP as coverage grows; never down
        # without an owner decision.
        - name: Run full suite (with coverage floor)
          run: python run_all_tests.py --cov=backend --cov=training --cov-report=term --cov-fail-under=70

    # Cross-OS determinism guard for the committed golden regression fixtures
    # (docs/GOLDEN_REGRESSION_FIXTURES.md). The CONTROL replay path is pure-stdlib
    # (only numpy, no cv2/torch), so this is a FAST, focused job ? it does NOT run
    # the full suite x3 OSes. It replays the Windows/Linux/macOS runner against the
    # committed (parse-compared) snapshots, catching any cross-OS float drift beyond
    # the 1e-6 tolerance before it can flake a contributor's PR on another machine.
    golden-crossos:
      name: Golden fixtures cross-OS (${{ matrix.os }})
      strategy:
        fail-fast: false
        matrix:
          os: [ubuntu-latest, windows-latest, macos-latest]
      runs-on: ${{ matrix.os }}
      timeout-minutes: 15
      steps:
        - uses: actions/checkout@v6
        - uses: actions/setup-python@v6
          with:
            python-version: "3.13"
        - name: Install minimal deps (no GPU stack ? the CONTROL replay needs only numpy)
          run: python -m pip install pytest fastapi uvicorn pydantic python-dotenv google-genai numpy
        - name: Golden-fixtures characterization (cross-OS determinism guard)
          # Also runs the freeze-real / time-origin-reset / M2 refusal-gate tests (synthetic /
          # tmp_path only ? no GPU/network) so the reset metric-invariance (fix #2) and the
```

```
            # de-attribution paths get cross-OS coverage and surface any win/mac float drift.
            run: |
              python -m backend.eval.golden_fixtures --check
```

116. assistant (2026-07-07T14:18:15.354Z)

```
CI deliberately runs **without** obsreport and calibrates the coverage floor against that lane ?
so my current run is the CI-faithful gate (reporting tests skipping is expected). Let me wait
for it to finish.
```

117. user (2026-07-07T14:18:18.180Z)

```
========================================================================
........................................................................ [  4%]
.............
```

118. assistant (2026-07-07T14:18:38.190Z)

```
The full suite is running (~1764 tests). I'll wait for it to complete. Let me monitor with a
lightweight loop until it finishes:
```

119. user (2026-07-07T14:22:27.531Z)

```
DONE
---last lines---
backend\storage\gdrive.py                                 61      3    95%
backend\storage\gdrive_api.py                            105      6    94%
backend\storage\local.py                                  49      3    94%
backend\storage\ref.py                                    49      0   100%
backend\storage\s3.py                                     47      5    89%
backend\tools\__init__.py                                  0      0   100%
backend\tools\export.py                                   75      2    97%
backend\tools\rally_slicer.py                            105     47    55%
backend\utils\__init__.py                                  0      0   100%
backend\utils\annotation_proxy.py                         64     11    83%
backend\utils\app_home.py                                 40      0   100%
backend\utils\ffmpeg.py                                   47      3    94%
backend\utils\freespace.py                                32      2    94%
backend\utils\geometry.py                                 29      0   100%
backend\utils\hashing.py                                   9      0   100%
backend\utils\logging_setup.py                            11      0   100%
backend\utils\paths.py                                    44      3    93%
backend\utils\redact.py                                   16      0   100%
backend\utils\run_telemetry.py                           126     12    90%
backend\utils\single_instance.py                          43     11    74%
backend\utils\video.py                                    34      3    91%
backend\utils\workspace.py                                49      0   100%
backend\worker\__init__.py                                 0      0   100%
backend\worker\__main__.py                               311     29    91%
training\__init__.py                                       0      0   100%
training\gen0\__init__.py                                  0      0   100%
training\gen0\harness.py                                 130      9    93%
----------------------------------------------------------------------
TOTAL                                                  12303   1895    85%
Required test coverage of 70% reached. Total coverage: 84.60%
1770 passed, 4 skipped, 25 warnings in 301.43s (0:05:01)
========================================================================
 RESULT: PASS - all tests green
========================================================================
EXIT=0
```

120. assistant (2026-07-07T14:23:04.910Z)

Backend gate is fully green:
- ? `ruff check` (all tracked files) ? clean
- ? `mypy backend training` ? Success, 148 files
- ? Full suite: **1770 passed, 4 skipped** (the 4 skips are the obsreport reporting tests, exactly as CI expects)
- ? Coverage **84.60%** (floor 70)

Both gates pass. Let me commit the completed work on each branch (local only ? no push/PR without your go-ahead), following the branching-safety checks. Backend first:

121. user (2026-07-07T14:23:08.337Z)

```
BRANCH=perf/validation-shared-decode-pass
---sibling commits on branch (must be empty)---
---staging explicit paths---
M  backend/orchestration.py
 M backend/pipeline/validators/base.py
 M backend/pipeline/validators/blur.py
 M backend/pipeline/validators/relevance.py
 M backend/pipeline/validators/scene_cut.py
M  tests/test_api_cloud_dispatch.py
?? backend/pipeline/validators/frame_sampler.py
?? score_golden_gate.py
?? score_local.py
```

122. assistant (2026-07-07T14:25:32.358Z)

?? **Concurrent-session hazard detected.** The branch has shifted under me ? HEAD is now `perf/validation-shared-decode-pass` (a sibling session's branch) with its own uncommitted validator changes in the shared checkout. My `git add` just polluted the *shared* index. I need to protect the sibling's work and mine. First, unstage immediately and gather read-only state:

123. user (2026-07-07T14:25:35.788Z)

```
=== index now clean of my staging ===
 M backend/orchestration.py
 M backend/pipeline/validators/base.py
 M backend/pipeline/validators/blur.py
 M backend/pipeline/validators/relevance.py
 M backend/pipeline/validators/scene_cut.py
 M tests/test_api_cloud_dispatch.py
?? backend/pipeline/validators/frame_sampler.py
?? score_golden_gate.py
?? score_local.py
=== branch shas ===
3be09357cf0783f1971e92774d6e8cc6830ff7c1
3be09357cf0783f1971e92774d6e8cc6830ff7c1
3be09357cf0783f1971e92774d6e8cc6830ff7c1
3be09357cf0783f1971e92774d6e8cc6830ff7c1
=== my files: is orchestration.py modified only by me? ===
 backend/orchestration.py         |  49 ++++++++++++++++++
 tests/test_api_cloud_dispatch.py | 108 ++++++++++++++++++++++++++++++++++++++++++++++++
 2 files changed, 157 insertions(+)
```

124. assistant (2026-07-07T14:25:58.304Z)

All branches sit at the same commit `3be0935`, and my 2 files are disjoint from the sibling's validator edits. I'll commit my work via a **separate git worktree** ? this never moves the main checkout's HEAD, eliminating any risk of the sibling committing onto my branch. Step 1: create the worktree and copy my files in.

125. user (2026-07-07T14:26:03.716Z)

```
worktree added at fix branch: fix/hosted-local-detection-guardrail @ 3be0935
```

```
=== worktree diff stat (should be my 2 files, +157) ===
 backend/orchestration.py         |  49 ++++++++++++++++++
 tests/test_api_cloud_dispatch.py | 108 ++++++++++++++++++++++++++++++++++++++++
 2 files changed, 157 insertions(+)
```

126. assistant (2026-07-07T14:26:11.752Z)

The worktree has exactly my two files. Now commit there (explicit paths, conventional message):

127. user (2026-07-07T14:26:23.134Z)

```
[fix/hosted-local-detection-guardrail 4565c74] fix(serving): fail fast when compute=local forces
weightless in-process detection
 2 files changed, 157 insertions(+)
=== fix branch head ===
4565c74 fix(serving): fail fast when compute=local forces weightless in-process detection
=== only my commit ahead of origin/master ===
4565c74 fix(serving): fail fast when compute=local forces weightless in-process detection
```

128. assistant (2026-07-07T14:26:29.829Z)

My commit `4565c74` is on the fix branch with **only my commit** ahead of origin/master ? no sibling commit rode in. Now remove the worktree and restore the main checkout to the sibling's expected state (my files are safely committed on fix):

129. user (2026-07-07T14:26:36.538Z)

```
=== main HEAD still on sibling branch (untouched)? ===
perf/validation-shared-decode-pass
3be0935
worktree removed
=== revert MY 2 files in main tree (safely committed on fix) ===
=== final main-tree status: only the sibling's changes remain ===
 M backend/pipeline/validators/base.py
 M backend/pipeline/validators/blur.py
 M backend/pipeline/validators/relevance.py
 M backend/pipeline/validators/scene_cut.py
?? backend/pipeline/validators/frame_sampler.py
?? score_golden_gate.py
?? score_local.py
?? tests/test_frame_sampler.py
=== worktrees ===
C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer
3be0935 [perf/validation-shared-decode-pass]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/cfg-wt        20b2d5a [claude/wasb-
config-validation]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/compile-wt    b09a1d4 [claude/compile-
reliability]
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-pre-wt  355dad6 (detached HEAD)
C:/Users/avidu/AppData/Local/Temp/claude/c--Users-avidu-Projects-badminton-highlight-
indexer/6e9d0958-a2ec-4a8e-bb2d-6cce9833c527/scratchpad/pr458-wt      5de1ef2 (detached HEAD)
```

130. assistant (2026-07-07T14:27:06.436Z)

My commit is safely on `fix/hosted-local-detection-guardrail` (`4565c74`), and the main checkout is restored to the sibling session's expected state (only their validator changes remain; my 2 files reverted to baseline since they now live on the fix branch). The temporary worktree is removed.

Now the SPA repo ? it could have shifted under me too. Let me check its state before committing:

131. user (2026-07-07T14:27:07.856Z)

```
Exit code 128
BRANCH=fix/hosted-compute-picker-defaults-cloud
HEAD=c7265a4
=== status ===
 M web/src/components/IngestPanel.test.tsx
 M web/src/components/IngestPanel.tsx
=== fix branch sha vs origin/master ===
fatal: Needed a single revision
```

132. assistant (2026-07-07T14:27:21.921Z)

The SPA repo is uncontended ? HEAD is still on my branch `fix/hosted-compute-picker-defaults-cloud` (`c7265a4` == origin/master) with only my 2 files modified. Let me verify and commit directly here:

133. user (2026-07-07T14:27:25.057Z)

```
=== branch + safety ===
fix/hosted-compute-picker-defaults-cloud
=== commits ahead of origin/master (must be empty pre-commit) ===
=== stage explicit paths ===
M  web/src/components/IngestPanel.test.tsx
M  web/src/components/IngestPanel.tsx
```

134. user (2026-07-07T14:27:36.148Z)

```
[fix/hosted-compute-picker-defaults-cloud 1470e2a] fix(ingest): default the compute picker to
CLOUD on a hosted control-plane
 2 files changed, 34 insertions(+), 1 deletion(-)
=== result ===
1470e2a fix(ingest): default the compute picker to CLOUD on a hosted control-plane
```

135. assistant (2026-07-07T14:27:54.817Z)

Both commits are in place, each with **only my commit** ahead of `origin/master`:
- **Backend** `4565c74` on `fix/hosted-local-detection-guardrail`
- **SPA** `1470e2a` on `fix/hosted-compute-picker-defaults-cloud`

Neither is pushed (holding for your go-ahead). Let me record this in-flight work to memory so it survives a restart, and note the contended-checkout hazard:

136. assistant (2026-07-07T14:28:11.617Z)

---
name: compute-local-hosted-guardrail
description: "Two local branches (backend + SPA) fixing compute=local on the hosted control-plane ? gated, unpushed, awaiting push/PR"
metadata:
  type: project

Fix for the confusing `compute="local"` failure on the hosted Cloud Run control-plane (job f29d41d0?, 2026-07-07: `[COMPUTE] requested=local actual=in_process ? WASB weights not found at /opt/models/?`). Root cause: `_maybe_dispatch_remote` returned None for `compute="local"` ? the in-process segmenter ran on the WEIGHTLESS control-plane (weights only mount on the worker Job).

Two-part fix, COMPLETE + GATED locally 2026-07-07, NOT pushed (awaiting owner push/PR):
1. **Backend** ? repo `khelsutra-guru/badminton-highlight-indexer`, branch
   `fix/hosted-local-detection-guardrail` @ commit **4565c74** (from origin/master 3be0935).
   `backend/orchestration.py`: new `_WEIGHTS_BACKED_SEGMENTERS` +
`_local_detector_weights_present`;
   guardrail in `_maybe_dispatch_remote` `choice=="local"` branch fails FAST (mirrors the

cloud-not-available `_failed(...)`) when a weights-backed detector (wasb/tracknet/fusion) is forced local on a `_cloud_available` server with no readable local weights. +4 tests in `tests/test_api_cloud_dispatch.py`. Gate: full suite **1770 passed / 4 skipped**, coverage **84.6%** (floor 70), ruff (tracked files) + `mypy backend training` clean.
2. **SPA** ? repo `avidullu/khelsutra` (source at `khelsutra-guru/khelsutra/web`; `backend/ui/` is the VENDORED built bundle via `scripts/bundle_ui.sh`), branch `fix/hosted-compute-picker-defaults-cloud` @ commit **1470e2a** (from origin/master c7265a4). `web/src/components/IngestPanel.tsx`: compute picker defaults to CLOUD when `capabilities.deployment.compute_target==="cloud_run"` (keeps free-LOCAL default on a local appliance that merely also offers cloud). +1 IngestPanel test. Gate: typecheck ?, **310 tests** ?, coverage ? floor, `vite build` ? (playwright e2e not run locally ? needs browsers + live engine).

Next steps: push both branches + open PRs (each repo has its own CI gate). The SPA fix reaches the deployed hosted build only after a re-vendor (`scripts/bundle_ui.sh` from an SPA release ? `backend/ui/` + `ui_version.json`) ? a separate RELEASE step, deliberately out of the code PRs.

? Contended checkout (2026-07-07): the badminton main working tree is shared by several live sessions ? mid-task the branch shifted under me to `perf/validation-shared-decode-pass` (a sibling's validator WIP: `backend/pipeline/validators/*` + `frame_sampler.py`) and extra `claude/*` worktrees exist. I committed via a throwaway `git worktree` so the main HEAD never moved. Re-apply [[efficient-timeouts-and-retries]] branching-safety: verify branch right before any add/commit.

137. user (2026-07-07T14:28:11.707Z)

File created successfully at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md (file state is current in your context ? no need to Read it back)

138. user (2026-07-07T14:28:16.073Z)

1 # Memory index
2
3 - ? **Start here (GPU-resident rollout):** session-handoff-gpu-resident.md ? #506 implemented, PR #507 open; next: on-box parity gate + 15-clip golden gate.
4
5 - [Actual repo location](actual-repo-location.md) ? primary working dir is an empty STUB; real code is under `khelsutra-guru/badminton-highlight-indexer`.
6 - [Efficient timeouts, retries & Windows gcloud SSH](efficient-timeouts-and-retries.md) ? no silent retry loops; plain `gcloud compute ssh` (plink/pscp reject `-o`, mishandle `~`); diagnose-first, run long work detached.
7 - [GPU VM setup gotchas](gpu-vm-setup-gotchas.md) ? NVDEC libs, apt-lock-at-boot, torch2.x/pynvc; what #501 should encode.
8 - [LGTM only when merge-ready](lgtm-only-when-merge-ready.md) ? post LGTM only when genuinely satisfied a PR is merge-ready (verified), for all PRs.
9 - [GPU-resident modernization status](gpu-resident-modernization.md) ? #506 validated clean; the resize must match cv2.warpAffine via grid_sample (not F.interpolate).
10

139. user (2026-07-07T14:28:23.520Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

140. assistant (2026-07-07T14:28:53.932Z)

Both parts are implemented, gated, and committed to local branches (not pushed). Here's the summary.

What was wrong

On the hosted control-plane, `_maybe_dispatch_remote` returned `None` for `compute="local"`, so the in-process segmenter ran **on the weightless control-plane** and died deep in the healthcheck with `WASB weights not found at /opt/models/?` ? a guaranteed, confusing failure.

The two-part fix

1. Backend guardrail ? backend/orchestration.py (branch `fix/hosted-local-detection-guardrail`, commit `4565c74`)
- New `_WEIGHTS_BACKED_SEGMENTERS = {wasb_hybrid, tracknet_hybrid, fusion_hybrid}` and `_local_detector_weights_present()` (mirrors the native/WSL runner + `/api/capabilities` weights resolution, incl. fusion's substrate).
- In the `choice == "local"` branch: when the detector is weights-backed **and** the server is cloud-capable (`_cloud_available`) **and** local weights aren't readable ? `_failed("this hosted server can't run detection locally ? Choose 'Run in cloud'.", False)`, mirroring the existing `compute="cloud"`-not-available reject. Honest provenance: `path=not_run, requested=local, dispatched=False` (so the `[COMPUTE]` line now reads `actual=not_run` instead of `actual=in_process`).
- Escape hatch: a truly-local box, or a cloud-wired box that actually has the weights on disk, still runs in-process unchanged; `motion` (weight-free) is out of scope.
- 4 new tests in tests/test_api_cloud_dispatch.py: fail-fast, weights-present escape hatch, motion unaffected, pure-local not guardrailed.

2. SPA compute-picker default ? `web/src/components/IngestPanel.tsx` in the `avidullu/khelsutra` repo (branch `fix/hosted-compute-picker-defaults-cloud`, commit `1470e2a`)
- The picker now defaults to **CLOUD** when `capabilities.deployment.compute_target === "cloud_run"`, so the hosted SPA never sends `compute=local` by default. The free-LOCAL default is kept for a local appliance that merely also offers cloud (`compute_target ? cloud_run`); the cloud cost-ack gate is unchanged. +1 IngestPanel test.

Pre-merge gate ? both green

	Backend (badminton-highlight-indexer)	SPA (khelsutra/web)
Tests	**1770 passed / 4 skipped** (obsreport skips, as CI expects)	**310 passed**
Coverage	**84.6%** (floor 70)	? floor (stmts 94.9 / br 88.8)
Lint	ruff clean (all tracked files)	? (no lint script)
Types	`mypy backend training` clean	`tsc --noEmit` clean
Build	?	`vite build` ?

(SPA Playwright `e2e:ci` not run locally ? needs browser binaries + a live engine; it runs in CI.)

?? One thing you should know

The badminton working tree is a **shared checkout with several live sessions**. Mid-task, the branch shifted under me to `perf/validation-shared-decode-pass` (a sibling's uncommitted validator WIP), and my `git add` had touched the *shared* index. I unstaged it, then committed my work through a **throwaway `git worktree`** so the main HEAD never moved and the sibling's WIP was left exactly as I found it. The main checkout is back on their branch with only their changes; my work is safely on the fix branch. (Noted in memory as `compute-local-hosted-guardrail`.)

Not done (your call)

- **Push + open PRs** on both branches (each repo has its own CI).
- The SPA fix only reaches the *deployed* hosted build after a **re-vendor** (`scripts/bundle_ui.sh` ? `backend/ui/` + `ui_version.json`) ? a separate release step I deliberately kept out of the code PRs.

Want me to push both branches and open the PRs?

141. user (2026-07-07T15:01:45.864Z)

please push and open PRs

142. assistant (2026-07-07T15:02:18.328Z)

The user authorized the push + PRs. I'll use the `gh` CLI (per environment guidance). Let me first confirm `gh` auth and both remotes, and re-verify each branch ref points only at my commit (contended checkout ? verify right before pushing).

143. user (2026-07-07T15:02:23.292Z)

```
=== gh auth ===
  ? Logged in to github.com account avidullu (keyring)
  - Active account: true
  - Token: gho_************************************
  - Token scopes: 'gist', 'read:org', 'repo', 'workflow'
=== backend remote ===
https://github.com/Khelsutra/badminton-highlight-indexer.git
=== backend fix branch: only my commit ahead? ===
4565c74 fix(serving): fail fast when compute=local forces weightless in-process detection
=== backend fix branch tip ===
4565c74
=== main HEAD unchanged (sibling) ===
perf/validation-shared-decode-pass
```

144. assistant (2026-07-07T15:02:30.130Z)

`gh` is authenticated (`avidullu`, `repo` scope). Pushing the backend branch by ref (no checkout ? keeps the sibling's HEAD untouched):

145. user (2026-07-07T15:02:35.269Z)

```
remote:
remote: Create a pull request for 'fix/hosted-local-detection-guardrail' on GitHub by visiting:
remote:      https://github.com/Khelsutra/badminton-highlight-indexer/pull/new/fix/hosted-local-
detection-guardrail
remote:
branch 'fix/hosted-local-detection-guardrail' set up to track 'origin/fix/hosted-local-
detection-guardrail'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch]      fix/hosted-local-detection-guardrail -> fix/hosted-local-detection-
guardrail
```

146. assistant (2026-07-07T15:02:41.500Z)

Backend branch pushed. Now open the PR (specifying `--head` since the shared checkout's HEAD is on the sibling branch):

147. user (2026-07-07T15:03:01.141Z)

https://github.com/Khelsutra/badminton-highlight-indexer/pull/511

148. assistant (2026-07-07T15:03:05.271Z)

Backend PR opened: [#511](https://github.com/Khelsutra/badminton-highlight-indexer/pull/511). Now the SPA repo:

149. user (2026-07-07T15:03:10.998Z)

```
=== branch + remote ===
fix/hosted-compute-picker-defaults-cloud
https://github.com/avidullu/khelsutra.git
=== only my commit ahead? ===
1470e2a fix(ingest): default the compute picker to CLOUD on a hosted control-plane
```

```
=== push ===
remote: Create a pull request for 'fix/hosted-compute-picker-defaults-cloud' on GitHub by
visiting:
remote:      https://github.com/avidullu/khelsutra/pull/new/fix/hosted-compute-picker-defaults-
cloud
remote:
branch 'fix/hosted-compute-picker-defaults-cloud' set up to track 'origin/fix/hosted-compute-
picker-defaults-cloud'.
To https://github.com/avidullu/khelsutra.git
 * [new branch]      fix/hosted-compute-picker-defaults-cloud -> fix/hosted-compute-picker-
defaults-cloud
```

150. user (2026-07-07T15:03:32.585Z)

```
https://github.com/avidullu/khelsutra/pull/169
```

151. assistant (2026-07-07T15:04:45.050Z)

```
Both PRs are open:
- Backend: [Khelsutra/badminton-highlight-indexer#511](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/511)
- SPA: [avidullu/khelsutra#169](https://github.com/avidullu/khelsutra/pull/169)

Since this lands on a hosted production control-plane across two repos, I'll run an adversarial
multi-agent review of the exact diffs before they merge. Let me first capture the authoritative
diffs to pass into the review.
```

152. user (2026-07-07T15:04:49.434Z)

```
backend.diff lines: 216
backend_context lines: 366
spa.diff lines: 88
```

153. user (2026-07-07T15:04:53.785Z)

```
File does not exist. Note: your current working directory is C:\Users\avidu\Projects\khelsutra-
guru\khelsutra.
```

154. assistant (2026-07-07T15:05:11.602Z)

```
The Bash `/tmp` isn't visible to the Read tool. Let me write the diffs into the scratchpad
(absolute path) so the workflow agents can read them directly:
```

155. user (2026-07-07T15:05:18.331Z)

```
11027 /c/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-
indexer/05c32f87-3bfb-4a7f-861c-0050ac4169ac/scratchpad/backend.diff
29098 /c/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-
indexer/05c32f87-3bfb-4a7f-861c-0050ac4169ac/scratchpad/IngestPanel_new.tsx
55735 /c/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-
indexer/05c32f87-3bfb-4a7f-861c-0050ac4169ac/scratchpad/orchestration_new.py
5007 /c/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-
indexer/05c32f87-3bfb-4a7f-861c-0050ac4169ac/scratchpad/spa.diff
42525 /c/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-
indexer/05c32f87-3bfb-4a7f-861c-0050ac4169ac/scratchpad/test_api_cloud_dispatch_new.py
```

156. assistant (2026-07-07T15:06:29.556Z)

```
All snapshots captured. Now I'll run an adversarial review workflow ? 5 review dimensions over
the authoritative diffs, each finding independently verified against the real code, then
synthesized.
```

157. user (2026-07-07T15:07:26.776Z)

Workflow launched in background. Task ID: wcs9m72gz
Summary: Adversarial multi-dimension review of the compute=local hosted-control-plane fix across two repos, each finding verified against the real code
Transcript dir: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\subagents\workflows\wf_a8f43370-4cb
Script file: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru-khelsutra\05c32f87-3bfb-4a7f-861c-0050ac4169ac\workflows\scripts\review-hosted-local-guardrail-wf_a8f43370-4cb.js
(Edit this file with Write/Edit and re-invoke Workflow with {scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru-khelsutra\05c32f87-3bfb-4a7f-861c-0050ac4169ac\workflows\scripts\review-hosted-local-guardrail-wf_a8f43370-4cb.js"} to iterate without resending the script.)
Run ID: wf_a8f43370-4cb
To resume after editing the script: Workflow({scriptPath: "C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru-khelsutra\05c32f87-3bfb-4a7f-861c-0050ac4169ac\workflows\scripts\review-hosted-local-guardrail-wf_a8f43370-4cb.js", resumeFromRunId: "wf_a8f43370-4cb"}) ? completed agents return cached results (cached results may themselves be empty ? inspect journal.jsonl before assuming there is something to recover).

You will be notified when it completes. Use /workflows to watch live progress.

158. assistant (2026-07-07T15:07:37.431Z)

The review workflow is running. While it works, let me update memory to reflect the now-pushed state (PRs open):

159. user (2026-07-07T15:07:39.943Z)

<tool_use_error>String to replace not found in file.
String: ---
name: compute-local-hosted-guardrail
description: "Two local branches (backend + SPA) fixing compute=local on the hosted control-plane ? gated, unpushed, awaiting push/PR"
metadata:
  type: project

(note: Edit also tried swapping \uXXXX escapes and their characters; neither form matched, so the mismatch is likely elsewhere in old_string. Re-read the file and copy the exact surrounding text.)</tool_use_error>

160. user (2026-07-07T15:07:45.636Z)

1 ---
2 name: compute-local-hosted-guardrail
3 description: "Two local branches (backend + SPA) fixing compute=local on the hosted control-plane ? gated, unpushed, awaiting push/PR"
4 metadata:
5   node_type: memory
6   type: project
7   originSessionId: 05c32f87-3bfb-4a7f-861c-0050ac4169ac
8 ---
9
10 Fix for the confusing `compute="local"` failure on the hosted Cloud Run control-plane (job
11 f29d41d0?, 2026-07-07: `[COMPUTE] requested=local actual=in_process ? WASB weights not found at
12 /opt/models/?`). Root cause: `_maybe_dispatch_remote` returned None for `compute="local"` ? the
13 in-process segmenter ran on the WEIGHTLESS control-plane (weights only mount on the worker Job).
14
15 **Two-part fix, COMPLETE + GATED locally 2026-07-07, NOT pushed (awaiting owner push/PR):**

```
   1. **Backend** ? repo `khelsutra-guru/badminton-highlight-indexer`, branch
      `fix/hosted-local-detection-guardrail` @ commit **4565c74** (from origin/master 3be0935).
      `backend/orchestration.py`: new `_WEIGHTS_BACKED_SEGMENTERS` + `_local_detector_weights_present`;
      guardrail in `_maybe_dispatch_remote` `choice=="local"` branch fails FAST (mirrors the
      cloud-not-available `_failed(...)`) when a weights-backed detector (wasb/tracknet/fusion) is
      forced local on a `_cloud_available` server with no readable local weights. +4 tests in
      `tests/test_api_cloud_dispatch.py`. Gate: full suite **1770 passed / 4 skipped**, coverage
      **84.6%** (floor 70), ruff (tracked files) + `mypy backend training` clean.
   2. **SPA** ? repo `avidullu/khelsutra` (source at `khelsutra-guru/khelsutra/web`; `backend/ui/` is
      the VENDORED built bundle via `scripts/bundle_ui.sh`), branch
      `fix/hosted-compute-picker-defaults-cloud` @ commit **1470e2a** (from origin/master c7265a4).
      `web/src/components/IngestPanel.tsx`: compute picker defaults to CLOUD when
      `capabilities.deployment.compute_target==="cloud_run"` (keeps free-LOCAL default on a local
      appliance that merely also offers cloud). +1 IngestPanel test. Gate: typecheck ?, **310 tests**
      ?, coverage ? floor, `vite build` ? (playwright e2e not run locally ? needs browsers + live engine).

   **Next steps:** push both branches + open PRs (each repo has its own CI gate). The SPA fix reaches
   the deployed hosted build only after a re-vendor (`scripts/bundle_ui.sh` from an SPA release ?
   `backend/ui/` + `ui_version.json`) ? a separate RELEASE step, deliberately out of the code PRs.

   **? Contended checkout (2026-07-07):** the badminton main working tree is shared by several live
   sessions ? mid-task the branch shifted under me to `perf/validation-shared-decode-pass` (a sibling's
   validator WIP: `backend/pipeline/validators/*` + `frame_sampler.py`) and extra `claude/*` worktrees
   exist. I committed via a throwaway `git worktree` so the main HEAD never moved. Re-apply
   [[efficient-timeouts-and-retries]] branching-safety: verify branch right before any add/commit.
```

161. user (2026-07-07T15:07:58.778Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

162. user (2026-07-07T15:08:06.070Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

163. user (2026-07-07T15:08:16.664Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

164. user (2026-07-07T15:08:24.888Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

165. user (2026-07-07T15:08:30.522Z)

<tool_use_error>File has been modified since read, either by the user or by a linter. Read it again before attempting to write it.</tool_use_error>

166. user (2026-07-07T15:08:37.496Z)

1 # Memory index
2
3 - ? **Start here (GPU-resident rollout):** session-handoff-gpu-resident.md ? #506 implemented, PR #507 open; next: on-box parity gate + 15-clip golden gate.
4
5 - [Actual repo location](actual-repo-location.md) ? primary working dir is an empty STUB; real code is under `khelsutra-guru/badminton-highlight-indexer`.
6 - [Efficient timeouts, retries & Windows gcloud SSH](efficient-timeouts-and-retries.md) ? no silent retry loops; plain `gcloud compute ssh` (plink/pscp reject `-o`, mishandle `~`); diagnose-first, run long work detached.
7 - [GPU VM setup gotchas](gpu-vm-setup-gotchas.md) ? NVDEC libs, apt-lock-at-boot, torch2.x/pynvc; what #501 should encode.
8 - [LGTM only when merge-ready](lgtm-only-when-merge-ready.md) ? post LGTM only when genuinely satisfied a PR is merge-ready (verified), for all PRs.
9 - [GPU-resident modernization status](gpu-resident-modernization.md) ? #506 validated clean; the resize must match cv2.warpAffine via grid_sample (not F.interpolate).
10 - [compute=local hosted guardrail](compute-local-hosted-guardrail.md) ? two gated-but-UNPUSHED branches (backend 4565c74 + SPA 1470e2a) fixing the "weights not found" trap; contended checkout warning.
11 - [Validation shared decode pass](validation-shared-decode-pass.md) ? validators now share ONE decode pass (branch `perf/validation-shared-decode-pass`, d53851b, **PR #512** open); pure grab()-through is SLOWER than seeks ? use forward-hybrid.
12 - [PR #513 worker skip-revalidation](pr513-worker-skip-revalidation.md) ? worker skips redundant re-validation on cloud dispatch (fixes the blur-threshold CUJ); PR open, done in an isolated worktree ? conflicts with `perf/validation-shared-decode-pass` (**PR #512**) on orchestration.py + test_api_cloud_dispatch.py.
13

167. assistant (2026-07-07T15:08:46.398Z)

A sibling session's memory reveals **PRs #512 and #513 also touch `orchestration.py` and `test_api_cloud_dispatch.py`** ? the same two files my PR #511 changes. Let me update the index and flag the merge-conflict overlap:

168. user (2026-07-07T15:08:50.221Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

169. user (2026-07-07T15:08:57.238Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

170. assistant (2026-07-07T15:09:16.120Z)

Both PRs are open and I've flagged an important cross-PR overlap. The adversarial review is still running (5 dimensions ? per-finding verification); I'll report its findings when it lands.

? Pushed + PRs open

- **Backend** ? [Khelsutra/badminton-highlight-indexer#511](https://github.com/Khelsutra/badminton-highlight-indexer/pull/511) (`fix/hosted-local-detection-guardrail`, commit `4565c74`)
- **SPA** ? [avidullu/khelsutra#169](https://github.com/avidullu/khelsutra/pull/169) (`fix/hosted-compute-picker-defaults-cloud`, commit `1470e2a`)

Both pushed by ref without disturbing the shared checkout (its HEAD is still on a sibling session's `perf/validation-shared-decode-pass`).

?? Heads-up: overlapping in-flight PRs

While updating memory I noticed sibling sessions have **PR #512** (`perf/validation-shared-decode-pass`) and **PR #513** (worker skip-revalidation) open, and **both edit the same two files as my #511** ? `backend/orchestration.py` and `tests/test_api_cloud_dispatch.py`. All three branch from the same `origin/master` (3be0935), and my edits are additive and in different regions (new helper above `_cloud_available`, a block inside the `choice=="local"` branch, appended test functions), so it's a mechanical rebase ? but **whoever merges second will need to resolve conflicts** in those two files. Worth sequencing the merges deliberately.

The review workflow is verifying correctness, weights-resolution fidelity, the cross-repo contract, test adequacy, and regressions/conventions ? findings to follow.

171. user (2026-07-07T15:26:47.443Z)

<task-notification>
<task-id>wcs9m72gz</task-id>
<tool-use-id>toolu_01QBgAUApHYznZeJhQqVMTHF</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\wcs9m72gz.output</output-file>
<status>completed</status>
<summary>Dynamic workflow "Adversarial multi-dimension review of the compute=local hosted-control-plane fix across two repos, each finding verified against the real code" completed</summary>
<result>{"totals":{"surfaced":4,"confirmed":0,"refuted":4},"confirmed":[],"refuted":[{"severity":"medium","category":"weights-resolution-fidelity","location":"backend/orchestration.py ? _local_detector_weights_present() tracknet branch: `if name == \"tracknet_hybrid\" or (name == \"fusion_hybrid\" and substrate == \"tracknet\")`","claim":"For tracknet_hybrid and fusion_hybrid(substrate=\"tracknet\") the guard's \"weights present ? run in-process locally\" escape hatch is UNSOUND on the hosted control-plane: it keys on indexing.tracknet.weights_path existence, but there is NO native TrackNet runner and Linux has no WSL, so weights-present does not imply the in-process detector can run. The docstring's claim to \"mirror the native/WSL runners' resolution\" is itself false for tracknet on Linux.","failure_scenario":"Hosted Linux CP (deployment.compute_target=cloud_run ? _cloud_available True). Operator sets indexing.tracknet.weights_path to a path that exists on the CP (any existing file flips the check to True). POST /api/process segmenter=tracknet_hybrid compute=local ? _local_detector_weights_present returns True ? the choice==\"local\" fast-fail is SKIPPED ? the in-process run proceeds and dies deep: with detector_impl=native the base _build_native_runner raises NotImplementedError (\"detector_impl='native' is not supported for the 'TrackNet' family\"), or with detector_impl=auto the WSL TrackNetRunner.healthcheck runs `wsl` ? FileNotFoundError (\"`wsl` not found\"). Result is exactly the cryptic deep-failure UX the fix set out to eliminate for one of the three segmenters it explicitly claims to cover, instead of the actionable \"Choose 'Run in cloud'.\" (Same for fusion_hybrid with substrate=tracknet.) Note: the DEFAULT indexing.tracknet.weights_path=\"\" makes the guard return False and fire correctly, so this only misfires with a configured, existing tracknet weights_path; it is a coverage/fidelity gap, not a regression versus pre-fix.","suggested_fix":"Make \"is there a runner that can run this detector locally on THIS box\" part of the predicate, independent of weights. Since there is no native TrackNet runner, for tracknet_hybrid and fusion(substrate=tracknet) return False (not-locally-runnable) on a cloud-capable server so the fast-fail always fires, regardless of indexing.tracknet.weights_path ? or gate on the real precondition (detector_impl / WSL availability) the way trajectory_hybrid._resolve_runner does.","dimension":"weights-resolution-fidelity","verdict":{"verdict":"REFUTED","confidence":"high","reasoning":"The finding's mechanism is technically real (there is no native_tracknet_runner.py; TrackNetRunner is WSL-only, tracknet_runner.py:95,117-121; and _local_detector_weights_present keys the tracknet branch on os.path.exists(indexing.tracknet.weights_path), orchestration_new.py:179-187), but the

described failure does NOT occur against the real deployed code/config, and its consequence is overstated. Three grounded reasons:\n\n1) NOT REACHABLE on the real control-plane. The guard only misfires if indexing.tracknet.weights_path is non-empty AND points at a file that EXISTS on the Linux CP. On the actual hosted CP it is empty: the model default is \"\" (backend/config/models.py:51), the served config.json sets \"tracknet\": {\"weights_path\": \"\"} (config.json:49), and NO deploy config sets it ? the Cloud Run image (deploy/cloudrun/Dockerfile:118-119) only sets WASB_WEIGHTS and RALLY_DETECTOR_IMPL=native; a grep of deploy/ shows zero tracknet weights_path. With path=\"\" the tracknet branch returns bool(\"\")?False, so _local_detector_weights_present is False, `not (...)` is True, and the choice==\"local\" guard FIRES with the actionable \"Choose 'Run in cloud'.\" message ? exactly what the fix intends. So for tracknet_hybrid in the realistic hosted config the fix DOES cover it; the finding even concedes the default \"\" fires correctly. To trigger the misfire an operator must deliberately set a WSL-only path (that field is consumed solely by the WSL TrackNetRunner via --model_weights) to some unrelated existing file on the Linux CP. That is non-default, operationally meaningless, and by the bug's own premise the CP mounts NO detector weights, so no weights-shaped path exists there to point at; even copying a dev config won't do it because a ~/models/... WSL path resolves to a non-existent path on the CP (os.path.exists?False?guard still fires). No real config produces the repro.\n\n2) CONSEQUENCE OVERSTATED ? no \"deep crash\". Even in that contrived case the in-process run does not die deep: trajectory_hybrid.process_video CATCHES the detector_impl='native' NotImplementedError (trajectory_hybrid.py:247-254 ? Failure \"detector_impl='native' is not supported for the 'TrackNet' family ? only WASB has a Linux-native runner\") and, for detector_impl='auto', surfaces the WSL healthcheck FileNotFoundError as a clean Failure (tracknet_runner.py:117-121 ? trajectory_hybrid.py:283-291 \"TrackNet detector environment not ready: `wsl` not found ?\"). Both become status=failed with an informative message via the same clean-Failure mechanism the fix relies on ? not an uncaught traceback. (On the real CP RALLY_DETECTOR_IMPL=native, so it's specifically the NotImplementedError?Failure path.)\n\n3) NOT A REGRESSION and not \"wrong behavior\". The request still correctly fails (tracknet genuinely cannot run in-process on the CP), so there is no wrong result; the only delta versus the ideal fixed path is message wording, and only under a self-inflicted misconfiguration, and it is strictly no worse than pre-fix (the finding itself labels it \"a coverage/fidelity gap, not a regression\"). The docstring's \"mirrors the /api/capabilities probe\" is loosely worded for tracknet (capabilities probes only wasb_weights_present, main.py), but that is documentation imprecision, not a defect that causes wrong behavior, a bad merge, a regression, or a materially worse UX in the deployed system. Per the mandate ? CONFIRMED only with a concrete repro grounded in the real code/config ? there is none, so this is REFUTED."}},{"severity":"low","category":"weights-resolution-fidelity","location":"backend/orchestration.py ? _local_detector_weights_present() (whole function: it consults only env WASB_WEIGHTS + indexing.<det>.weights_path and never reads indexing.detector_impl / RALLY_DETECTOR_IMPL)","claim":"The guard hard-codes the native-runner weights resolution and ignores detector_impl, so it does not mirror the two other runners: the stub runner (needs NO weights; healthcheck returns True unconditionally) and the WSL runner (weights live INSIDE WSL and it ignores WASB_WEIGHTS). This yields a false-BLOCK of a compute=local run that would actually succeed.","failure_scenario":"(a) Stub: a cloud-capable box (compute_target=cloud_run ? _cloud_available True) configured with indexing.detector_impl=\"stub\" (or RALLY_DETECTOR_IMPL=stub ? a CI/demo/staging box that mimics hosted config to avoid GPU cost). POST /api/process segmenter=wasb_hybrid compute=local ? StubDetectorRunner.healthcheck() returns True and run_predict writes a deterministic CSV with no weights/GPU/WSL, so the run WOULD succeed; but _local_detector_weights_present sees no wasb weights on disk ? returns False ? the guard fast-fails \"this hosted server can't run detection locally\", wrongly blocking a working run (and pushing the user to paid cloud). (b) WSL divergence: a cloud-wired Windows+WSL box with detector_impl=auto and WASB weights at the default WSL path ~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar (present inside WSL, where WasbRunner reads them): the local run would succeed, but the guard expanduser's that path against the Windows HOST home (C:\\Users\\<user>\\models\\WASB-SBDT\\...) where it does not exist ? returns False ? false-block. Both require non-default configs (stub+cloud, or WSL+cloud-wired), hence low severity.","suggested_fix":"Consult the impl the run wil
... (truncated 14634 chars, full result in C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\wcs9m72gz.output)</result>
<diagnostics>Per-agent results: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\subagents\workflows\wf_a8f43370-4cb/journal.jsonl ? one {"type":"result",...} line per completed agent with its full return value.
If the result above is empty or unexpected, Read this file BEFORE diagnosing ? do not assume

agents returned non-empty results.
To re-run with edited post-processing: Workflow({scriptPath: 'C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-khelsutra-guru-khelsutra\05c32f87-3bfb-4a7f-861c-0050ac4169ac\workflows\scripts\review-hosted-local-guardrail-wf_a8f43370-4cb.js', resumeFromRunId: 'wf_a8f43370-4cb', args: "{\"scratch\":\"C:/Users/avidu/AppData/Local/Temp/claude/C--Users-avidu-Projects-badminton-highlight-indexer/05c32f87-3bfb-4a7f-861c-0050ac4169ac/scratchpad\",\"backendRepo\":\"C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer\",\"spaRepo\":\"C:/Users/avidu/Projects/khelsutra-guru/khelsutra\",\"backendBranch\":\"fix/hosted-local-detection-guardrail\",\"spaBranch\":\"fix/hosted-compute-picker-defaults-cloud\",\"backendPR\":511,\"spaPR\":169}"}) ? agents whose (prompt, opts) are unchanged replay from cache.</diagnostics>
<usage><agent_count>9</agent_count><agents_done>9</agents_done><agents_error>0</agents_error><agents_skipped>0</agents_skipped><agents_empty_result>3</agents_empty_result><subagent_tokens>1131896</subagent_tokens><tool_uses>199</tool_uses><duration_ms>1160610</duration_ms></usage>
</task-notification>

172. assistant (2026-07-07T15:27:11.474Z)

The review is complete: **4 findings surfaced, 0 confirmed, 4 refuted** after adversarial verification. And the memory note reveals a critical development ? **PR #513 just merged first**, so `origin/master` has moved and my #511 needs a rebase. Let me read the full review output and check the new master state.

173. user (2026-07-07T15:27:13.550Z)


```
{
  "summary": "Adversarial multi-dimension review of the compute=local hosted-control-plane fix across two repos, each finding verified against the real code",
  "agentCount": 9,
  "logs": [
    "Reviewing the hosted-local guardrail across 5 dimensions, then adversarially verifying each finding.",
    "Review complete: 4 findings surfaced, 0 CONFIRMED after adversarial verification."
  ],
  "result": {
    "totals": {
      "surfaced": 4,
      "confirmed": 0,
      "refuted": 4
    },
    "confirmed": [],
    "refuted": [
      {
        "severity": "medium",
        "category": "weights-resolution-fidelity",
        "location": "backend/orchestration.py ? _local_detector_weights_present() tracknet branch: `if name == \"tracknet_hybrid\" or (name == \"fusion_hybrid\" and substrate == \"tracknet\")`",
        "claim": "For tracknet_hybrid and fusion_hybrid(substrate=\"tracknet\") the guard's \"weights present ? run in-process locally\" escape hatch is UNSOUND on the hosted control-plane: it keys on indexing.tracknet.weights_path existence, but there is NO native TrackNet runner and Linux has no WSL, so weights-present does not imply the in-process detector can run. The docstring's claim to \"mirror the native/WSL runners' resolution\" is itself false for tracknet on Linux.",
        "failure_scenario": "Hosted Linux CP (deployment.compute_target=cloud_run ? _cloud_available True). Operator sets indexing.tracknet.weights_path to a path that exists on the CP (any existing file flips the check to True). POST /api/process segmenter=tracknet_hybrid compute=local ? _local_detector_weights_present returns True ? the choice==\"local\" fast-fail is SKIPPED ? the in-process run proceeds and dies deep: with detector_impl=native the base _build_native_runner raises NotImplementedError (\"detector_impl='native' is not supported for the 'TrackNet' family\"), or with detector_impl=auto the WSL TrackNetRunner.healthcheck runs `wsl` ? FileNotFoundError (\"`wsl` not found\"). Result is exactly the cryptic deep-failure UX the fix set out to eliminate for one of the three segmenters it explicitly claims to cover,
```

instead of the actionable \"Choose 'Run in cloud'.\" (Same for fusion_hybrid with substrate=tracknet.) Note: the DEFAULT indexing.tracknet.weights_path=\"\" makes the guard return False and fire correctly, so this only misfires with a configured, existing tracknet weights_path; it is a coverage/fidelity gap, not a regression versus pre-fix.",
            "suggested_fix": "Make \"is there a runner that can run this detector locally on THIS box\" part of the predicate, independent of weights. Since there is no native TrackNet runner, for tracknet_hybrid and fusion(substrate=tracknet) return False (not-locally-runnable) on a cloud-capable server so the fast-fail always fires, regardless of indexing.tracknet.weights_path ? or gate on the real precondition (detector_impl / WSL availability) the way trajectory_hybrid._resolve_runner does.",
            "dimension": "weights-resolution-fidelity",
            "verdict": {
              "verdict": "REFUTED",
              "confidence": "high",
              "reasoning": "The finding's mechanism is technically real (there is no native_tracknet_runner.py; TrackNetRunner is WSL-only, tracknet_runner.py:95,117-121; and _local_detector_weights_present keys the tracknet branch on os.path.exists(indexing.tracknet.weights_path), orchestration_new.py:179-187), but the described failure does NOT occur against the real deployed code/config, and its consequence is overstated. Three grounded reasons:\n\n1) NOT REACHABLE on the real control-plane. The guard only misfires if indexing.tracknet.weights_path is non-empty AND points at a file that EXISTS on the Linux CP. On the actual hosted CP it is empty: the model default is \"\" (backend/config/models.py:51), the served config.json sets \"tracknet\": {\"weights_path\": \"\"} (config.json:49), and NO deploy config sets it ? the Cloud Run image (deploy/cloudrun/Dockerfile:118-119) only sets WASB_WEIGHTS and RALLY_DETECTOR_IMPL=native; a grep of deploy/ shows zero tracknet weights_path. With path=\"\" the tracknet branch returns bool(\"\")?False, so _local_detector_weights_present is False, `not (...)` is True, and the choice==\"local\" guard FIRES with the actionable \"Choose 'Run in cloud'.\" message ? exactly what the fix intends. So for tracknet_hybrid in the realistic hosted config the fix DOES cover it; the finding even concedes the default \"\" fires correctly. To trigger the misfire an operator must deliberately set a WSL-only path (that field is consumed solely by the WSL TrackNetRunner via --model_weights) to some unrelated existing file on the Linux CP. That is non-default, operationally meaningless, and by the bug's own premise the CP mounts NO detector weights, so no weights-shaped path exists there to point at; even copying a dev config won't do it because a ~/models/... WSL path resolves to a non-existent path on the CP (os.path.exists?False?guard still fires). No real config produces the repro.\n\n2) CONSEQUENCE OVERSTATED ? no \"deep crash\". Even in that contrived case the in-process run does not die deep: trajectory_hybrid.process_video CATCHES the detector_impl='native' NotImplementedError (trajectory_hybrid.py:247-254 ? Failure \"detector_impl='native' is not supported for the 'TrackNet' family ? only WASB has a Linux-native runner\") and, for detector_impl='auto', surfaces the WSL healthcheck FileNotFoundError as a clean Failure (tracknet_runner.py:117-121 ? trajectory_hybrid.py:283-291 \"TrackNet detector environment not ready: `wsl` not found ?\"). Both become status=failed with an informative message via the same clean-Failure mechanism the fix relies on ? not an uncaught traceback. (On the real CP RALLY_DETECTOR_IMPL=native, so it's specifically the NotImplementedError?Failure path.)\n\n3) NOT A REGRESSION and not \"wrong behavior\". The request still correctly fails (tracknet genuinely cannot run in-process on the CP), so there is no wrong result; the only delta versus the ideal fixed path is message wording, and only under a self-inflicted misconfiguration, and it is strictly no worse than pre-fix (the finding itself labels it \"a coverage/fidelity gap, not a regression\"). The docstring's \"mirrors the /api/capabilities probe\" is loosely worded for tracknet (capabilities probes only wasb_weights_present, main.py), but that is documentation imprecision, not a defect that causes wrong behavior, a bad merge, a regression, or a materially worse UX in the deployed system. Per the mandate ? CONFIRMED only with a concrete repro grounded in the real code/config ? there is none, so this is REFUTED."
            }
          },
          {
            "severity": "low",
            "category": "weights-resolution-fidelity",
            "location": "backend/orchestration.py ? _local_detector_weights_present() (whole function: it consults only env WASB_WEIGHTS + indexing.<det>.weights_path and never reads indexing.detector_impl / RALLY_DETECTOR_IMPL)",
            "claim": "The guard hard-codes the native-runner weights resolution and ignores detector_impl, so it does not mirror the two other runners: the stub runner (needs NO weights; healthcheck returns True unconditionally) and the WSL runner (weights live INSIDE WSL and it

ignores WASB_WEIGHTS). This yields a false-BLOCK of a compute=local run that would actually succeed.",
35 "failure_scenario": "(a) Stub: a cloud-capable box (compute_target=cloud_run ? _cloud_available True) configured with indexing.detector_impl=\"stub\" (or RALLY_DETECTOR_IMPL=stub ? a CI/demo/staging box that mimics hosted config to avoid GPU cost). POST /api/process segmenter=wasb_hybrid compute=local ? StubDetectorRunner.healthcheck() returns True and run_predict writes a deterministic CSV with no weights/GPU/WSL, so the run WOULD succeed; but _local_detector_weights_present sees no wasb weights on disk ? returns False ? the guard fast-fails \"this hosted server can't run detection locally\", wrongly blocking a working run (and pushing the user to paid cloud). (b) WSL divergence: a cloud-wired Windows+WSL box with detector_impl=auto and WASB weights at the default WSL path ~/models/WASB-SBDT/pretrained_weights/wasb_badminton_best.pth.tar (present inside WSL, where WasbRunner reads them): the local run would succeed, but the guard expanduser's that path against the Windows HOST home (C:\\Users\\<user>\\models\\WASB-SBDT\\...) where it does not exist ? returns False ? false-block. Both require non-default configs (stub+cloud, or WSL+cloud-wired), hence low severity.",
36 "suggested_fix": "Consult the impl the run will actually use: short-circuit to True (not-guardrailed) when RALLY_DETECTOR_IMPL/indexing.detector_impl == \"stub\"; and either scope the guard to detector_impl==\"native\" (the only impl whose resolution it faithfully mirrors) or resolve weights the way the selected runner does (WSL runner = config weights_path only, checked inside WSL), so the guard mirrors the runner that will execute rather than assuming the native path.",
37 "dimension": "weights-resolution-fidelity",
38 "verdict": {
39 "verdict": "REFUTED",
40 "confidence": "high",
41 "reasoning": "The finding's technical observations are correct, but its \"false-BLOCK of a compute=local run that would actually succeed\" does not occur in any coherent/realistic deployment ? it is reachable only via a self-inconsistent, off-preset hand-mixed config, and even then with a working escape.\n\nVerified against the real code:\n1) The guard `_local_detector_weights_present` (orchestration_new.py:166-187) resolves weights via `WASB_WEIGHTS ? indexing.wasb.weights_path` + `os.path.exists(os.path.expanduser(path))`. This EXACTLY mirrors the NATIVE runner's healthcheck (native_wasb_runner.py:120 `env(\"WASB_WEIGHTS\") or w.weights_path`, and :202-208 `os.path.exists(os.path.expanduser(...))`). The stub healthcheck returns True unconditionally (stub_runner.py:84-85) and the WSL runner reads weights inside WSL via `wsl_tilde_quote(cfg.weights_path)`, never `WASB_WEIGHTS` (wasb_runner.py:183) ? so the finding is right that the guard mirrors only the native runner.\n2) BUT the guard fires only when `_cloud_available(cfg)` is True. I ran the real `_cloud_available` + the snapshot guard predicate across all three designed presets (presets.py): cloud-serving ? compute_target=cloud_run, detector_impl=NATIVE ? cloud_available=True, guard fires and is CORRECT (native resolution faithfully mirrored; this is the exact bug it fixes); truly-local ? compute_target=local_gpu, auto/WSL ? cloud_available=FALSE ? guard SKIPPED (WSL runs); cpu-mock ? compute_target=cpu_mock, stub ? cloud_available=FALSE ? guard SKIPPED (stub runs). So under EVERY coherent preset the guard never false-blocks.\n3) The finding's stub (a) and WSL (b) false-blocks require pairing compute_target=cloud_run with a NON-native detector_impl (stub or auto/WSL) ? a per-knob \"drift\" the preset system (presets.py docstring: modes must not \"drift apart one knob at a time\") is explicitly built to prevent. I confirmed the mechanism only by hand-mixing such an off-preset config (cloud_run+stub ? guard_fires=True). Scenario (b) is additionally self-defeating: the WSL runner is the Windows-local-GPU path, yet on a compute_target=cloud_run box the default/compute=cloud paths dispatch to a Linux Cloud Run worker (native), and CLOUD_RUN_JOB (the other _cloud_available trigger) is a Cloud Run runtime var absent on a Windows dev box.\n4) Even in the contrived config, the guard's own precondition _cloud_available=True guarantees the box is cloud-capable, so the \"Choose 'Run in cloud'\" message it returns actually works ? no dead-end, no broken product; only a niche manual compute=local override is blocked, against the SPA's cloud default.\n\nNet: a real fidelity gap (guard mirrors native, not stub/WSL) but not a defect that causes wrong behavior in a realistic deployment. The guard correctly mirrors the native runner ? the detector_impl a coherent compute_target=cloud_run deployment actually uses. Per the default-REFUTED bar and the earlier _cloud_available/preset-coherence guard the finding under-weights, this is REFUTED.",
42 "corrected_scenario": "N/A ? refuted. The described false-block is unreachable under any coherent preset (verified: truly-local and cpu-mock both yield _cloud_available=False, so the guard is never reached; cloud-serving uses detector_impl=native, which the guard mirrors faithfully). It surfaces only if an operator hand-wires an off-preset config that pairs deployment.compute_target=cloud_run with indexing.detector_impl=stub (or leaves auto/WSL on a

deliberately cloud-wired Windows box) ? the exact per-knob drift the preset bundles exist to prevent ? and even then _cloud_available=True means the 'Run in cloud' escape the guard points to actually works."
        }
      },
      {
        "severity": "medium",
        "category": "test-coverage",
        "location": "tests/test_api_cloud_dispatch.py (new guardrail block, snapshot lines ~731-829)",
        "claim": "The guardrail is declared for three weights-backed detectors (wasb_hybrid, tracknet_hybrid, fusion_hybrid) but the four new tests only exercise wasb_hybrid and motion; tracknet_hybrid and fusion_hybrid ? and thus the entire branchy half of _local_detector_weights_present (the tracknet weights_path resolution and the fusion substrate switch) ? have ZERO coverage (neither fail-fast nor escape-hatch).",
        "failure_scenario": "A deployment runs segmenter=fusion_hybrid with indexing.fusion.substrate='tracknet' (or segmenter=tracknet_hybrid). A later refactor breaks the substrate routing in _local_detector_weights_present (orchestration.py:179-186) ? e.g. it reads indexing.wasb.weights_path for all three, or TrackNetConfig.weights_path (models.py:51) is renamed. Every new test still passes (they only use wasb_hybrid/motion), yet on the hosted cloud_run control-plane the guardrail now probes the WRONG weights file: WasbIndexConfig.weights_path defaults to a real path (~/models/WASB-SBDT/...), so if that resolves 'present' the escape hatch is wrongly taken and the in-process run falls through and dies deep with the exact 'WASB weights not found' cryptic failure this fix exists to eliminate ? silently, with a green suite.",
        "suggested_fix": "Add at least two tests mirroring test_local_on_cloud_server_without_weights_fails_fast / _with_weights_present_runs_in_process but for (a) segmenter='tracknet_hybrid' driving indexing.tracknet.weights_path, and (b) segmenter='fusion_hybrid' with indexing.fusion.substrate='tracknet' set (tracknet weights missing ? fail-fast; tracknet weights present ? in_process) plus a fusion default-substrate('wasb') case. This pins the substrate switch and the tracknet config path so a routing regression can't ship green.",
        "dimension": "test-adequacy",
        "verdict": {
          "verdict": "REFUTED",
          "confidence": "medium",
          "reasoning": "I read the authoritative snapshots (orchestration_new.py:163-187, 480-498; test_api_cloud_dispatch_new.py:731-829), backend/config/models.py (TrackNetConfig.weights_path=\"\" @L51; WasbIndexConfig.weights_path defaults to \"~/models/WASB-SBDT/...pth.tar\" @L80; FusionConfig.substrate default \"wasb\" @L180), and executed the actual routing logic (current vs. a substrate-switch-dropping mutation) across every branch.\n\nThe finding's factual coverage claim is TRUE: grepping the new test file for tracknet_hybrid|fusion_hybrid|substrate returns ZERO matches; the four new guardrail tests only drive wasb_hybrid (x3) and motion (x1). My mutation harness confirms a routing regression ships green ? T1/T2/T3 give identical results under current and mutated code (False/False, True/True, True/True).\n\nBut the finding is REFUTED as a defect \"against the real code as written,\" for three grounded reasons:\n\n1) NO wrong result exists against the current code. Executed, the current function routes correctly for every input (tracknet tn-present?True, tn-absent?False; fusion+tracknet?reads tracknet; fusion-default?reads wasb). The finding's failure_scenario is explicitly predicated on \"A later refactor breaks the substrate routing\" ? a hypothetical mutation, not a defect in this change. The task requires \"a concrete repro grounded in the real code\"; the only repro requires introducing a refactor, i.e., it is not the real code.\n\n2) The headline outcome contradicts the fix's own deployment premise. On the canonical hosted control-plane the fix targets, NO detector weights are baked on disk. I executed the fusion+tracknet default case with nothing on disk: BOTH current and mutated routing return False ? both still fail-fast. So the described \"escape hatch wrongly taken ? silently dies deep with a green suite\" does NOT occur on the targeted deployment; it needs an atypical partial-weights box (WASB weights physically present on disk while TrackNet weights absent). Also, a tracknet/fusion+tracknet in-process death would raise a TrackNet-weights error, not the \"WASB weights not found\" the finding names.\n\n3) The uncovered surface is overstated. The else/WASB branch (orchestration.py:181-186) IS exercised by the three wasb_hybrid tests, and fusion_hybrid on its DEFAULT wasb substrate takes that same covered branch ? so the finding's requested \"fusion default-substrate('wasb') case\" is already covered by proxy. Only the single tracknet line (L180) is genuinely uncovered (reached by tracknet_hybrid always, and fusion+substrate==\"tracknet\"), and by default TrackNetConfig.weights_path=\"\" makes that path

always fail-fast (its escape hatch is a non-default operator opt-in). So the real gap is one
narrow, low-materiality line, not \"the entire branchy half.\"\n\nNet: a legitimate but narrow
coverage nicety, materially overstated; the concrete failure it describes does not occur against
the real code as written (it requires a future mutation plus a deployment state the fix
excludes). Medium confidence because a pure coverage-adequacy lens could still value pinning
L180."
        }
      },
      {
        "severity": "low",
        "category": "test-coverage",
        "location": "tests/test_api_cloud_dispatch.py::test_local_weights_backed_on_pure
_local_server_is_not_guardrailed (snapshot lines ~801-829)",
        "claim": "The guardrail fires only when _cloud_available(cfg) is True, which has
two routes: compute_target=='cloud_run' (config) and
CLOUD_RUN_JOB+GOOGLE_CLOUD_PROJECT+storage.output_root (env). Every guardrail-FIRES test uses
the config route (cloud_env sets compute_target=cloud_run); the env route into the guardrail is
never exercised ? the 'not guardrailed' test explicitly deletes both env vars AND uses
compute_target='', so compute_target=''-but-cloud-wired-via-env is untested.",
        "failure_scenario": "A local-default box wired for cloud via env
(deployment.compute_target='', CLOUD_RUN_JOB + GOOGLE_CLOUD_PROJECT + storage.output_root set)
is cloud-available through _cloud_available's env branch (orchestration.py:202-206). A
compute='local' + wasb_hybrid request with no local weights should fail-fast there too; if that
env branch of the guardrail's fire condition regressed, the box would silently fall through to
the deep 'weights not found' failure and no guardrail test would catch it. Mitigant:
_cloud_available's env branch is independently covered by
test_per_request_cloud_forces_dispatch_on_local_server /
test_cloud_override_survives_api_boundary_on_local_server, so real-world risk is low ? this is a
completeness gap, not a likely latent bug.",
        "suggested_fix": "Add one guardrail test on a compute_target='' server that sets
CLOUD_RUN_JOB + GOOGLE_CLOUD_PROJECT + a storage.output_root bucket, segmenter='wasb_hybrid',
compute='local', no local weights, and asserts the fast-fail (status=failed, message 'can't run
detection locally', fake.calls==[], path='not_run'), proving the env-wired cloud_available route
also triggers the guardrail.",
        "dimension": "test-adequacy",
        "verdict": {
          "verdict": "REFUTED",
          "confidence": "high",
          "reasoning": "The finding's underlying observation is factually accurate ? no
test drives the guardrail FIRING via the env route: the only fast-fail test
(test_local_on_cloud_server_without_weights_fails_fast, snapshot 731-753) uses the cloud_env
fixture (compute_target=cloud_run, the config route of _cloud_available), and the \"not
guardrailed\" test (801-829) sets compute_target='' AND deletes both env vars. So a
compute_target='' box wired for cloud via env (CLOUD_RUN_JOB+GOOGLE_CLOUD_PROJECT+output_root)
hitting the guardrail is genuinely not directly tested.\n\nBut the DESCRIBED FAILURE does not
occur against the real code, so this is a REFUTED as a reportable defect:\n\n1) The code is
correct on that path. I traced orchestration.py _maybe_dispatch_remote for compute_target='' +
CLOUD_RUN_JOB='rally-worker' + GOOGLE_CLOUD_PROJECT='proj' + output_root='gs://bucket' +
compute='local' + seg='wasb_hybrid' + no weights: choice=='local' block (480); 'wasb_hybrid' in
_WEIGHTS_BACKED_SEGMENTERS True; _cloud_available (199-206) skips the config return
(compute_target=='') then returns True via the env branch (has_job=True, out_root truthy);
_local_detector_weights_present returns False (path '/nonexistent' fails os.path.exists) so `not
...`=True ? all three True ? the guardrail FIRES, returning _failed(\"...can't run detection
locally...\", ran=False) with path=not_run, no dispatch. That is the correct, intended behavior
? no wrong result, no bad merge, no regression, no worse UX.\n\n2) The finding assumes a code
path that does not exist. There is no separate \"env branch of the guardrail's fire condition.\"
The guardrail (488-497) calls _cloud_available(cfg) as a single uniform boolean; whether that
returns True via the config route or the env route produces NO distinct branch within the
guardrail. The route distinction lives entirely inside _cloud_available. So the failure scenario
is explicitly conditional on a hypothetical future regression (\"if that env branch of the
guardrail's fire condition regressed\"), not a present defect.\n\n3) A realistic regression
cannot silently slip through. The env route of _cloud_available is independently regression-
protected: test_per_request_cloud_forces_dispatch_on_local_server (setenv CLOUD_RUN_JOB at
snapshot line 523) and test_cloud_override_survives_api_boundary_on_local_server (line 698) both
run compute='cloud' on a compute_target='' box and REQUIRE dispatch ? if that env branch

returned False, both would fail with \"isn't configured\" (the choice=='cloud' guard at 500-505). The guardrail reuses that exact tested function. capabilities() in main.py:404 also consumes the same _cloud_available, further pinning it. The only regression the missing test would uniquely catch is a hypothetical refactor that inlines a compute_target=='cloud_run' check directly into the guardrail instead of calling _cloud_available ? a speculative future edit, not a defect in the change under review.\n\nThe finding itself concedes \"real-world risk is low ? this is a completeness gap, not a likely latent bug.\" Against the review mandate (real, concrete defects with a concrete config?wrong-result), and the adversarial instruction to default to REFUTED unless the failure reproduces concretely, there is no reproducible present-code failure: the guardrail behaves correctly via the env route."
```
            }
          }
        ]
      },
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        },
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          "type": "workflow_agent",
          "index": 1,
          "label": "review:guardrail-correctness",
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          "phaseTitle": "Review",
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          "state": "done",
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          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST /api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at /gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437428046,
          "tokens": 158567,
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          "resultPreview": "{\"findings\":[]}"
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          "model": "claude-opus-4-8",
          "state": "done",
          "startedAt": 1783436846826,
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          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST /api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which
```

```
bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at
/gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437537508,
          "tokens": 147768,
          "toolCalls": 24,
          "durationMs": 690682,
          "resultPreview": "{\"findings\":[{\"severity\":\"medium\",\"category\":\"weights-
resolution-fidelity\",\"location\":\"backend/orchestration.py ?
_local_detector_weights_present() tracknet branch: `if name == \\\"tracknet_hybrid\\\" or (name
== \\\"fusion_hybrid\\\" and substrate == \\\"tracknet\\\")`\",\"claim\":\"For tracknet_hybrid
and fusion_hybrid(substrate=\\\"tracknet\\\") the guard's \\\"weights present ? run in-process
locally\\\" escape hatch?"
        },
        {
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          "phaseTitle": "Review",
          "agentId": "a483bdc3be00cae5b",
          "model": "claude-opus-4-8",
          "state": "done",
          "startedAt": 1783436846827,
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          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a
HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST
/api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which
bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at
/gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437450216,
          "tokens": 145732,
          "toolCalls": 24,
          "durationMs": 603389,
          "resultPreview": "{\"findings\":[]}"
        },
        {
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          "index": 4,
          "label": "review:test-adequacy",
          "phaseIndex": 1,
          "phaseTitle": "Review",
          "agentId": "ab8ad46e9cb84dca5",
          "model": "claude-opus-4-8",
          "state": "done",
          "startedAt": 1783436846829,
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          "attempt": 1,
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          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a
HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST
/api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which
bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at
/gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437186990,
          "tokens": 103722,
          "toolCalls": 18,
          "durationMs": 340161,
          "resultPreview": "{\"findings\":[{\"severity\":\"medium\",\"category\":\"test-
coverage\",\"location\":\"tests/test_api_cloud_dispatch.py (new guardrail block, snapshot lines
~731-829)\",\"claim\":\"The guardrail is declared for three weights-backed detectors
(wasb_hybrid, tracknet_hybrid, fusion_hybrid) but the four new tests only exercise wasb_hybrid
and motion; tracknet_hybrid and fusion_hybrid ? and thus the entire branchy half ?"
        },
```

```
        {
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          "phaseTitle": "Review",
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          "queuedAt": 1783436846783,
          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST /api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at /gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437768803,
          "tokens": 149697,
          "toolCalls": 42,
          "durationMs": 921973,
          "resultPreview": "{\"findings\":[]}"
        },
        {
          "type": "workflow_agent",
          "index": 6,
          "label": "verify:test-adequacy#1",
          "phaseIndex": 2,
          "phaseTitle": "Verify",
          "agentId": "a714804389c3a5e06",
          "model": "claude-opus-4-8",
          "state": "done",
          "startedAt": 1783437187030,
          "queuedAt": 1783437186999,
          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "lastToolSummary": "REFUTED",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST /api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at /gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437698711,
          "tokens": 102760,
          "toolCalls": 20,
          "durationMs": 511680,
          "resultPreview": "{\"verdict\":\"REFUTED\",\"confidence\":\"medium\",\"reasoning\":\"I read the authoritative snapshots (orchestration_new.py:163-187, 480-498; test_api_cloud_dispatch_new.py:731-829), backend/config/models.py (TrackNetConfig.weights_path=\\\"\\\" @L51; WasbIndexConfig.weights_path defaults to \\\"~/models/WASB-SBDT/...pth.tar\\\" @L80; FusionConfig.substrate default \\\"wasb\\\" @L180), and executed the actual routing lo?"
        },
        {
          "type": "workflow_agent",
          "index": 7,
          "label": "verify:test-adequacy#2",
          "phaseIndex": 2,
          "phaseTitle": "Verify",
          "agentId": "ad7916fab73825244",
          "model": "claude-opus-4-8",
          "state": "done",
          "startedAt": 1783437187032,
          "queuedAt": 1783437186999,
```

```
          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "lastToolSummary": "REFUTED",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a
HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST
/api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which
bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at
/gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437525741,
          "tokens": 101240,
          "toolCalls": 10,
          "durationMs": 338709,
          "resultPreview":
"{\"verdict\":\"REFUTED\",\"confidence\":\"high\",\"reasoning\":\"The finding's underlying
observation is factually accurate ? no test drives the guardrail FIRING via the env route: the
only fast-fail test (test_local_on_cloud_server_without_weights_fails_fast, snapshot 731-753)
uses the cloud_env fixture (compute_target=cloud_run, the config route of _cloud_available), and
the \\\"not guardrailed\\\" test (801-8?"
        },
        {
          "type": "workflow_agent",
          "index": 8,
          "label": "verify:weights-resolution-fidelity#1",
          "phaseIndex": 2,
          "phaseTitle": "Verify",
          "agentId": "ad6d53cc935c97d77",
          "model": "claude-opus-4-8",
          "state": "done",
          "startedAt": 1783437537521,
          "queuedAt": 1783437537518,
          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "lastToolSummary": "REFUTED",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a
HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST
/api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which
bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at
/gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
          "lastProgressAt": 1783437896382,
          "tokens": 94849,
          "toolCalls": 18,
          "durationMs": 358861,
          "resultPreview":
"{\"verdict\":\"REFUTED\",\"confidence\":\"high\",\"reasoning\":\"The finding's mechanism is
technically real (there is no native_tracknet_runner.py; TrackNetRunner is WSL-only,
tracknet_runner.py:95,117-121; and _local_detector_weights_present keys the tracknet branch on
os.path.exists(indexing.tracknet.weights_path), orchestration_new.py:179-187), but the described
failure does NOT occur against the real de?"
        },
        {
          "type": "workflow_agent",
          "index": 9,
          "label": "verify:weights-resolution-fidelity#2",
          "phaseIndex": 2,
          "phaseTitle": "Verify",
          "agentId": "ae6d36e2a5d15c57e",
          "model": "claude-opus-4-8",
          "state": "done",
          "startedAt": 1783437537522,
          "queuedAt": 1783437537518,
          "attempt": 1,
          "lastToolName": "StructuredOutput",
          "lastToolSummary": "REFUTED",
          "promptPreview": "You are reviewing a two-repo bug fix BEFORE it merges into a
HOSTED PRODUCTION Cloud Run control-plane.\n\nTHE BUG: On the hosted control-plane, POST
```

```
/api/process with compute=\"local\" ran the in-process\nsegmenter ON THE CONTROL-PLANE, which
bakes NO detector weights (only the worker Cloud Run Job\nGCSFuse-mounts weights at
/gcs/models/...). It died deep with a cryptic\n\"WASB weights not found at /opt/?",
264            "lastProgressAt": 1783438007364,
265            "tokens": 127561,
266            "toolCalls": 19,
267            "durationMs": 469842,
268            "resultPreview":
"{\"verdict\":\"REFUTED\",\"confidence\":\"high\",\"reasoning\":\"The finding's technical
observations are correct, but its \\\"false-BLOCK of a compute=local run that would actually
succeed\\\" does not occur in any coherent/realistic deployment ? it is reachable only via a
self-inconsistent, off-preset hand-mixed config, and even then with a working
escape.\\n\\nVerified against the real code:\\n1) The guard `_loc?"
269          }
270        ],
271        "totalTokens": 1131896,
272        "totalToolCalls": 199
273      }
```

174. assistant (2026-07-07T15:28:08.297Z)

```
The review is thorough and reassuring: **5 dimensions, 4 findings surfaced, all 4 REFUTED**
under adversarial verification ? **zero confirmed defects**. The refuted findings are
coverage/fidelity *nuances* (e.g. the guard mirrors the native runner's weight resolution, which
is exactly the impl a coherent `cloud_run` deployment uses), not reachable bugs.

But there's now a **required action**: PR #513 merged first and touched the same two files, so
#511 needs a rebase. Let me assess the conflict ? fetch the new master and see what #513
changed.
```

175. user (2026-07-07T15:28:14.147Z)

```
=== new origin/master ===
23c1868 fix(serving): worker skips redundant re-validation on the cloud dispatch path (#513)
3be0935 docs(runlog): DEPLOY_REPORT addendum ? control-plane redeployed to the forwarding image
(#510)
66756ad chore(serving): forward WASB decode/tuning knobs to worker + release hardening (#509)
=== does my branch's base (3be0935) still == origin/master? ===
23c1868
=== #513 (merged) touched these files: ===
commit 23c186813b980b421a471aca1c0fd292f408ae26
Author: Avi <avi.dullu@gmail.com>
Date:   Tue Jul 7 20:52:15 2026 +0530

    fix(serving): worker skips redundant re-validation on the cloud dispatch path (#513)

    On the cloud-serving path the control-plane is the AUTHORITATIVE validation
    gate: backend/orchestration.py runs the validators with its resolved config and
    only dispatches to the Cloud Run worker on PASS (a failure returns "failed"
    before _maybe_dispatch_remote). The worker's cmd_infer then RE-RAN validation
    with its OWN config ? and since the control-plane's resolved validation.* keys
    are not forwarded on dispatch (#509 forwards only skip_ai_handoff + the wasb.*
    knobs), the worker used its default thresholds. Contradictory decisions result.

    Live 2026-07-07 CUJ: a 5312x2988 10-bit HEVC clip (sharpness 11.9) PASSED the
    control-plane at validation.min_blur_threshold=8.0, then the worker REJECTED it
    at its default 20.0 ? "[worker] validation FAILED: blur_check ... 11.9 < 20.0",
    rc=2. A successful GPU dispatch failed after the fact.

    Fix (skip-on-dispatch ? also saves the redundant second full-video decode):
    - ComputeJobSpec.skip_validation (new bool, default False)
    - CloudRunCompute emits a "--skip-validation" worker flag when set
    - worker cmd_infer honors it: skips validator_registry.run_all_with_context
      (val_results=[], val_context={}); the direct `worker infer` CLI still
      validates by default (no control-plane there)
```

```
    - _maybe_dispatch_remote sets skip_validation=True on every dispatch (only a
      validated PASS ever reaches it)
    - _dispatch_remote propagates the flag through the worker->worker recursion path

    val_context feeds only the observability report (ffprobe_meta), not
```

176. assistant (2026-07-07T15:28:29.347Z)

PR #513 modified `_maybe_dispatch_remote` (skip_validation on dispatch) and the test file. Let me see its exact hunks to understand the overlap with my additive changes, then rebase in a safe worktree.

177. user (2026-07-07T15:28:33.939Z)

```
=== #513 hunks in orchestration.py (context lines only) ===
@@ -568,11 +568,18 @@ def _maybe_dispatch_remote(
+        skip_validation=True,

=== #513 hunks in test file (@@ headers + added test defs) ===
@@ -236,6 +236,22 @@ def test_cloud_dispatch_omits_unset_wasb_tuning(cloud_env):
+def test_cloud_dispatch_skips_worker_revalidation(cloud_env):
```

178. assistant (2026-07-07T15:29:12.777Z)

Overlap is clean and disjoint ? #513 touches `_maybe_dispatch_remote` at line ~568 (`skip_validation=True`) and adds a test at line ~236; my changes live at lines ~158/~478 and append tests near line ~724. A rebase should replay with no conflict. I'll rebase in a fresh worktree (keeps the contended main checkout untouched), and while re-gating I'll fold in the tracknet/fusion test coverage the review flagged (it pins the one genuinely-uncovered branch).

179. user (2026-07-07T15:29:19.577Z)

```
worktree on fix/hosted-local-detection-guardrail
=== rebase onto origin/master (23c1868) ===
Rebasing (1/1)

Successfully rebased and updated refs/heads/fix/hosted-local-detection-guardrail.
=== post-rebase: my commit replayed on top of #513? ===
3ff538f fix(serving): fail fast when compute=local forces weightless in-process detection
3ff538f fix(serving): fail fast when compute=local forces weightless in-process detection
23c1868 fix(serving): worker skips redundant re-validation on the cloud dispatch path (#513)
3be0935 docs(runlog): DEPLOY_REPORT addendum ? control-plane redeployed to the forwarding image
(#510)
```

180. assistant (2026-07-07T15:29:29.639Z)

Clean rebase ? my commit replayed as `3ff538f` on top of #513, one commit ahead. Now let me add the tracknet/fusion coverage. Let me read the guardrail test section in the worktree to place them correctly:

181. user (2026-07-07T15:29:30.931Z)

```
        compute='local' on a cloud server is unaffected ? it runs in-process, never fails
fast."""
        _db, _bucket, _mp = cloud_env
        m.global_config.indexing.wasb.weights_path =
"/nonexistent/should_not_matter.pth.tar"
        monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _InProcSeg)
        fake = _FakeCloudClient()
        _inject_fake(monkeypatch, fake)

        out = m._execute_processing(
            m.ProcessRequest(
                video_path="gs://b/clip.mp4", segmenter="motion", compute="local"
```

```
        )
    )
    assert out["status"] == "success"
    assert fake.calls == []
    assert out["compute"]["path"] == "in_process"


def test_local_weights_backed_on_pure_local_server_is_not_guardrailed(
    tmp_path, monkeypatch
):
    """The guardrail only fires on a cloud-capable server. On a pure-local server (no cloud), a
    weights-backed compute='local' request falls through to the in-process segmenter as before ? the
    'run in the cloud' message would be nonsense there (no cloud to fall back to)."""
    db = Database(str(tmp_path / "t.db"))  # noqa: F841
    monkeypatch.setattr(m, "db_instance", db)
    cfg = AppConfig.model_validate({"deployment": {"compute_target": ""}})  # noqa: F841
    monkeypatch.setattr(m, "global_config", cfg)
    monkeypatch.setattr(
        m.registry, "run_all_with_context", lambda *a, **k: (_pass(), {})
    )
    monkeypatch.delenv("CLOUD_RUN_JOB", raising=False)
    monkeypatch.delenv("GOOGLE_CLOUD_PROJECT", raising=False)
    monkeypatch.delenv("WASB_WEIGHTS", raising=False)
    cfg.indexing.wasb.weights_path = "/nonexistent/wasb.pth.tar"
    monkeypatch.setattr(m.segmenter_registry, "get", lambda name: _InProcSeg)
    video = tmp_path / "v.mp4"
    video.write_bytes(b"hashable-bytes")

    out = m._execute_processing(
        m.ProcessRequest(
```

182. user (2026-07-07T15:29:39.121Z)

```
        m.ProcessRequest(
            video_path=str(video), segmenter="wasb_hybrid", compute="local"
        )
    )
    # Not the hosted fast-fail ? it ran in-process (the local box owns its own weights story).
    assert out["status"] == "success"
    assert out["compute"]["path"] == "in_process"


# --- compute-path PROVENANCE (the validation method: prove which backend ran) ------------------


def test_cloud_dispatch_stamps_compute_provenance(cloud_env):
    """A cloud run stamps result['compute'] with path=cloud_run + the REAL Cloud Run execution name
    (the GCP-verifiable proof) + the requested choice + job/region/outputs ? so the intended path is
    provable from the job result alone (no guessing from segment metadata)."""
    _db, _bucket, monkeypatch = cloud_env
    _inject_fake(monkeypatch, _FakeCloudClient())

    out = m._execute_processing(
```

183. assistant (2026-07-07T15:29:45.390Z)

Now I'll insert the two tracknet/fusion coverage tests before the provenance section:

184. user (2026-07-07T15:30:01.276Z)

The file C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\scratchpad\rebasewt\tests\test_api_cloud_dispatch.py has been updated successfully. (file state is current in your context ? no need to Read it back)

185. assistant (2026-07-07T15:30:34.594Z)

Important subtlety: the editable install may make `import backend` resolve to the **main** checkout (on the sibling's branch) rather than the worktree. Let me verify which `backend` gets imported from the worktree before trusting any test run:

186. user (2026-07-07T15:30:44.002Z)

```
FILE: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\scratchpad\rebasewt\backend\orchestration.py
HAS_GUARDRAIL: True
--- without PYTHONPATH override (plain, from worktree cwd) ---
FILE: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\scratchpad\rebasewt\backend\orchestration.py
HAS_GUARDRAIL: True
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

187. assistant (2026-07-07T15:30:52.067Z)

Confirmed ? from the worktree cwd, `import backend` resolves to the worktree's code with my guardrail. Let me run the guardrail tests (incl. the 3 new ones), then ruff/mypy, then the full suite ? all from the worktree:

188. user (2026-07-07T15:31:15.766Z)

```
................                                                         [100%]
============================== warnings summary ===============================
tests/test_api_cloud_dispatch.py::test_local_override_survives_api_boundary
tests/test_api_cloud_dispatch.py::test_local_override_survives_api_boundary
tests/test_api_cloud_dispatch.py::test_cloud_override_survives_api_boundary_on_local_server
tests/test_api_cloud_dispatch.py::test_cloud_override_survives_api_boundary_on_local_server
  C:\Users\avidu\AppData\Local\Programs\Python\Python313\Lib\site-packages\google\auth\_default.py:113: UserWarning: Your application has authenticated using end user credentials from Google Cloud SDK without a quota project. You might receive a "quota exceeded" or "API not enabled" error. See the following page for troubleshooting: https://cloud.google.com/docs/authentication/adc-troubleshooting/user-creds.
    warnings.warn(_CLOUD_SDK_CREDENTIALS_WARNING)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
16 passed, 22 deselected, 4 warnings in 18.74s
=== RUFF (changed test file) ===
All checks passed!
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

189. assistant (2026-07-07T15:31:23.273Z)

16 passed (my 3 new tests + #513's test + all guardrail/local tests), ruff clean. Now the full gate on the rebased+augmented code ? mypy plus the complete suite:

190. user (2026-07-07T15:32:07.186Z)

```
backend\pipeline\validators\base.py:44: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs  [annotation-unchecked]
backend\providers\__init__.py:9: note: By default the bodies of untyped functions are not checked, consider using --check-untyped-defs  [annotation-unchecked]
Success: no issues found in 148 source files
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

191. user (2026-07-07T15:32:13.535Z)

```
Command running in background with ID: blvj2shjd. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\blvj2shjd.output. You will be notified when
it completes. To check interim output, use Read on that file path.
```

192. assistant (2026-07-07T15:32:39.661Z)

```
Full suite is running on the rebased+augmented code (mypy already clean). I'll wait for it, then
amend the commit to fold in the new tests and force-push. Let me wait for completion:
```

193. user (2026-07-07T15:36:13.963Z)

```
DONE
--- tail ---
Required test coverage of 70% reached. Total coverage: 85.51%
1787 passed, 8 skipped, 25 warnings in 220.70s (0:03:40)
========================================================================
 RESULT: PASS - all tests green
========================================================================
EXIT=0
```

194. assistant (2026-07-07T15:36:28.331Z)

```
**Rebased + augmented gate is green:** 1787 passed / 8 skipped, coverage **85.51%** (floor 70),
mypy + ruff clean. Now amend the commit to fold in the 3 new tests (updated message), then
force-push with lease:
```

195. user (2026-07-07T15:36:38.955Z)

```
=== staged (should be only the test file) ===
M  tests/test_api_cloud_dispatch.py
[fix/hosted-local-detection-guardrail e6213f0] fix(serving): fail fast when compute=local forces
weightless in-process detection
 Date: Tue Jul 7 19:56:22 2026 +0530
 2 files changed, 228 insertions(+)
=== amended commit ===
e6213f0 fix(serving): fail fast when compute=local forces weightless in-process detection
commit e6213f05e0a1b6be65dc6c7302c39e4717baa62f
Author: Avi Dullu <avi.dullu@gmail.com>
Date:   Tue Jul 7 19:56:22 2026 +0530

    fix(serving): fail fast when compute=local forces weightless in-process detection

    A hosted cloud-serving control-plane bakes NO detector weights into its image
    (only the worker Cloud Run Job GCSFuse-mounts them at /gcs/models/...). So a
    POST /api/process with compute="local" for a weights-backed trajectory detector
    (wasb_hybrid / tracknet_hybrid / fusion_hybrid) returned None from
    _maybe_dispatch_remote and ran the in-process segmenter ON THE CONTROL-PLANE,
    which then died deep in the healthcheck with the confusing
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

196. assistant (2026-07-07T15:36:44.519Z)

```
Amended commit `e6213f0` (one commit ahead of the new master). Now force-push with lease
(guarding that the remote is still at my original `4565c74`):
```

197. user (2026-07-07T15:36:51.185Z)

```
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 + 4565c74...e6213f0 fix/hosted-local-detection-guardrail -> fix/hosted-local-detection-
guardrail (forced update)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

198. assistant (2026-07-07T15:36:57.552Z)

Force-push succeeded (`4565c74...e6213f0`). Let me verify PR #511 is now mergeable, clean up the worktree, and add a note for reviewers:

199. user (2026-07-07T15:37:04.543Z)

```
=== PR #511 state ===
{"base":"master","commits":1,"head":"e6213f05e0a1b6be65dc6c7302c39e4717baa62f","mergeable":"MERG
EABLE","state":"OPEN"}
=== remove rebase worktree ===
removed
=== remaining worktrees (mine gone; siblings untouched) ===
rebasewt count: 0
=== main checkout still on sibling branch, untouched ===
fix/cloud-run-exec-status-fastfail
```

200. assistant (2026-07-07T15:37:12.793Z)

PR #511 is now **MERGEABLE** (rebased onto master, 1 commit `e6213f0`). Worktree removed; the contended main checkout was never touched. Let me check CI status and add a reviewer note documenting the rebase + added coverage:

201. user (2026-07-07T15:37:22.951Z)

```
=== CI checks on #511 (may still be queued) ===
Golden fixtures cross-OS (windows-latest)       pending 0
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942744
Markdown link check (live docs) pass    8s      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942823
Ruff lint       pass    11s     https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942705
Golden fixtures cross-OS (macos-latest) pending 0       https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942857
Leak scan (PII / attribution guard)     pass    9s      https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942723
Full test suite (offline, mocked)       pending 0       https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942684
Mypy (lenient baseline) pending 0       https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942727
Syntax sweep + import smoke (no GPU stack)      pending 0
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942837
Golden fixtures cross-OS (ubuntu-latest)        pass    20s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942889
Built wheel smoke (install + launch)    pending 0       https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942735
Dependency license scan (no copyleft in runtime deps)   pending 0
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942790
=== add reviewer note ===
https://github.com/Khelsutra/badminton-highlight-indexer/pull/511#issuecomment-4905567519
```

202. assistant (2026-07-07T15:37:36.848Z)

CI on #511 is running (ruff/leak/golden-ubuntu/markdown already ?; heavy jobs pending). Let me check the SPA PR #169's status and whether it also needs a rebase:

203. user (2026-07-07T15:37:43.559Z)

```
=== #169 state ===
{"head":"1470e2acdea4f2b478476150da61e844dce0a8e9","mergeable":"MERGEABLE","state":"OPEN"}
```

=== #169 CI checks ===
Workers Builds: hidden-king-khelsutra   pass   0
https://dash.cloudflare.com/213c12cf60de9041dd8a3521aa206472/workers/services/view/hidden-king-khelsutra/production/builds/bbc2e1b1-8c51-47db-a885-890b81ba46b1
layout pass   1m33s
https://github.com/avidullu/khelsutra/actions/runs/28876482649/job/85652833200
site   pass   16s
https://github.com/avidullu/khelsutra/actions/runs/28876482649/job/85652833153
web   pass   2m11s
https://github.com/avidullu/khelsutra/actions/runs/28876482649/job/85652833102

204. assistant (2026-07-07T15:37:53.210Z)

SPA PR #169: CI fully green (`web` + `layout` + `site` + Cloudflare build all pass) and MERGEABLE ? no rebase needed there. Let me update memory with the final state and start watching #511's CI to green:

205. user (2026-07-07T15:37:57.634Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

206. user (2026-07-07T15:38:07.581Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

207. assistant (2026-07-07T15:38:13.721Z)

Now let me watch #511's CI to completion in the background so I can confirm it goes green:

208. user (2026-07-07T15:38:16.258Z)

Command running in background with ID: b20k05gbt. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\b20k05gbt.output. You will be notified when it completes. To check interim output, use Read on that file path.

209. assistant (2026-07-07T15:38:40.498Z)

Everything is done and both PRs are in good shape. Here's the full picture since your "push and open PRs".

? Both PRs open + mergeable
	PR	Head	Mergeable	CI
Backend	[Khelsutra/badminton-highlight-indexer#511](https://github.com/Khelsutra/badminton-highlight-indexer/pull/511)	`e6213f0`	?	running (ruff/leak/golden-ubuntu ?; heavy jobs pending ? watching to green)
SPA	[avidullu/khelsutra#169](https://github.com/avidullu/khelsutra/pull/169)	`1470e2a`	?	? **all green** (web + layout + site + Cloudflare build)

? Adversarial review ? no bugs found
The multi-agent review (5 dimensions ? per-finding verification, ~1.1M tokens) surfaced **4 findings, verified 0 CONFIRMED / 4 REFUTED**. All four were coverage/fidelity *nuances* that the verifiers refuted against the real code ? e.g. "the guard mirrors only the native runner's weight resolution," which is exactly the detector impl a coherent `cloud_run` deployment uses; and the tracknet edge case can't be reached because the hosted config leaves `tracknet.weights_path` empty (so it fires correctly). No reachable defect, no regression.

? Required rebase + coverage hardening

I discovered **PR #513 merged first**, touching the same two files. I:
- **Rebased #511 onto the new `master`** (`23c1868`) ? the edits are region-disjoint (my guardrail at lines ~158/~478, #513's `skip_validation` at ~568), so it replayed **conflict-free**.
- **Added the tracknet/fusion test coverage** the review flagged (its one substantive point): `tracknet_hybrid` fail-fast, the fusion `substrate=tracknet` switch (proven to gate on TrackNet weights even when WASB weights are present), and the fusion default-`wasb` escape hatch ? pinning the previously-uncovered branch.
- **Re-ran the full CI-faithful gate:** 1787 passed / 8 skipped, coverage **85.51%**, ruff + `mypy backend training` clean. Force-pushed with lease and left a reviewer note on the PR.

? Safety notes
- All git writes went through **throwaway worktrees** ? the shared/contended main checkout (now on a sibling's `fix/cloud-run-exec-status-fastfail`) was never disturbed.
- Memory updated (`compute-local-hosted-guardrail`), including the ?? that #511 overlaps #512/#513.
- **Rollout reminder:** the SPA fix reaches the *deployed* hosted build only after a re-vendor (`scripts/bundle_ui.sh` ? `backend/ui/`) ? a separate release step.

I'm watching #511's CI to green now and will confirm when it lands. One sequencing note: **#512** (`perf/validation-shared-decode-pass`) also touches these two files and hasn't merged yet ? after #511 lands, #512 will need the same rebase.

210. user (2026-07-07T15:38:42.209Z)

please merge this PR after due rebasing and proceed

211. assistant (2026-07-07T15:39:18.820Z)

Understood ? I'll ensure #511 is rebased on the latest master, confirm CI is green, then merge. Let me first check the CI watch result and whether master has moved again (sibling merges are frequent).

212. user (2026-07-07T15:39:26.646Z)

```
=== #511 CI watch result ===
Full test suite (offline, mocked)	pending	0	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942684
Built wheel smoke (install + launch)	pass	1m4s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942735
Dependency license scan (no copyleft in runtime deps)	pass	38s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942790
Golden fixtures cross-OS (ubuntu-latest)	pass	20s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942889
Refreshing checks status every 30 seconds. Press Ctrl+C to quit.

Golden fixtures cross-OS (macos-latest)	pass	31s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942857
Golden fixtures cross-OS (windows-latest)	pass	1m9s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942744
Leak scan (PII / attribution guard)	pass	9s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942723
Markdown link check (live docs)	pass	8s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942823
Mypy (lenient baseline)	pass	35s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942727
Ruff lint	pass	11s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942705
Syntax sweep + import smoke (no GPU stack)	pass	23s	https://github.com/Khelsutra/badminton-highlight-indexer/actions/runs/28878774590/job/85660942837
```

Full test suite (offline, mocked)   pending 0   https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942684
Built wheel smoke (install + launch)   pass   1m4s   https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942735
Dependency license scan (no copyleft in runtime deps)   pass   38s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942790
Golden fixtures cross-OS (ubuntu-latest)   pass   20s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942889
Refreshing checks status every 30 seconds. Press Ctrl+C to quit.

Golden fixtures cross-OS (macos-latest) pass   31s   https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942857
Golden fixtures cross-OS (windows-latest)   pass   1m9s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942744
Leak scan (PII / attribution guard)   pass   9s   https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942723
Markdown link check (live docs) pass   8s   https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942823
Mypy (lenient baseline) pass   35s   https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942727
Ruff lint   pass   11s   https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942705
Syntax sweep + import smoke (no GPU stack)   pass   23s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942837
Full test suite (offline, mocked)   pending 0   https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942684
Built wheel smoke (install + launch)   pass   1m4s   https://github.com/Khelsutra/badminton-
highlight-indexer/actions/runs/28878774590/job/85660942735
Dependency license scan (no copyleft in runtime deps)   pass   38s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942790
Golden fixtures cross-OS (ubuntu-latest)   pass   20s
https://github.com/Khelsutra/badminton-highlight-
indexer/actions/runs/28878774590/job/85660942889

=== current origin/master (did it move past 23c1868?) ===
20862d7 perf(validators): decode scene_cut/blur/relevance frames in one shared pass (#512)
23c1868 fix(serving): worker skips redundant re-validation on the cloud dispatch path (#513)
3be0935 docs(runlog): DEPLOY_REPORT addendum ? control-plane redeployed to the forwarding image
(#510)
=== is my PR head e6213f0 still 1 commit ahead of master? ===
20862d7
e6213f0 fix(serving): fail fast when compute=local forces weightless in-process detection

213. assistant (2026-07-07T15:39:59.692Z)

Two developments: **master moved again** ? PR #512 just merged (`20862d7`), so #511 needs
another rebase; and #511's "Full test suite" CI job is still pending (now moot ? a fresh rebase
will retrigger CI). Let me stop the stale CI watch and inspect what #512 changed in my two
files.

214. user (2026-07-07T15:40:04.558Z)

```
=== #512 (20862d7) changed files ===
 backend/pipeline/validators/base.py          |  98 ++++++
 backend/pipeline/validators/blur.py          |  56 ++--
 backend/pipeline/validators/frame_sampler.py | 143 +++++++++
 backend/pipeline/validators/relevance.py     |  93 +++---
 backend/pipeline/validators/scene_cut.py     |  99 +++---
 docs/CODE_MAP.md                             |   9 +-
 tests/test_frame_sampler.py                  | 444 +++++++++++++++++++++++++++
```

 7 files changed, 812 insertions(+), 130 deletions(-)

=== #512 hunk headers in MY two files (if any) ===

215. assistant (2026-07-07T15:40:33.646Z)

#512 only touches validators/`test_frame_sampler.py` ? **disjoint from my two files**, so the rebase onto `20862d7` will be clean (the earlier memory note conflated #512 with #513). Let me stop the stale CI watch and rebase in a fresh worktree.

216. assistant (2026-07-07T15:41:30.052Z)

The CI watch actually completed **green** (exit 0 ? all checks passed, including the full suite) on `e6213f0`. But since master advanced to `20862d7` (#512), I'll rebase onto it as you asked, re-verify, then merge. #512 is disjoint from my files, so this rebase is clean. Let me set it up:

217. user (2026-07-07T15:41:37.034Z)

```
worktree ready
Rebasing (1/1)

Successfully rebased and updated refs/heads/fix/hosted-local-detection-guardrail.
=== post-rebase: my 1 commit on top of 20862d7 (#512) ===
394d79c fix(serving): fail fast when compute=local forces weightless in-process detection
394d79c fix(serving): fail fast when compute=local forces weightless in-process detection
20862d7 perf(validators): decode scene_cut/blur/relevance frames in one shared pass (#512)
23c1868 fix(serving): worker skips redundant re-validation on the cloud dispatch path (#513)
3be0935 docs(runlog): DEPLOY_REPORT addendum ? control-plane redeployed to the forwarding image
(#510)
```

218. assistant (2026-07-07T15:41:41.957Z)

Clean rebase ? my commit is now `394d79c` atop `20862d7`. Let me run the full local suite on this exact rebased tree (verifies my change + #512 + #513 together), then force-push and merge once green:

219. user (2026-07-07T15:41:44.854Z)

Command running in background with ID: b27y2ziqn. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\b27y2ziqn.output. You will be notified when it completes. To check interim output, use Read on that file path.

220. assistant (2026-07-07T15:42:04.744Z)

Full suite is running on the rebased tree. Let me wait for it, then force-push and merge both PRs.

221. user (2026-07-07T15:47:41.779Z)

```
DONE
--- tail ---
1824 passed, 8 skipped, 25 warnings in 343.71s (0:05:43)
========================================================================
 RESULT: PASS - all tests green
========================================================================
EXIT=0
```

222. assistant (2026-07-07T15:47:53.955Z)

1824 passed / 8 skipped on the exact rebased tree (includes #512's + #513's changes alongside mine), green. Let me confirm ruff clean, force-push, and get the coverage number:

223. user (2026-07-07T15:48:01.313Z)

```
=== coverage ===
Required test coverage of 70% reached. Total coverage: 85.60%
=== ruff (my files) ===
All checks passed!
=== force-push 394d79c (lease guards remote==e6213f0) ===
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 + e6213f0...394d79c fix/hosted-local-detection-guardrail -> fix/hosted-local-detection-
guardrail (forced update)
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

224. assistant (2026-07-07T15:49:14.237Z)

Force-push clean (`e6213f0...394d79c`), coverage 85.60%, ruff clean. Now let me remove the worktree and merge. I'll merge the SPA PR (already green) and enable auto-merge on #511 (merges when its fresh CI passes):

225. user (2026-07-07T15:49:20.855Z)

```
rebasewt2 removed
=== #511 mergeStateStatus after force-push ===
{"mergeStateStatus":"UNSTABLE","mergeable":"MERGEABLE"}
=== #169 state ===
{"mergeStateStatus":"CLEAN","mergeable":"MERGEABLE"}
```

226. assistant (2026-07-07T15:49:42.541Z)

#169 is **CLEAN** (ready); #511 is **UNSTABLE** (mergeable, fresh CI still running on the force-pushed head). Let me merge the SPA PR now, and enable auto-merge on #511 so it lands the moment its CI passes:

227. user (2026-07-07T15:49:55.864Z)

```
=== merge SPA #169 (squash) ===
=== delete #169 remote branch (leaves local checkout untouched) ===
To https://github.com/avidullu/khelsutra.git
 - [deleted]         fix/hosted-compute-picker-defaults-cloud

=== enable auto-merge (squash) on #511 ? merges when CI passes ===
```

228. assistant (2026-07-07T15:50:01.734Z)

Let me verify both merge operations landed as intended:

229. user (2026-07-07T15:50:06.596Z)

```
=== #169 (SPA) state ===
{"mergeCommit":"ba5f45777441bb288c5b6a8868c3dbc736d8c2fa","mergedAt":"2026-07-
07T15:49:48Z","state":"MERGED"}
=== #511 (backend) state + auto-merge ===
{"autoMerge":null,"mergeStateStatus":"UNKNOWN","state":"MERGED"}
```

230. assistant (2026-07-07T15:50:18.526Z)

Both PRs are **MERGED**:
- **SPA #169** ? merge commit `ba5f457`
- **Backend #511** ? merged (the `--auto` merged it immediately since it was mergeable, and my local full-suite gate on the exact tree was green)

Let me confirm both landed on master and the branches are cleaned up:

231. user (2026-07-07T15:50:31.543Z)

```
 - [deleted]         (none)     -> origin/fix/hosted-local-detection-guardrail
 - [deleted]         (none)     -> origin/perf/validation-shared-decode-pass
   20862d7..ec2d080  master     -> origin/master
=== backend origin/master top (my squash should be HEAD) ===
ec2d080 fix(serving): fail fast when compute=local forces weightless in-process detection (#511)
20862d7 perf(validators): decode scene_cut/blur/relevance frames in one shared pass (#512)
23c1868 fix(serving): worker skips redundant re-validation on the cloud dispatch path (#513)
=== my guardrail present on master? ===
_WEIGHTS_BACKED_SEGMENTERS occurrences: 3
=== backend branch deleted on remote? ===
remaining refs (0=deleted): 0

 - [deleted]         (none)     -> origin/pr/162
   c7265a4..ba5f457  master     -> origin/master
=== SPA origin/master top ===
ba5f457 fix(ingest): default the compute picker to CLOUD on a hosted control-plane (#169)
c7265a4 Add Instagram + YouTube + Discord to the site footer (closes #159) (#167)
=== SPA picker default-to-cloud present on master? ===
cloud_run default occurrences: 1
```

232. assistant (2026-07-07T15:51:16.161Z)

Both fixes are now **on master**:
- **Backend** `ec2d080` ? `#511` merged; guardrail present (`_WEIGHTS_BACKED_SEGMENTERS` ×3). Branch deleted.
- **SPA** `ba5f457` ? `#169` merged; `compute_target === "cloud_run"` default present. Branch deleted.

Let me update memory to the terminal state, then lay out what remains for this to take effect in production.

233. user (2026-07-07T15:51:18.971Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

234. user (2026-07-07T15:51:33.855Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

235. user (2026-07-07T15:51:37.520Z)

1   # Memory index
2
3   - ? **Start here (GPU-resident rollout):** session-handoff-gpu-resident.md ? #506 implemented, PR #507 open; next: on-box parity gate + 15-clip golden gate.
4
5   - [Actual repo location](actual-repo-location.md) ? primary working dir is an empty STUB; real code is under `khelsutra-guru/badminton-highlight-indexer`.
6   - [Efficient timeouts, retries & Windows gcloud SSH](efficient-timeouts-and-retries.md) ? no silent retry loops; plain `gcloud compute ssh` (plink/pscp reject `-o`, mishandle `~`); diagnose-first, run long work detached.
7   - [GPU VM setup gotchas](gpu-vm-setup-gotchas.md) ? NVDEC libs, apt-lock-at-boot, torch2.x/pynvc; what #501 should encode.
8   - [LGTM only when merge-ready](lgtm-only-when-merge-ready.md) ? post LGTM only when genuinely satisfied a PR is merge-ready (verified), for all PRs.
9   - [GPU-resident modernization status](gpu-resident-modernization.md) ? #506 validated clean; the resize must match cv2.warpAffine via grid_sample (not F.interpolate).
10   - [compute=local hosted guardrail](compute-local-hosted-guardrail.md) ? PRs #511 (backend) + #169 (SPA) fixing the "weights not found" trap, gated + pushed. ? #511 touches

orchestration.py + test_api_cloud_dispatch.py ? OVERLAPS PRs #512/#513 (merge-conflict; whoever lands 2nd rebases).
11   - [Validation shared decode pass](validation-shared-decode-pass.md) ? validators now share ONE decode pass (**PR #512 MERGED**, squash 20862d7); pure grab()-through is SLOWER than seeks ? use forward-hybrid. True-4K sub-second follow-up = issue #514.
12   - [PR #513 worker skip-revalidation](pr513-worker-skip-revalidation.md) ? worker skips redundant re-validation on cloud dispatch (fixes the blur-threshold CUJ). **MERGED** 2026-07-07 (squash 23c1868). ? landed first on orchestration.py + test_api_cloud_dispatch.py, so PR #511 must now REBASE onto master (#512 was file-disjoint ? merged clean as 20862d7, no rebase needed). Follow-up (crash-before-marker fast-fail) now running on branch `fix/cloud-run-exec-status-fastfail`.
13   - [Cloud-run exec-status fast-fail](cloud-run-exec-status-fastfail.md) ? the #513 follow-up: `run_infer` now probes the Executions API so a worker that crashes before the done-marker fails fast (rc 5) instead of a ~65-min hang. Branch `fix/cloud-run-exec-status-fastfail` (145854e, **rebased onto current master 20862d7**; gate 1833?), **local/unpushed**.
14

236. user (2026-07-07T15:51:44.402Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

237. assistant (2026-07-07T15:52:23.475Z)

Both PRs are merged and master is updated ? the code fix is complete. What remains is **deployment**, which I've intentionally not done unilaterally (it's outward-facing, costs money, and a control-plane redeploy would now also ship the sibling-merged #512/#513). Let me confirm how you want to proceed.

238. user (2026-07-07T15:53:01.461Z)

Your questions have been answered: "Both PRs are merged to master. How should I proceed with getting the fix live on the hosted control-plane?"="Full go-live now". You can now continue with these answers in mind.

239. assistant (2026-07-07T15:54:03.859Z)

You chose **full go-live**. That's a multi-step release: (1) re-vendor the merged SPA into `backend/ui/` ? PR ? merge, then (2) rebuild the Cloud Run image from master and (3) redeploy `rally-control-plane`. It ships the current master (so #511 + re-vendored SPA + the sibling-merged #512/#513) to prod.

Let me first verify I can actually deploy (gcloud auth) and capture the current live image for rollback, before investing in the build.

240. user (2026-07-07T15:54:16.090Z)

```
=== gcloud auth + project ===
quota_project = gen-lang-client-0306026826
account = avi.dullu@gmail.com
project = gen-lang-client-0306026826
=== active account ===
avi.dullu@gmail.com
=== current live control-plane (rollback ref) ===
rally-control-plane-00017-qbd   asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest
```

241. assistant (2026-07-07T15:54:44.787Z)

gcloud is authed (`avi.dullu@gmail.com`, project `gen-lang-client-0306026826`). Live control-plane: revision **`rally-control-plane-00017-qbd`** (rollback ref), image `rally-worker:latest`. Let me capture the pinned digest for precise rollback and read the deploy/bundle configs:

242. user (2026-07-07T15:54:58.971Z)

```
=== current control-plane image DIGEST (precise rollback) ===
rally-control-plane-00017-qbd
asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker@sha256:09d2edf7fe8d62e5ec5acdf56690d03f55e89b4e300599ad6e73c1be6e20797b
=== worker job current image ===
asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker:latest
=== khelsutra main checkout state (for building the SPA) ===
HEAD=1470e2a branch=fix/hosted-compute-picker-defaults-cloud
```

243. assistant (2026-07-07T15:55:26.852Z)

Rollback refs captured (control-plane digest `sha256:09d2edf7?`, revision `00017-qbd`). Now the re-vendor. Let me update the khelsutra checkout to the merged master, set up a badminton worktree for the vendor commit, and read the build + bundle-validation configs.

244. user (2026-07-07T15:55:36.262Z)

```
=== update khelsutra checkout to master (ba5f457) ===
  (use "git pull" to update your local branch)
Switched to branch 'master'
 web/src/components/IngestPanel.test.tsx | 24 ++++++++++++++++++++++++
 web/src/components/IngestPanel.tsx      | 11 ++++++++++-
 2 files changed, 34 insertions(+), 1 deletion(-)
khelsutra HEAD now: ba5f457 (master)
=== SPA fix present in checkout? ===
1
=== set up badminton revendor worktree from origin/master ec2d080 ===
eng worktree on chore/revendor-hosted-compute-picker @ ec2d080
```

245. assistant (2026-07-07T15:55:53.711Z)

khelsutra is on master @ `ba5f457` (fix present), and the badminton revendor worktree is on `chore/revendor-hosted-compute-picker` @ `ec2d080`. Now run the bundle script (builds the SPA from khelsutra master and vendors it into the worktree's `backend/ui/`):

246. user (2026-07-07T15:55:56.025Z)

Command running in background with ID: b4hqgysui. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\b4hqgysui.output. You will be notified when it completes. To check interim output, use Read on that file path.

247. assistant (2026-07-07T15:56:00.817Z)

While the bundle builds, let me read the image-build config and the bundle-validation test for the next phases:

248. user (2026-07-07T15:56:01.288Z)

```
# Cloud Build config for the Cloud Run GPU worker image (COMPUTE_DECOUPLED_SERVING M7).
#
# `gcloud builds submit --tag ? -f ?` does NOT work ? `gcloud builds submit` has no `-f` flag, so a
# non-root Dockerfile needs a build CONFIG like this one. Build with:
#
#   gcloud builds submit --config=deploy/cloudrun/cloudbuild.yaml --project <PROJECT> .
#
# Override the region/repo/image with --substitutions=_REGION=?,_REPO=?,_IMAGE=? if needed.
# PROJECT_ID is provided by Cloud Build automatically.
substitutions:
  _REGION: asia-southeast1   # L4 GPU is region-limited (asia-southeast1 / europe-west1
```

```
/ europe-west4 / us-central1 / us-east4)
  _REPO: rally
  _IMAGE: rally-worker
steps:
  - name: 'gcr.io/cloud-builders/docker'
    args:
      - build
      - -t
      - '${_REGION}-docker.pkg.dev/${PROJECT_ID}/${_REPO}/${_IMAGE}:latest'
      - -f
      - deploy/cloudrun/Dockerfile
      - .
images:
  - '${_REGION}-docker.pkg.dev/${PROJECT_ID}/${_REPO}/${_IMAGE}:latest'
options:
  machineType: E2_HIGHCPU_8   # the CUDA + micromamba image is heavy; a bigger builder keeps it ~8?10 min
timeout: 3600s

```

249. user (2026-07-07T15:56:10.981Z)

```
"""The bundled SPA is served single-origin + the SPA<->engine contract (PACKAGING_PLAN.md P1b)."""

import json
import os

import pytest

pytest.importorskip("httpx")
from fastapi.testclient import TestClient

import backend.main as m

client = TestClient(m.app)
UI_DIR = os.path.join(os.path.dirname(os.path.abspath(m.__file__)), "ui")


def test_bundled_ui_snapshot_is_committed():
    assert os.path.isfile(os.path.join(UI_DIR, "index.html")), (
        "backend/ui/index.html missing ? the SPA snapshot must be committed (run scripts/bundle_ui.sh)."
    )
    assert os.path.isfile(os.path.join(UI_DIR, "ui_version.json"))


def test_serves_spa_at_root():
    r = client.get("/")
    assert r.status_code == 200
    # The bundled index.html references its single-origin /assets bundle (Vite base='/').
    assert "/assets/" in r.text


def test_serves_spa_assets_byte_for_byte():
    assets = os.path.join(UI_DIR, "assets")
    js = next(f for f in os.listdir(assets) if f.endswith(".js"))
    r = client.get(f"/assets/{js}")
    assert r.status_code == 200
    with open(os.path.join(assets, js), "rb") as f:
        assert (
            r.content == f.read()
        )  # served verbatim (mimetype varies by OS, so compare bytes)

```

```
def test_api_version_reports_bundled_ui():
    body = client.get("/api/version").json()
    assert body["api_version"] == m.API_VERSION
    assert body["ui"] is not None
    assert body["ui"]["spa_commit"]  # provenance recorded


def test_spa_engine_contract_version_matches():
    """Skew guard: the committed snapshot records the engine API_VERSION it was built against. If
    someone bumps API_VERSION without re-running scripts/bundle_ui.sh, this FAILS ? forcing a
    re-bundle so the shipped SPA can't silently target a stale API contract."""
    with open(os.path.join(UI_DIR, "ui_version.json"), encoding="utf-8") as f:
        uiv = json.load(f)
    assert uiv["engine_api_version"] == m.API_VERSION, (
        f"bundled SPA built against engine api_version {uiv['engine_api_version']} but engine is now "
        f"{m.API_VERSION} ? re-run scripts/bundle_ui.sh and commit backend/ui/."
    )


def _recompute_content_hash() -> str:
    """Deterministic Python mirror of scripts/bundle_ui.sh's content-hash recipe: per-file
    sha256 lines in the POSIX text form ("<hash>  ./<relpath>\\n" ? two spaces, LC_ALL=C path
    sort), hashed again. The shell recipe normalizes MSYS sha256sum's "*" binary-mode marker
    to this same form (the #495 review finding: the marker silently changed the OUTER hash on
    Windows over byte-identical payloads). Stable across OSes because backend/ui/** is
    gitattributes ``-text`` (bytes identical in every checkout)."""
    import hashlib

    excluded = {"ui_version.json", "THIRD_PARTY_NOTICES.md"}
    rels = []
    for dirpath, _dirs, files in os.walk(UI_DIR):
        for name in files:
            if name in excluded:
                continue
            rel = os.path.relpath(os.path.join(dirpath, name), UI_DIR)
            rels.append("./" + rel.replace(os.sep, "/"))
    lines = []
    for rel in sorted(rels):
        with open(os.path.join(UI_DIR, rel[2:]), "rb") as f:
            digest = hashlib.sha256(f.read()).hexdigest()
        lines.append(f"{digest}  {rel}\n")
    return hashlib.sha256("".join(lines).encode("ascii")).hexdigest()


def test_ui_manifest_content_hash_matches_the_committed_payload():
    """Provenance drift guard (#495 review): the manifest's content_hash must equal a recompute
    over the committed payload ? a hand-edited/drifted snapshot (or a platform-skewed bundling
    run) can't ship silently. The P9 deploy verification pins the served UI against exactly
    this manifest."""
    with open(os.path.join(UI_DIR, "ui_version.json"), encoding="utf-8") as f:
        uiv = json.load(f)
    assert uiv["content_hash"] == _recompute_content_hash(), (
        "backend/ui/ui_version.json.content_hash does not match the committed SPA payload ? "
```

```
            "re-run scripts/bundle_ui.sh (it rewrites the manifest) instead of hand-editing either side."
        )


    def test_ui_manifest_provenance_shape():
        with open(os.path.join(UI_DIR, "ui_version.json"), encoding="utf-8") as f:
            uiv = json.load(f)
        assert len(uiv["spa_commit"]) == 40 and int(uiv["spa_commit"], 16) >= 0
        assert uiv["spa_repo"].startswith("https://")

```

250. assistant (2026-07-07T15:56:20.167Z)

The bundle build finished (exit 0). Let me check the result and the vendored diff:

251. user (2026-07-07T15:56:24.340Z)

```
=== bundle log tail ===
assets/index-BjemzRyy.css
assets/index-kJhH4dVE.js
assets/noto-sans-devanagari-devanagari-wght-normal-DWI_L5Cu.woff2
assets/noto-sans-devanagari-latin-ext-wght-normal-CLkupaxV.woff2
assets/noto-sans-devanagari-latin-wght-normal-Bxfr4UJH.woff2
assets/noto-sans-kannada-kannada-wght-normal-DuwB-P8d.woff2
assets/noto-sans-kannada-latin-ext-wght-normal-MdBBF41x.woff2
assets/noto-sans-kannada-latin-wght-normal-DeLZCgFT.woff2
index.html
THIRD_PARTY_NOTICES.md
ui_version.json
EXIT=0
=== git diff stat in revendor worktree ===
 backend/ui/THIRD_PARTY_NOTICES.md    |  2 +-
 backend/ui/assets/index-5O3c-FCB.js  | 62 ------------------------------------
 backend/ui/assets/index-dvIzm83F.css |  1 -
 backend/ui/index.html                |  4 +--
 backend/ui/ui_version.json           |  6 ++--
 5 files changed, 6 insertions(+), 69 deletions(-)
=== ui_version.json (provenance) ===
{
  "spa_repo": "https://github.com/avidullu/khelsutra.git",
  "spa_commit": "ba5f45777441bb288c5b6a8868c3dbc736d8c2fa",
  "engine_api_version": 1,
  "content_hash": "f41f829f959eca1ff21695fd2aef058e3f423b6b7172e32e47b0f026b2d7e9be",
  "built_at": "2026-07-07T15:56:09Z"
}
```

252. assistant (2026-07-07T15:56:46.909Z)

Bundle vendored cleanly ? new JS/CSS hashed bundles, updated `index.html`, and `ui_version.json` correctly records `spa_commit: ba5f457` (the merged master) with `engine_api_version: 1`. Let me stage `backend/ui/`, validate it with the bundle test, and confirm only expected files changed:

253. user (2026-07-07T15:56:55.577Z)

```
=== full status (should be only backend/ui/*) ===
 M backend/ui/THIRD_PARTY_NOTICES.md
 D backend/ui/assets/index-5O3c-FCB.js
 D backend/ui/assets/index-dvIzm83F.css
 M backend/ui/index.html
 M backend/ui/ui_version.json
?? backend/ui/assets/index-BjemzRyy.css
?? backend/ui/assets/index-kJhH4dVE.js
=== staged summary ===
```

```
 backend/ui/THIRD_PARTY_NOTICES.md    |  2 +-
 backend/ui/assets/index-5O3c-FCB.js  | 62 ------------------------------------
 backend/ui/assets/index-BjemzRyy.css |  1 +
 backend/ui/assets/index-dvIzm83F.css |  1 -
 backend/ui/assets/index-kJhH4dVE.js  | 62 ++++++++++++++++++++++++++++++++++++
 backend/ui/index.html                |  4 +--
 backend/ui/ui_version.json           |  6 ++--
 7 files changed, 69 insertions(+), 69 deletions(-)
=== validate the vendored bundle (test_ui_bundle.py) ===
.......                                                                  [100%]
7 passed in 3.38s
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

254. assistant (2026-07-07T15:57:04.571Z)

Bundle test: 7 passed ? content hash, API-version skew guard, and byte-for-byte serving all validate. Let me sanity-check that the fix's logic is actually in the new minified bundle before committing:

255. user (2026-07-07T15:57:08.291Z)

```
=== does the new bundle contain the cloud_run compute-picker default? ===
compute_target)==="cloud_run
cloud_run occurrences in bundle: 1
```

256. assistant (2026-07-07T15:57:18.164Z)

The bundle contains `compute_target)==="cloud_run"` ? the fix is really in it. Let me commit, push, and open the re-vendor PR:

257. user (2026-07-07T15:57:27.034Z)

```
=== branch safety ===
chore/revendor-hosted-compute-picker
[chore/revendor-hosted-compute-picker 9a60ca9] chore(ui): re-vendor SPA bundle with the hosted
compute-picker cloud default
 7 files changed, 69 insertions(+), 69 deletions(-)
 delete mode 100644 backend/ui/assets/index-5O3c-FCB.js
 create mode 100644 backend/ui/assets/index-BjemzRyy.css
 delete mode 100644 backend/ui/assets/index-dvIzm83F.css
 create mode 100644 backend/ui/assets/index-kJhH4dVE.js
=== push ===
remote:
branch 'chore/revendor-hosted-compute-picker' set up to track 'origin/chore/revendor-hosted-
compute-picker'.
To https://github.com/Khelsutra/badminton-highlight-indexer.git
 * [new branch]      chore/revendor-hosted-compute-picker -> chore/revendor-hosted-compute-
picker
Shell cwd was reset to C:\Users\avidu\Projects\badminton-highlight-indexer
```

258. assistant (2026-07-07T15:57:34.620Z)

Committed `9a60ca9` and pushed. Now open the re-vendor PR and enable auto-merge:

259. user (2026-07-07T15:57:45.826Z)

https://github.com/Khelsutra/badminton-highlight-indexer/pull/516

260. assistant (2026-07-07T15:57:54.326Z)

Re-vendor PR opened: [#516](https://github.com/Khelsutra/badminton-highlight-indexer/pull/516). Let me remove its worktree and merge it (auto-merge when CI passes):

261. user (2026-07-07T15:58:04.908Z)

```
revendor worktree removed
=== #516 state ===
{"mergeStateStatus":"UNSTABLE","mergeable":"MERGEABLE"}
=== enable squash auto-merge + delete branch ===
```

262. user (2026-07-07T15:58:20.009Z)

```
=== #516 state ===
{"autoMerge":null,"mergedAt":"2026-07-07T15:58:02Z","state":"MERGED"}
=== wait briefly + re-check if auto-merge pending ===
{"mergeStateStatus":"UNKNOWN","state":"MERGED"}
```

263. assistant (2026-07-07T15:58:36.440Z)

#516 MERGED ? master now carries the re-vendored `backend/ui/`. Now Phase 2: build the Cloud Run image from the new master. Let me set up a clean build worktree and verify contents before kicking off the build:

264. user (2026-07-07T15:58:45.455Z)

```
From https://github.com/Khelsutra/badminton-highlight-indexer
   ec2d080..f5c9dd9  master     -> origin/master
=== new master ===
f5c9dd9 chore(ui): re-vendor SPA bundle with the hosted compute-picker cloud default (#516)
ec2d080 fix(serving): fail fast when compute=local forces weightless in-process detection (#511)
build worktree @ f5c9dd9
=== verify re-vendor + guardrail both on this tree ===
1
3
=== .gcloudignore present (keeps build context lean)? ===
yes
```

265. assistant (2026-07-07T15:58:58.067Z)

Master is `f5c9dd9` (re-vendor + guardrail both present), `.gcloudignore` in place. Now build the Cloud Run image from this tree (~8?10 min, ~$0.3). Running in the background:

266. user (2026-07-07T15:59:00.426Z)

```
Command running in background with ID: b68riim4h. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\b68riim4h.output. You will be notified when
it completes. To check interim output, use Read on that file path.
```

267. assistant (2026-07-07T15:59:26.865Z)

The image build is running (~8?10 min). I've mapped out the deploy: per the handoff, one `rally-worker` image drives both the `rally-control-plane` service and the `rally-worker` job, so I'll update both to the new digest, then verify the new revision boots healthy. Let me wait for the build:

268. user (2026-07-07T16:09:30.420Z)

```
Command running in background with ID: bfaqcra8b. Output is being written to:
C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-
indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\bfaqcra8b.output. You will be notified when
it completes. To check interim output, use Read on that file path.
```

269. assistant (2026-07-07T16:10:50.134Z)

My wait-loop got auto-backgrounded (foreground sleep is blocked here). Let me check the build progress directly:

270. user (2026-07-07T16:10:51.572Z)

```
<system-reminder>[Truncated: PARTIAL view ? showing lines 1-499 of 1952 total (82965 tokens, cap
25000). Call Read with offset=500 limit=499 for the next page, or Grep to find a specific
section. Do NOT answer from this page alone if the answer may be further in the file.]</system-
reminder>

1       Creating temporary archive of 277 file(s) totalling 6.0 MiB before compression.
2       Uploading tarball of [.] to [gs://gen-lang-
client-0306026826_cloudbuild/source/1783439943.389161-ffad55d3c59f49beb6e60592587a59b2.tgz]
3       Created [https://cloudbuild.googleapis.com/v1/projects/gen-lang-
client-0306026826/locations/global/builds/17ab46b4-629a-428e-a234-64f18296a41f].
4       Logs are available at [ https://console.cloud.google.com/cloud-
build/builds/17ab46b4-629a-428e-a234-64f18296a41f?project=492532266377 ].
5       Waiting for build to complete. Polling interval: 1 second(s).
6       ----------------------------- REMOTE BUILD OUTPUT ------------------------------
7       starting build "17ab46b4-629a-428e-a234-64f18296a41f"
8
9       FETCHSOURCE
10      Fetching storage object: gs://gen-lang-client-
0306026826_cloudbuild/source/1783439943.389161-
ffad55d3c59f49beb6e60592587a59b2.tgz#1783439950367493
11      Copying gs://gen-lang-client-0306026826_cloudbuild/source/1783439943.389161-
ffad55d3c59f49beb6e60592587a59b2.tgz#1783439950367493...
12      / [0 files][    0.0 B/  2.5 MiB]
/ [1 files][  2.5 MiB/  2.5 MiB]
13      Operation completed over 1 objects/2.5 MiB.
14      BUILD
15      Already have image (with digest): gcr.io/cloud-builders/docker
16      Sending build context to Docker daemon  6.572MB
17      Step 1/20 : ARG NVIDIA_DRIVER_VERSION=580.159.03
18      Step 2/20 : FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04 AS nvlibs
19      12.4.1-runtime-ubuntu22.04: Pulling from nvidia/cuda
20      3c645031de29: Pulling fs layer
21      0d6448aff889: Pulling fs layer
22      0a7674e3e8fe: Pulling fs layer
23      b71b637b97c5: Pulling fs layer
24      56dc85502937: Pulling fs layer
25      ec6d5f6c9ed9: Pulling fs layer
26      47b8539d532f: Pulling fs layer
27      fd9cc1ad8dee: Pulling fs layer
28      83525caeeb35: Pulling fs layer
29      fd9cc1ad8dee: Waiting
30      47b8539d532f: Waiting
31      83525caeeb35: Waiting
32      b71b637b97c5: Verifying Checksum
33      b71b637b97c5: Download complete
34      56dc85502937: Verifying Checksum
35      56dc85502937: Download complete
36      0d6448aff889: Verifying Checksum
37      0d6448aff889: Download complete
38      fd9cc1ad8dee: Download complete
39      83525caeeb35: Verifying Checksum
40      83525caeeb35: Download complete
41      47b8539d532f: Verifying Checksum
42      47b8539d532f: Download complete
43      3c645031de29: Verifying Checksum
44      3c645031de29: Download complete
45      3c645031de29: Pull complete
46      0d6448aff889: Pull complete
47      ec6d5f6c9ed9: Verifying Checksum
48      ec6d5f6c9ed9: Download complete
49      0a7674e3e8fe: Verifying Checksum
```

```
50      0a7674e3e8fe: Download complete
51      0a7674e3e8fe: Pull complete
52      b71b637b97c5: Pull complete
53      56dc85502937: Pull complete
54      ec6d5f6c9ed9: Pull complete
55      47b8539d532f: Pull complete
56      fd9cc1ad8dee: Pull complete
57      83525caeeb35: Pull complete
58      Digest: sha256:517da2300c184c9999ec203c2665244bdebd3578d12fcc7065e83667932643d9
59      Status: Downloaded newer image for nvidia/cuda:12.4.1-runtime-ubuntu22.04
60       ---> a112cf8dd3c2
61      Step 3/20 : ARG NVIDIA_DRIVER_VERSION
62       ---> Running in 7812d9a06379
63      Removing intermediate container 7812d9a06379
64       ---> 7ad7e6861b32
65      Step 4/20 : RUN apt-get update && apt-get install -y --no-install-recommends curl ca-
certificates     && rm -rf /var/lib/apt/lists/*     && curl -fsSL -o /tmp/drv.run
"https://us.download.nvidia.com/tesla/${NVIDIA_DRIVER_VERSION}/NVIDIA-
Linux-x86_64-${NVIDIA_DRIVER_VERSION}.run"     && sh /tmp/drv.run --extract-only --target
/tmp/drv     && mkdir -p /nvlibs     && cp -a /tmp/drv/libnvcuvid.so.* /tmp/drv/libnvidia-
encode.so.* /nvlibs/     && rm -rf /tmp/drv /tmp/drv.run
66       ---> Running in fea57df099aa
67      Get:1 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
InRelease [1581 B]
68      Get:2 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
69      Get:3 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
70      Get:4 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
Packages [2806 kB]
71      Get:5 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
72      Get:6 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
73      Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [4083 kB]
74      Get:8 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
75      Get:9 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
76      Get:10 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [7332
kB]
77      Get:11 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1308
kB]
78      Get:12 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
79      Get:13 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
80      Get:14 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [77.8
kB]
81      Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4416 kB]
82      Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1613 kB]
83      Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [86.4
kB]
84      Get:18 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7600
kB]
85      Get:19 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.6
kB]
86      Get:20 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [82.8 kB]
87      Fetched 49.8 MB in 2s (24.0 MB/s)
88      Reading package lists...
89      Reading package lists...
90      Building dependency tree...
91      Reading state information...
92      The following additional packages will be installed:
93        libbrotli1 libcurl4 libnghttp2-14 libpsl5 librtmp1 libssh-4
94      Recommended packages:
95        publicsuffix
96      The following NEW packages will be installed:
97        curl libbrotli1 libcurl4 libnghttp2-14 libpsl5 librtmp1 libssh-4
98      The following packages will be upgraded:
99        ca-certificates
100     1 upgraded, 7 newly installed, 0 to remove and 67 not upgraded.
101     Need to get 1321 kB of archives.
```

```
102     After this operation, 2975 kB of additional disk space will be used.
103     Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ca-certificates all
20260601~22.04.1 [141 kB]
104     Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnghttp2-14 amd64
1.43.0-1ubuntu0.4 [76.9 kB]
105     Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 libpsl5 amd64 0.21.0-1.2build2
[58.4 kB]
106     Get:4 http://archive.ubuntu.com/ubuntu jammy/main amd64 libbrotli1 amd64 1.0.9-2build6
[315 kB]
107     Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 librtmp1 amd64
2.4+20151223.gitfa8646d.1-2build4 [58.2 kB]
108     Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libssh-4 amd64
0.9.6-2ubuntu0.22.04.7 [187 kB]
109     Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcurl4 amd64
7.81.0-1ubuntu1.25 [291 kB]
110     Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64
7.81.0-1ubuntu1.25 [194 kB]
111     debconf: delaying package configuration, since apt-utils is not installed
112     Fetched 1321 kB in 0s (4525 kB/s)
113     (Reading database ...
(Reading database ... 5%
(Reading database ... 10%
(Reading database ... 15%
(Reading database ... 20%
(Reading database ... 25%
(Reading database ... 30%
(Reading database ... 35%
(Reading database ... 40%
(Reading database ... 45%
(Reading database ... 50%
(Reading database ... 55%
(Reading database ... 60%
(Reading database ... 65%
(Reading database ... 70%
(Reading database ... 75%
(Reading database ... 80%
(Reading database ... 85%
(Reading database ... 90%
(Reading database ... 95%
(Reading database ... 100%
(Reading database ... 5316 files and directories currently installed.)
114     Preparing to unpack .../0-ca-certificates_20260601~22.04.1_all.deb ...
115     Unpacking ca-certificates (20260601~22.04.1) over (20230311ubuntu0.22.04.1) ...
116     Selecting previously unselected package libnghttp2-14:amd64.
117     Preparing to unpack .../1-libnghttp2-14_1.43.0-1ubuntu0.4_amd64.deb ...
118     Unpacking libnghttp2-14:amd64 (1.43.0-1ubuntu0.4) ...
119     Selecting previously unselected package libpsl5:amd64.
120     Preparing to unpack .../2-libpsl5_0.21.0-1.2build2_amd64.deb ...
121     Unpacking libpsl5:amd64 (0.21.0-1.2build2) ...
122     Selecting previously unselected package libbrotli1:amd64.
123     Preparing to unpack .../3-libbrotli1_1.0.9-2build6_amd64.deb ...
124     Unpacking libbrotli1:amd64 (1.0.9-2build6) ...
125     Selecting previously unselected package librtmp1:amd64.
126     Preparing to unpack .../4-librtmp1_2.4+20151223.gitfa8646d.1-2build4_amd64.deb ...
127     Unpacking librtmp1:amd64 (2.4+20151223.gitfa8646d.1-2build4) ...
128     Selecting previously unselected package libssh-4:amd64.
129     Preparing to unpack .../5-libssh-4_0.9.6-2ubuntu0.22.04.7_amd64.deb ...
130     Unpacking libssh-4:amd64 (0.9.6-2ubuntu0.22.04.7) ...
131     Selecting previously unselected package libcurl4:amd64.
132     Preparing to unpack .../6-libcurl4_7.81.0-1ubuntu1.25_amd64.deb ...
133     Unpacking libcurl4:amd64 (7.81.0-1ubuntu1.25) ...
134     Selecting previously unselected package curl.
135     Preparing to unpack .../7-curl_7.81.0-1ubuntu1.25_amd64.deb ...
136     Unpacking curl (7.81.0-1ubuntu1.25) ...
137     Setting up libpsl5:amd64 (0.21.0-1.2build2) ...
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138     Setting up libbrotli1:amd64 (1.0.9-2build6) ...
139     Setting up libnghttp2-14:amd64 (1.43.0-1ubuntu0.4) ...
140     Setting up ca-certificates (20260601~22.04.1) ...
141     debconf: unable to initialize frontend: Dialog
142     debconf: (TERM is not set, so the dialog frontend is not usable.)
143     debconf: falling back to frontend: Readline
144     debconf: unable to initialize frontend: Readline
145     debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the
Term::ReadLine module) (@INC contains: /etc/perl /usr/local/lib/x86_64-linux-gnu/perl/5.34.0
/usr/local/share/perl/5.34.0 /usr/lib/x86_64-linux-gnu/perl5/5.34 /usr/share/perl5
/usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.34 /usr/share/perl/5.34
/usr/local/lib/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
146     debconf: falling back to frontend: Teletype
147     Updating certificates in /etc/ssl/certs...
148     rehash: warning: skipping ca-certificates.crt,it does not contain exactly one
certificate or CRL
149     24 added, 40 removed; done.
150     Setting up librtmp1:amd64 (2.4+20151223.gitfa8646d.1-2build4) ...
151     Setting up libssh-4:amd64 (0.9.6-2ubuntu0.22.04.7) ...
152     Setting up libcurl4:amd64 (7.81.0-1ubuntu1.25) ...
153     Setting up curl (7.81.0-1ubuntu1.25) ...
154     Processing triggers for libc-bin (2.35-0ubuntu3.6) ...
155     Processing triggers for ca-certificates (20260601~22.04.1) ...
156     Updating certificates in /etc/ssl/certs...
157     0 added, 0 removed; done.
158     Running hooks in /etc/ca-certificates/update.d...
159     done.
160     Creating directory /tmp/drv
161     Verifying archive integrity... OK
162     Uncompressing NVIDIA Accelerated Graphics Driver for Linux-x86_64 580.159.03............
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163     Removing intermediate container fea57df099aa
164      ---> d8aa9927e506
165     Step 5/20 : FROM nvidia/cuda:12.4.1-runtime-ubuntu22.04
166      ---> a112cf8dd3c2
167     Step 6/20 : ENV DEBIAN_FRONTEND=noninteractive     PYTHONUNBUFFERED=1
PIP_NO_CACHE_DIR=1     CONDA_DIR=/opt/conda     WASB_REPO_DIR=/opt/models/WASB-SBDT
168      ---> Running in 7e33ea1504d6
169     Removing intermediate container 7e33ea1504d6
170      ---> 456e2e162a60
171     Step 7/20 : RUN apt-get update && apt-get install -y --no-install-recommends
ffmpeg fonts-dejavu-core git curl ca-certificates build-essential         python3 python3-pip
python3-venv     && rm -rf /var/lib/apt/lists/*
172      ---> Running in b7177e0be4b7
173     Get:1 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
InRelease [1581 B]
174     Get:2 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
175     Get:3 http://archive.ubuntu.com/ubuntu jammy InRelease [270 kB]
176     Get:4 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu2204/x86_64
Packages [2806 kB]
177     Get:5 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1308 kB]
178     Get:6 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [77.8
kB]
```

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Get:7 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [4083 kB]
Get:8 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [7332 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [266 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 Packages [1792 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [17.5 MB]
Get:14 http://archive.ubuntu.com/ubuntu jammy/restricted amd64 Packages [164 kB]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [7600 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [4416 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1613 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [86.4 kB]
Get:19 http://archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [82.8 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [35.6 kB]
Fetched 49.8 MB in 2s (23.2 MB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
The following additional packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu bzip2 cpp cpp-11 dpkg-dev
  fontconfig fontconfig-config g++ g++-11 gcc gcc-11 gcc-11-base gcc-12-base
  git-man libaom3 libapparmor1 libasan6 libasound2 libasound2-data libass9
  libasyncns0 libatomic1 libavc1394-0 libavcodec58 libavdevice58 libavfilter7
  libavformat58 libavutil56 libbinutils libblas3 libbluray2 libbrotli1
  libbs2b0 libbsd0 libc-dev-bin libc6 libc6-dev libcaca0 libcairo-gobject2
  libcairo2 libcc1-0 libcdio-cdda2 libcdio-paranoia2 libcdio19 libchromaprint1
  libcodec2-1.0 libcrypt-dev libctf-nobfd0 libctf0 libcurl3-gnutls libcurl4
  libdatrie1 libdav1d5 libdbus-1-3 libdc1394-25 libdecor-0-0 libdeflate0
  libdpkg-perl libdrm-amdgpu1 libdrm-common libdrm-intel1 libdrm-nouveau2
  libdrm-radeon1 libdrm2 libedit2 libelf1 liberror-perl libexpat1 libflac8
  libflite1 libfontconfig1 libfreetype6 libfribidi0 libgbm1 libgcc-11-dev
  libgcc-s1 libgdbm-compat4 libgdbm6 libgdk-pixbuf-2.0-0
  libgdk-pixbuf2.0-common libgfortran5 libgl1 libgl1-mesa-dri libglapi-mesa
  libglib2.0-0 libglvnd0 libglx-mesa0 libglx0 libgme0 libgomp1 libgraphite2-3
  libgsm1 libharfbuzz0b libicu70 libiec61883-0 libisl23 libitm1
  libjack-jackd2-0 libjbig0 libjpeg-turbo8 libjpeg8 liblapack3 liblilv-0-0
  libllvm15 liblsan0 libmd0 libmfx1 libmp3lame0 libmpc3 libmpdec3 libmpfr6
  libmpg123-0 libmysofa1 libnghttp2-14 libnorm1 libnsl-dev libnuma1 libogg0
  libopenal-data libopenal1 libopenjp2-7 libopenmpt0 libopus0 libpango-1.0-0
  libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libperl5.34 libpgm-5.3-0
  libpixman-1-0 libpng16-16 libpocketsphinx3 libpostproc55 libpsl5 libpulse0
  libpython3-stdlib libpython3.10-minimal libpython3.10-stdlib libquadmath0
  librabbitmq4 libraw1394-11 librsvg2-2 librtmp1 librubberband2 libsamplerate0
  libsdl2-2.0-0 libsensors-config libsensors5 libserd-0-0 libshine3 libslang2
  libsnappy1v5 libsndfile1 libsndio7.0 libsodium23 libsord-0-0 libsoxr0
  libspeex1 libsphinxbase3 libsratom-0-0 libsrt1.4-gnutls libssh-4
  libssh-gcrypt-4 libstdc++-11-dev libstdc++6 libswresample3 libswscale5
  libthai-data libthai0 libtheora0 libtiff5 libtirpc-dev libtsan0 libtwolame0
  libubsan1 libudfread0 libusb-1.0-0 libva-drm2 libva-x11-2 libva2 libvdpau1
  libvidstab1.1 libvorbis0a libvorbisenc2 libvorbisfile3 libvpx7
  libwayland-client0 libwayland-cursor0 libwayland-egl1 libwayland-server0
  libwebp7 libwebpmux3 libx11-6 libx11-data libx11-xcb1 libx264-163
  libx265-199 libxau6 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0
  libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1
  libxcb-xfixes0 libxcb1 libxcursor1 libxdmcp6 libxext6 libxfixes3 libxi6
  libxinerama1 libxkbcommon0 libxml2 libxrandr2 libxrender1 libxshmfence1
  libxss1 libxv1 libxvidcore4 libxxf86vm1 libzimg2 libzmq5 libzvbi-common
  libzvbi0 linux-libc-dev lto-disabled-list make media-types
  ocl-icd-libopencl1 patch perl perl-base perl-modules-5.34 python3-distutils
  python3-lib2to3 python3-minimal python3-pip-whl python3-pkg-resources
```

```
  python3-setuptools python3-setuptools-whl python3-wheel python3.10
  python3.10-minimal python3.10-venv rpcsvc-proto shared-mime-info ucf
  x11-common xkb-data xz-utils
Suggested packages:
  binutils-doc bzip2-doc cpp-doc gcc-11-locales debian-keyring ffmpeg-doc
  g++-multilib g++-11-multilib gcc-11-doc gcc-multilib manpages-dev autoconf
  automake libtool flex bison gdb gcc-doc gcc-11-multilib gettext-base
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn libasound2-plugins alsa-utils libcuda1
  libnvcuvid1 libnvidia-encode1 libbluray-bdj glibc-doc locales bzr gdbm-l10n
  jackd2 libportaudio2 opus-tools pciutils pulseaudio libraw1394-doc
  librsvg2-bin xdg-utils lm-sensors serdi sndiod sordi speex libstdc++-11-doc
  make-doc opencl-icd ed diffutils-doc perl-doc libterm-readline-gnu-perl
  | libterm-readline-perl-perl libtap-harness-archive-perl python3-doc
  python3-tk python-setuptools-doc python3.10-doc binfmt-support
Recommended packages:
  fakeroot libalgorithm-merge-perl less ssh-client alsa-ucm-conf
  alsa-topology-conf libaacs0 manpages manpages-dev libc-devtools libnss-nis
  libnss-nisplus dbus libdecor-0-plugin-1-cairo | libdecor-0-plugin-1
  libfile-fcntllock-perl liblocale-gettext-perl libgdk-pixbuf2.0-bin
  libgl1-amber-dri libglib2.0-data xdg-user-dirs pocketsphinx-en-us
  publicsuffix librsvg2-common va-driver-all | va-driver vdpau-driver-all
  | vdpau-driver netbase python3-dev
The following NEW packages will be installed:
  binutils binutils-common binutils-x86-64-linux-gnu build-essential bzip2 cpp
  cpp-11 curl dpkg-dev ffmpeg fontconfig fontconfig-config fonts-dejavu-core
  g++ g++-11 gcc gcc-11 gcc-11-base git git-man libaom3 libapparmor1 libasan6
  libasound2 libasound2-data libass9 libasyncns0 libatomic1 libavc1394-0
  libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56
  libbinutils libblas3 libbluray2 libbrotli1 libbs2b0 libbsd0 libc-dev-bin
  libc6-dev libcaca0 libcairo-gobject2 libcairo2 libcc1-0 libcdio-cdda2
  libcdio-paranoia2 libcdio19 libchromaprint1 libcodec2-1.0 libcrypt-dev
  libctf-nobfd0 libctf0 libcurl3-gnutls libcurl4 libdatrie1 libdav1d5
  libdbus-1-3 libdc1394-25 libdecor-0-0 libdeflate0 libdpkg-perl
  libdrm-amdgpu1 libdrm-common libdrm-intel1 libdrm-nouveau2 libdrm-radeon1
  libdrm2 libedit2 libelf1 liberror-perl libexpat1 libflac8 libflite1
  libfontconfig1 libfreetype6 libfribidi0 libgbm1 libgcc-11-dev
  libgdbm-compat4 libgdbm6 libgdk-pixbuf-2.0-0 libgdk-pixbuf2.0-common
  libgfortran5 libgl1 libgl1-mesa-dri libglapi-mesa libglib2.0-0 libglvnd0
  libglx-mesa0 libglx0 libgme0 libgomp1 libgraphite2-3 libgsm1 libharfbuzz0b
  libicu70 libiec61883-0 libisl23 libitm1 libjack-jackd2-0 libjbig0
  libjpeg-turbo8 libjpeg8 liblapack3 liblilv-0-0 libllvm15 liblsan0 libmd0
  libmfx1 libmp3lame0 libmpc3 libmpdec3 libmpfr6 libmpg123-0 libmysofa1
  libnghttp2-14 libnorm1 libnsl-dev libnuma1 libogg0 libopenal-data libopenal1
  libopenjp2-7 libopenmpt0 libopus0 libpango-1.0-0 libpangocairo-1.0-0
  libpangoft2-1.0-0 libpciaccess0 libperl5.34 libpgm-5.3-0 libpixman-1-0
  libpng16-16 libpocketsphinx3 libpostproc55 libpsl5 libpulse0
  libpython3-stdlib libpython3.10-minimal libpython3.10-stdlib libquadmath0
  librabbitmq4 libraw1394-11 librsvg2-2 librtmp1 librubberband2 libsamplerate0
  libsdl2-2.0-0 libsensors-config libsensors5 libserd-0-0 libshine3 libslang2
  libsnappy1v5 libsndfile1 libsndio7.0 libsodium23 libsord-0-0 libsoxr0
  libspeex1 libsphinxbase3 libsratom-0-0 libsrt1.4-gnutls libssh-4
  libssh-gcrypt-4 libstdc++-11-dev libswresample3 libswscale5 libthai-data
  libthai0 libtheora0 libtiff5 libtirpc-dev libtsan0 libtwolame0 libubsan1
  libudfread0 libusb-1.0-0 libva-drm2 libva-x11-2 libva2 libvdpau1
  libvidstab1.1 libvorbis0a libvorbisenc2 libvorbisfile3 libvpx7
  libwayland-client0 libwayland-cursor0 libwayland-egl1 libwayland-server0
  libwebp7 libwebpmux3 libx11-6 libx11-data libx11-xcb1 libx264-163
  libx265-199 libxau6 libxcb-dri2-0 libxcb-dri3-0 libxcb-glx0 libxcb-present0
  libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1
  libxcb-xfixes0 libxcb1 libxcursor1 libxdmcp6 libxext6 libxfixes3 libxi6
  libxinerama1 libxkbcommon0 libxml2 libxrandr2 libxrender1 libxshmfence1
  libxss1 libxv1 libxvidcore4 libxxf86vm1 libzimg2 libzmq5 libzvbi-common
  libzvbi0 linux-libc-dev lto-disabled-list make media-types
  ocl-icd-libopencl1 patch perl perl-modules-5.34 python3 python3-distutils
```

```
  python3-lib2to3 python3-minimal python3-pip python3-pip-whl
  python3-pkg-resources python3-setuptools python3-setuptools-whl python3-venv
  python3-wheel python3.10 python3.10-minimal python3.10-venv rpcsvc-proto
  shared-mime-info ucf x11-common xkb-data xz-utils
The following packages will be upgraded:
  ca-certificates gcc-12-base libc6 libgcc-s1 libstdc++6 perl-base
6 upgraded, 259 newly installed, 0 to remove and 62 not upgraded.
Need to get 208 MB of archives.
After this operation, 720 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 perl-base amd64 5.34.0-3ubuntu1.7 [1763 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-12-base amd64 12.3.0-1ubuntu1~22.04.3 [216 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgcc-s1 amd64 12.3.0-1ubuntu1~22.04.3 [53.9 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libstdc++6 amd64 12.3.0-1ubuntu1~22.04.3 [699 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6 amd64 2.35-0ubuntu3.13 [3235 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpython3.10-minimal amd64 3.10.12-1~22.04.16 [817 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libexpat1 amd64 2.4.7-1ubuntu0.7 [92.1 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3.10-minimal amd64 3.10.12-1~22.04.16 [2254 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-minimal amd64 3.10.6-1~22.04.1 [24.3 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy/main amd64 media-types all 7.0.0 [25.5 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmpdec3 amd64 2.5.1-2build2 [86.8 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpython3.10-stdlib amd64 3.10.12-1~22.04.16 [1850 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3.10 amd64 3.10.12-1~22.04.16 [508 kB]
Get:14 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpython3-stdlib amd64 3.10.6-1~22.04.1 [6812 B]
Get:15 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3 amd64 3.10.6-1~22.04.1 [22.8 kB]
Get:16 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 perl-modules-5.34 all 5.34.0-3ubuntu1.7 [2977 kB]
Get:17 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgdbm6 amd64 1.23-1 [33.9 kB]
Get:18 http://archive.ubuntu.com/ubuntu jammy/main amd64 libgdbm-compat4 amd64 1.23-1 [6606 B]
Get:19 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libperl5.34 amd64 5.34.0-3ubuntu1.7 [4817 kB]
Get:20 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 perl amd64 5.34.0-3ubuntu1.7 [232 kB]
Get:21 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 ca-certificates all 20260601~22.04.1 [141 kB]
Get:22 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapparmor1 amd64 3.0.4-2ubuntu2.5 [39.6 kB]
Get:23 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmd0 amd64 1.0.4-1build1 [23.0 kB]
Get:24 http://archive.ubuntu.com/ubuntu jammy/main amd64 libbsd0 amd64 0.11.5-1 [44.8 kB]
Get:25 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libdbus-1-3 amd64 1.12.20-2ubuntu4.1 [189 kB]
Get:26 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libelf1 amd64 0.186-1ubuntu0.1 [51.1 kB]
Get:27 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libfribidi0 amd64 1.0.8-2ubuntu3.1 [26.1 kB]
Get:28 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libglib2.0-0 amd64 2.72.4-0ubuntu2.9 [1467 kB]
Get:29 http://archive.ubuntu.com/ubuntu jammy/main amd64 libicu70 amd64 70.1-2 [10.6 MB]
Get:30 http://archive.ubuntu.com/ubuntu jammy/main amd64 libslang2 amd64 2.3.2-5build4
```

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[468 kB]
344     Get:31 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libxml2 amd64
2.9.13+dfsg-1ubuntu0.12 [765 kB]
345     Get:32 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 python3-pkg-resources
all 59.6.0-1.2ubuntu0.22.04.3 [133 kB]
346     Get:33 http://archive.ubuntu.com/ubuntu jammy/main amd64 shared-mime-info amd64 2.1-2
[454 kB]
347     Get:34 http://archive.ubuntu.com/ubuntu jammy/main amd64 ucf all 3.0043 [56.1 kB]
348     Get:35 http://archive.ubuntu.com/ubuntu jammy/main amd64 xkb-data all 2.33-1 [394 kB]
349     Get:36 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libdrm-common all
2.4.113-2~ubuntu0.22.04.1 [5450 B]
350     Get:37 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libdrm2 amd64
2.4.113-2~ubuntu0.22.04.1 [38.1 kB]
351     Get:38 http://archive.ubuntu.com/ubuntu jammy/main amd64 libedit2 amd64
3.1-20210910-1build1 [96.8 kB]
352     Get:39 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libnghttp2-14 amd64
1.43.0-1ubuntu0.4 [76.9 kB]
353     Get:40 http://archive.ubuntu.com/ubuntu jammy/main amd64 libnuma1 amd64 2.0.14-3ubuntu2
[22.5 kB]
354     Get:41 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpng16-16 amd64
1.6.37-3ubuntu0.5 [192 kB]
355     Get:42 http://archive.ubuntu.com/ubuntu jammy/main amd64 libpsl5 amd64 0.21.0-1.2build2
[58.4 kB]
356     Get:43 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libusb-1.0-0 amd64
2:1.0.25-1ubuntu2 [52.7 kB]
357     Get:44 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxau6 amd64 1:1.0.9-1build5
[7634 B]
358     Get:45 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxdmcp6 amd64
1:1.1.3-0ubuntu5 [10.9 kB]
359     Get:46 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb1 amd64 1.14-3ubuntu3
[49.0 kB]
360     Get:47 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libx11-data all
2:1.7.5-1ubuntu0.3 [120 kB]
361     Get:48 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libx11-6 amd64
2:1.7.5-1ubuntu0.3 [667 kB]
362     Get:49 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxext6 amd64 2:1.3.4-1build1
[31.8 kB]
363     Get:50 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 xz-utils amd64
5.2.5-2ubuntu1.1 [84.8 kB]
364     Get:51 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 binutils-common amd64
2.38-4ubuntu2.12 [223 kB]
365     Get:52 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libbinutils amd64
2.38-4ubuntu2.12 [663 kB]
366     Get:53 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libctf-nobfd0 amd64
2.38-4ubuntu2.12 [108 kB]
367     Get:54 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libctf0 amd64
2.38-4ubuntu2.12 [103 kB]
368     Get:55 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 binutils-x86-64-linux-
gnu amd64 2.38-4ubuntu2.12 [2324 kB]
369     Get:56 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 binutils amd64
2.38-4ubuntu2.12 [3184 B]
370     Get:57 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc-dev-bin amd64
2.35-0ubuntu3.13 [20.3 kB]
371     Get:58 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 linux-libc-dev amd64
5.15.0-185.195 [1315 kB]
372     Get:59 http://archive.ubuntu.com/ubuntu jammy/main amd64 libcrypt-dev amd64 1:4.4.27-1
[112 kB]
373     Get:60 http://archive.ubuntu.com/ubuntu jammy/main amd64 rpcsvc-proto amd64
1.4.2-0ubuntu6 [68.5 kB]
374     Get:61 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtirpc-dev amd64
1.3.2-2ubuntu0.1 [192 kB]
375     Get:62 http://archive.ubuntu.com/ubuntu jammy/main amd64 libnsl-dev amd64 1.3.0-2build2
[71.3 kB]
376     Get:63 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libc6-dev amd64
2.35-0ubuntu3.13 [2101 kB]
```

```
377     Get:64 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-11-base amd64
11.4.0-1ubuntu1~22.04.3 [216 kB]
378     Get:65 http://archive.ubuntu.com/ubuntu jammy/main amd64 libisl23 amd64 0.24-2build1
[727 kB]
379     Get:66 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmpfr6 amd64 4.1.0-3build3
[1425 kB]
380     Get:67 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmpc3 amd64 1.2.1-2build1
[46.9 kB]
381     Get:68 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 cpp-11 amd64
11.4.0-1ubuntu1~22.04.3 [10.0 MB]
382     Get:69 http://archive.ubuntu.com/ubuntu jammy/main amd64 cpp amd64 4:11.2.0-1ubuntu1
[27.7 kB]
383     Get:70 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcc1-0 amd64
12.3.0-1ubuntu1~22.04.3 [48.3 kB]
384     Get:71 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgomp1 amd64
12.3.0-1ubuntu1~22.04.3 [127 kB]
385     Get:72 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libitm1 amd64
12.3.0-1ubuntu1~22.04.3 [30.2 kB]
386     Get:73 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libatomic1 amd64
12.3.0-1ubuntu1~22.04.3 [10.5 kB]
387     Get:74 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasan6 amd64
11.4.0-1ubuntu1~22.04.3 [2283 kB]
388     Get:75 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 liblsan0 amd64
12.3.0-1ubuntu1~22.04.3 [1069 kB]
389     Get:76 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtsan0 amd64
11.4.0-1ubuntu1~22.04.3 [2260 kB]
390     Get:77 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libubsan1 amd64
12.3.0-1ubuntu1~22.04.3 [976 kB]
391     Get:78 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libquadmath0 amd64
12.3.0-1ubuntu1~22.04.3 [154 kB]
392     Get:79 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgcc-11-dev amd64
11.4.0-1ubuntu1~22.04.3 [2517 kB]
393     Get:80 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 gcc-11 amd64
11.4.0-1ubuntu1~22.04.3 [20.1 MB]
394     Get:81 http://archive.ubuntu.com/ubuntu jammy/main amd64 gcc amd64 4:11.2.0-1ubuntu1
[5112 B]
395     Get:82 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libstdc++-11-dev amd64
11.4.0-1ubuntu1~22.04.3 [2101 kB]
396     Get:83 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 g++-11 amd64
11.4.0-1ubuntu1~22.04.3 [11.4 MB]
397     Get:84 http://archive.ubuntu.com/ubuntu jammy/main amd64 g++ amd64 4:11.2.0-1ubuntu1
[1412 B]
398     Get:85 http://archive.ubuntu.com/ubuntu jammy/main amd64 make amd64 4.3-4.1build1 [180
kB]
399     Get:86 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libdpkg-perl all
1.21.1ubuntu2.6 [237 kB]
400     Get:87 http://archive.ubuntu.com/ubuntu jammy/main amd64 bzip2 amd64 1.0.8-5build1 [34.8
kB]
401     Get:88 http://archive.ubuntu.com/ubuntu jammy/main amd64 patch amd64 2.7.6-7build2 [109
kB]
402     Get:89 http://archive.ubuntu.com/ubuntu jammy/main amd64 lto-disabled-list all 24 [12.5
kB]
403     Get:90 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 dpkg-dev all
1.21.1ubuntu2.6 [922 kB]
404     Get:91 http://archive.ubuntu.com/ubuntu jammy/main amd64 build-essential amd64
12.9ubuntu3 [4744 B]
405     Get:92 http://archive.ubuntu.com/ubuntu jammy/main amd64 libbrotli1 amd64 1.0.9-2build6
[315 kB]
406     Get:93 http://archive.ubuntu.com/ubuntu jammy/main amd64 librtmp1 amd64
2.4+20151223.gitfa8646d.1-2build4 [58.2 kB]
407     Get:94 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libssh-4 amd64
0.9.6-2ubuntu0.22.04.7 [187 kB]
408     Get:95 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcurl4 amd64
7.81.0-1ubuntu1.25 [291 kB]
409     Get:96 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 curl amd64
```

7.81.0-1ubuntu1.25 [194 kB]
410 Get:97 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libaom3 amd64 3.3.0-1ubuntu0.1 [1748 kB]
411 Get:98 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libva2 amd64 2.14.0-1 [65.0 kB]
412 Get:99 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libmfx1 amd64 22.3.0-1 [3105 kB]
413 Get:100 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libva-drm2 amd64 2.14.0-1 [7502 B]
414 Get:101 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxfixes3 amd64 1:6.0.0-1 [11.7 kB]
415 Get:102 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libva-x11-2 amd64 2.14.0-1 [12.6 kB]
416 Get:103 http://archive.ubuntu.com/ubuntu jammy/main amd64 libvdpau1 amd64 1.4-3build2 [27.0 kB]
417 Get:104 http://archive.ubuntu.com/ubuntu jammy/universe amd64 ocl-icd-libopencl1 amd64 2.2.14-3 [39.1 kB]
418 Get:105 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libavutil56 amd64 7:4.4.2-0ubuntu0.22.04.1 [290 kB]
419 Get:106 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libfreetype6 amd64 2.11.1+dfsg-1ubuntu0.3 [388 kB]
420 Get:107 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-dejavu-core all 2.37-2build1 [1041 kB]
421 Get:108 http://archive.ubuntu.com/ubuntu jammy/main amd64 fontconfig-config all 2.13.1-4.2ubuntu5 [29.1 kB]
422 Get:109 http://archive.ubuntu.com/ubuntu jammy/main amd64 libfontconfig1 amd64 2.13.1-4.2ubuntu5 [131 kB]
423 Get:110 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpixman-1-0 amd64 0.40.0-1ubuntu0.22.04.1 [264 kB]
424 Get:111 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-render0 amd64 1.14-3ubuntu3 [16.4 kB]
425 Get:112 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxcb-shm0 amd64 1.14-3ubuntu3 [5780 B]
426 Get:113 http://archive.ubuntu.com/ubuntu jammy/main amd64 libxrender1 amd64 1:0.9.10-1build4 [19.7 kB]
427 Get:114 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcairo2 amd64 1.16.0-5ubuntu2.1 [628 kB]
428 Get:115 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libcodec2-1.0 amd64 1.0.1-3 [8435 kB]
429 Get:116 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libdav1d5 amd64 0.9.2-1 [463 kB]
430 Get:117 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libgsm1 amd64 1.0.19-1 [27.7 kB]
431 Get:118 http://archive.ubuntu.com/ubuntu jammy/main amd64 libmp3lame0 amd64 3.100-3build2 [141 kB]
432 Get:119 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libopenjp2-7 amd64 2.4.0-6ubuntu0.5 [158 kB]
433 Get:120 http://archive.ubuntu.com/ubuntu jammy/main amd64 libopus0 amd64 1.3.1-0.1build2 [203 kB]
434 Get:121 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcairo-gobject2 amd64 1.16.0-5ubuntu2.1 [19.5 kB]
435 Get:122 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgdk-pixbuf2.0-common all 2.42.8+dfsg-1ubuntu0.5 [5560 B]
436 Get:123 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjpeg-turbo8 amd64 2.1.2-0ubuntu1 [134 kB]
437 Get:124 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjpeg8 amd64 8c-2ubuntu10 [2264 B]
438 Get:125 http://archive.ubuntu.com/ubuntu jammy/main amd64 libdeflate0 amd64 1.10-2 [70.9 kB]
439 Get:126 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libjbig0 amd64 2.1-3.1ubuntu0.22.04.1 [29.2 kB]
440 Get:127 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libwebp7 amd64 1.2.2-2ubuntu0.22.04.2 [206 kB]
441 Get:128 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libtiff5 amd64 4.3.0-6ubuntu0.13 [185 kB]

442   Get:129 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgdk-pixbuf-2.0-0
amd64 2.42.8+dfsg-1ubuntu0.5 [148 kB]
443   Get:130 http://archive.ubuntu.com/ubuntu jammy/main amd64 fontconfig amd64
2.13.1-4.2ubuntu5 [177 kB]
444   Get:131 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgraphite2-3 amd64
1.3.14-1ubuntu0.1 [71.6 kB]
445   Get:132 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libharfbuzz0b amd64
2.7.4-1ubuntu3.2 [353 kB]
446   Get:133 http://archive.ubuntu.com/ubuntu jammy/main amd64 libthai-data all
0.1.29-1build1 [162 kB]
447   Get:134 http://archive.ubuntu.com/ubuntu jammy/main amd64 libdatrie1 amd64 0.2.13-2
[19.9 kB]
448   Get:135 http://archive.ubuntu.com/ubuntu jammy/main amd64 libthai0 amd64 0.1.29-1build1
[19.2 kB]
449   Get:136 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpango-1.0-0 amd64
1.50.6+ds-2ubuntu1 [230 kB]
450   Get:137 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpangoft2-1.0-0
amd64 1.50.6+ds-2ubuntu1 [54.0 kB]
451   Get:138 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpangocairo-1.0-0
amd64 1.50.6+ds-2ubuntu1 [39.8 kB]
452   Get:139 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 librsvg2-2 amd64
2.52.5+dfsg-3ubuntu0.2 [2974 kB]
453   Get:140 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libshine3 amd64 3.1.1-2
[23.2 kB]
454   Get:141 http://archive.ubuntu.com/ubuntu jammy/main amd64 libsnappy1v5 amd64
1.1.8-1build3 [17.5 kB]
455   Get:142 http://archive.ubuntu.com/ubuntu jammy/main amd64 libspeex1 amd64
1.2~rc1.2-1.1ubuntu3 [57.9 kB]
456   Get:143 http://archive.ubuntu.com/ubuntu jammy/main amd64 libsoxr0 amd64 0.1.3-4build2
[79.8 kB]
457   Get:144 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libswresample3
amd64 7:4.4.2-0ubuntu0.22.04.1 [62.2 kB]
458   Get:145 http://archive.ubuntu.com/ubuntu jammy/main amd64 libogg0 amd64 1.3.5-0ubuntu3
[22.9 kB]
459   Get:146 http://archive.ubuntu.com/ubuntu jammy/main amd64 libtheora0 amd64
1.1.1+dfsg.1-15ubuntu4 [209 kB]
460   Get:147 http://archive.ubuntu.com/ubuntu jammy/main amd64 libtwolame0 amd64
0.4.0-2build2 [52.5 kB]
461   Get:148 http://archive.ubuntu.com/ubuntu jammy/main amd64 libvorbis0a amd64
1.3.7-1build2 [99.2 kB]
462   Get:149 http://archive.ubuntu.com/ubuntu jammy/main amd64 libvorbisenc2 amd64
1.3.7-1build2 [82.6 kB]
463   Get:150 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libvpx7 amd64
1.11.0-2ubuntu2.5 [1078 kB]
464   Get:151 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libwebpmux3 amd64
1.2.2-2ubuntu0.22.04.2 [20.5 kB]
465   Get:152 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libx264-163 amd64
2:0.163.3060+git5db6aa6-2build1 [591 kB]
466   Get:153 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libx265-199 amd64 3.5-2
[1170 kB]
467   Get:154 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libxvidcore4 amd64
2:1.3.7-1 [201 kB]
468   Get:155 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzvbi-common all
0.2.35-19 [35.5 kB]
469   Get:156 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzvbi0 amd64 0.2.35-19
[262 kB]
470   Get:157 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libavcodec58 amd64
7:4.4.2-0ubuntu0.22.04.1 [5567 kB]
471   Get:158 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasound2-data all
1.2.6.1-1ubuntu1.1 [19.3 kB]
472   Get:159 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasound2 amd64
1.2.6.1-1ubuntu1.1 [391 kB]
473   Get:160 http://archive.ubuntu.com/ubuntu jammy/main amd64 libraw1394-11 amd64
2.1.2-2build2 [27.0 kB]
474   Get:161 http://archive.ubuntu.com/ubuntu jammy/main amd64 libavc1394-0 amd64

0.5.4-5build2 [17.0 kB]
475 Get:162 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libass9 amd64 1:0.15.2-1 [97.5 kB]
476 Get:163 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libudfread0 amd64 1.1.2-1 [16.2 kB]
477 Get:164 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libbluray2 amd64 1:1.3.1-1 [159 kB]
478 Get:165 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libchromaprint1 amd64 1.5.1-2 [28.4 kB]
479 Get:166 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libgme0 amd64 0.6.3-2 [127 kB]
480 Get:167 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libmpg123-0 amd64 1.29.3-1ubuntu0.1 [172 kB]
481 Get:168 http://archive.ubuntu.com/ubuntu jammy/main amd64 libvorbisfile3 amd64 1.3.7-1build2 [17.1 kB]
482 Get:169 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libopenmpt0 amd64 0.6.1-1 [592 kB]
483 Get:170 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 librabbitmq4 amd64 0.10.0-1ubuntu2.1 [39.5 kB]
484 Get:171 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libsrt1.4-gnutls amd64 1.4.4-4 [309 kB]
485 Get:172 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libssh-gcrypt-4 amd64 0.9.6-2ubuntu0.22.04.7 [224 kB]
486 Get:173 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libnorm1 amd64 1.5.9+dfsg-2 [221 kB]
487 Get:174 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpgm-5.3-0 amd64 5.3.128~dfsg-2 [161 kB]
488 Get:175 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libsodium23 amd64 1.0.18-1ubuntu0.22.04.1 [164 kB]
489 Get:176 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libzmq5 amd64 4.3.4-2 [256 kB]
490 Get:177 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 libavformat58 amd64 7:4.4.2-0ubuntu0.22.04.1 [1103 kB]
491 Get:178 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libbs2b0 amd64 3.1.0+dfsg-2.2build1 [10.2 kB]
492 Get:179 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libflite1 amd64 2.2-3 [13.7 MB]
493 Get:180 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libserd-0-0 amd64 0.30.10-2 [40.8 kB]
494 Get:181 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libsord-0-0 amd64 0.16.8-2 [21.2 kB]
495 Get:182 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libsratom-0-0 amd64 0.6.8-1 [17.0 kB]
496 Get:183 http://archive.ubuntu.com/ubuntu jammy/universe amd64 liblilv-0-0 amd64 0.24.12-2 [42.8 kB]
497 Get:184 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libmysofa1 amd64 1.2.1~dfsg0-1 [1157 kB]
498 Get:185 http://archive.ubuntu.com/ubuntu jammy/main amd64 libblas3 amd64 3.10.0-2ubuntu1 [228 kB]
499 Get:186 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgfortran5 amd64 12.3.0-1ubuntu1~22.04.3 [879 kB]

271. assistant (2026-07-07T16:11:13.571Z)

The image build (`b68riim4h`) is running ? currently at Docker step 7/20 (installing the CUDA/ffmpeg base). It's a ~8?10 min build and I'll be re-invoked when it completes, then I'll capture the new image digest and do the deploy + health verification.

Progress so far on the full go-live:
- ? SPA re-vendored into `backend/ui/` ? **PR #516 merged** (master `f5c9dd9`); the built bundle verifiably contains `compute_target==="cloud_run"`.
- ? Rollback refs captured: control-plane revision `rally-control-plane-00017-qbd`, image digest `sha256:09d2edf7?`.
- ? **Building** the Cloud Run image from master `f5c9dd9` (has #511 guardrail + re-vendored SPA + #512/#513).

- ? Next: point `rally-control-plane` (and the `rally-worker` job) at the new image, verify the new revision boots healthy.

I'll continue automatically when the build finishes.

272. user (2026-07-07T16:14:37.810Z)

<task-notification>
<task-id>b68riim4h</task-id>
<tool-use-id>toolu_01TqPFdgrCBpqhMNCpRvgseq</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\b68riim4h.output</output-file>
<status>completed</status>
<summary>Background command "Build Cloud Run image from master (background)" completed (exit code 0)</summary>
</task-notification>

273. assistant (2026-07-07T16:14:41.944Z)

Build completed (exit 0). Let me extract the SUCCESS status and the new image digest from the build log:

274. user (2026-07-07T16:14:46.046Z)

```
=== final status lines ===
  Building wheel for trackers (pyproject.toml): finished with status 'done'
Successfully built trackers
Successfully installed certifi-2026.6.17 charset_normalizer-3.4.9 contourpy-1.3.2 cycler-0.12.1
defusedxml-0.7.1 fonttools-4.63.0 idna-3.18 kiwisolver-1.5.0 markdown-it-py-4.2.0
matplotlib-3.10.9 mdurl-0.1.2 numpy-2.2.6 opencv-python-5.0.0.93 packaging-26.2 pillow-12.3.0
pydeprecate-0.9.0 pygments-2.20.0 pyparsing-3.3.2 python-dateutil-2.9.0.post0 pyyaml-6.0.3
requests-2.34.2 rich-15.0.0 scipy-1.15.3 six-1.17.0 supervision-0.29.1 tqdm-4.68.4
trackers-2.5.0 urllib3-2.7.0
  Installing build dependencies: finished with status 'done'
  Getting requirements to build wheel: finished with status 'done'
  Preparing metadata (pyproject.toml): finished with status 'done'
Collecting grpcio-status<2.0.0,>=1.41.0 (from google-api-core[grpc]<3.0.0,>=2.17.1->google-
cloud-run<1,>=0.10->badminton-highlight-indexer==0.1.0)
  Downloading grpcio_status-1.82.0-py3-none-any.whl.metadata (1.2 kB)
Downloading grpcio_status-1.82.0-py3-none-any.whl (14 kB)
  Building wheel for badminton-highlight-indexer (pyproject.toml): finished with status 'done'
Successfully built badminton-highlight-indexer
Installing collected packages: websockets, typing-extensions, tenacity, sniffio, python-dotenv,
pycparser, pyasn1, protobuf, MarkupSafe, h11, google-crc32c, distro, click, annotated-types,
annotated-doc, uvicorn, typing-inspection, pyjwt, pydantic-core, pyasn1-modules, proto-plus,
jinja2, httpcore, grpcio, googleapis-common-protos, google-resumable-media, exceptiongroup,
cffi, pydantic, grpcio-status, cryptography, anyio, starlette, httpx, grpc-google-iam-v1,
google-auth, google-api-core, fastapi, google-genai, google-cloud-core, google-cloud-storage,
google-cloud-run, badminton-highlight-indexer
Successfully installed MarkupSafe-3.0.3 annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.14.1
badminton-highlight-indexer-0.1.0 cffi-2.1.0 click-8.4.2 cryptography-49.0.0 distro-1.9.0
exceptiongroup-1.3.1 fastapi-0.139.0 google-api-core-2.31.0 google-auth-2.55.1 google-cloud-
core-2.6.0 google-cloud-run-0.16.1 google-cloud-storage-3.12.0 google-crc32c-1.8.0 google-
genai-2.10.0 google-resumable-media-2.10.0 googleapis-common-protos-1.75.0 grpc-google-
iam-v1-0.14.4 grpcio-1.82.0 grpcio-status-1.82.0 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1
jinja2-3.1.6 proto-plus-1.28.0 protobuf-7.35.1 pyasn1-0.6.3 pyasn1-modules-0.4.2 pycparser-3.0
pydantic-2.13.4 pydantic-core-2.46.4 pyjwt-2.13.0 python-dotenv-1.2.2 sniffio-1.3.1
starlette-1.3.1 tenacity-9.1.4 typing-extensions-4.16.0 typing-inspection-0.4.2 uvicorn-0.50.2
websockets-16.0
Successfully built e03b14b64076
Successfully tagged asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-
worker:latest
latest: digest: sha256:8045f54ea9fcbf52426305ab987f122f4fc8bed6b13be93283d21c5cab57a655 size:
4309
```

DONE
ID   CREATE_TIME   DURATION SOURCE
IMAGES   STATUS
17ab46b4-629a-428e-a234-64f18296a41f 2026-07-07T15:59:11+00:00 12M48S   gs://gen-lang-client-0306026826_cloudbuild/source/1783439943.389161-ffad55d3c59f49beb6e60592587a59b2.tgz
asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-worker (+1 more) SUCCESS
EXIT=0
=== digest / pushed image ===
Digest: sha256:517da2300c184c9999ec203c2665244bdebd3578d12fcc7065e83667932643d9
latest: digest: sha256:8045f54ea9fcbf52426305ab987f122f4fc8bed6b13be93283d21c5cab57a655 size: 4309

275. assistant (2026-07-07T16:15:07.136Z)

Build SUCCESS (12m48s). New image digest:
`sha256:8045f54ea9fcbf52426305ab987f122f4fc8bed6b13be93283d21c5cab57a655`.

Now Phase 3 ? deploy. I'll cut the `rally-control-plane` service to the new image by digest (creates a new revision, shifts traffic), which is the critical piece for this fix:

276. user (2026-07-07T16:17:10.036Z)

Command running in background with ID: b62d1qaev. Output is being written to: C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\b62d1qaev.output. You will be notified when it completes. To check interim output, use Read on that file path.

277. assistant (2026-07-07T16:17:18.651Z)

The control-plane deploy (`b62d1qaev`) is running ? Cloud Run is creating the new revision and health-checking it before shifting traffic (~1?2 min). I'll be notified when it finishes, then verify the revision booted healthy and cut the worker job over too. Waiting for it to complete.

278. user (2026-07-07T16:17:40.383Z)

<task-notification>
<task-id>b62d1qaev</task-id>
<tool-use-id>toolu_01MF7AjBh6TB6McCiXFk4Kgb</tool-use-id>
<output-file>C:\Users\avidu\AppData\Local\Temp\claude\C--Users-avidu-Projects-badminton-highlight-indexer\05c32f87-3bfb-4a7f-861c-0050ac4169ac\tasks\b62d1qaev.output</output-file>
<status>completed</status>
<summary>Background command "Deploy control-plane service to new image" completed (exit code 0)</summary>
</task-notification>

279. assistant (2026-07-07T16:17:43.116Z)

Control-plane deploy completed. Let me read the result and verify the new revision is healthy and serving:

280. user (2026-07-07T16:17:53.718Z)

=== deploy output ===
=== deploy control-plane to new image (by digest) ===
Deploying...
Creating Revision...
....
....
....
....
....
....
....
....

```
.................................................................................................
.................................................................................................
.................................................................................................
.................................................................................................
.................................................................................................
.................................................................................................
..............................done
Routing traffic.....done
Done.
Service [rally-control-plane] revision [rally-control-plane-00018-qvz] has been
deployed and is serving 100 percent of traffic.
Service URL: https://rally-control-plane-492532266377.asia-southeast1.run.app
rally-control-plane-00018-qvz   https://rally-control-plane-2bosgeukpa-as.a.run.app

=== control-plane status: latest ready revision + 100% traffic + image ===
rally-control-plane-00018-qvz   rally-control-plane-00018-qvz   100
=== running image digest on the live revision ===
asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-
worker@sha256:8045f54ea9fcbf52426305ab987f122f4fc8bed6b13be93283d21c5cab57a655
```

281. assistant (2026-07-07T16:18:11.289Z)

Control-plane is live on the new image ? revision **`rally-control-plane-00018-qvz`**, Ready, serving 100% traffic, running digest `sha256:8045f54e?`. Cloud Run only shifts traffic after the startup health check passes, so the container is healthy. Let me verify app-level health (new UI provenance + boot logs) and then cut the worker job over:

282. user (2026-07-07T16:18:19.553Z)

```
=== /api/version (is the new SPA bundle served? spa_commit should be ba5f457) ===
{"engine_version":"0.1.0","api_version":1,"ui":{"spa_repo":"https://github.com/avidullu/khelsutr
a.git","spa_commit":"ba5f45777441bb288c5b6a8868c3dbc736d8c2fa","engine_api_version":1,"content_h
ash":"f41f829f959eca1ff21695fd2aef058e3f423b6b7172e32e47b0f026b2d7e9be","built_at":"2026-07-
07T15:56:09Z"}}

=== recent boot logs for the new revision (health) ===
2026-07-07 16:17:27 16:17:27 [INFO] backend.config.loader: [CONFIG] deployment profile ACTIVE:
cloud-serving (compute_target=cloud_run)
2026-07-07 16:17:27 [SERVER] ? Binding 0.0.0.0 exposes the API on the network ? ensure an auth
gate (edge_auth_enabled / a reverse proxy) is in front, and set RALLY_ALLOWED_HOSTS to your
public host(s) (the Host-header guard defaults to loopback-only and will 400 forwarded requests
otherwise).
2026-07-07 16:17:27 INFO:     Started server process [1]
2026-07-07 16:17:27 INFO:     Waiting for application startup.
2026-07-07 16:17:27 INFO:     Application startup complete.
2026-07-07 16:17:27 INFO:     Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
```

283. assistant (2026-07-07T16:18:37.592Z)

The fix is LIVE on the control-plane. `/api/version` confirms it now serves the re-vendored SPA (`spa_commit: ba5f457?`), and the boot logs are healthy (`cloud-serving profile ACTIVE`, `Application startup complete`, uvicorn up). The #511 guardrail is in the same image, so it's live too.

Now cut the `rally-worker` job to the same image (keeps them in sync + picks up the #512/#513 worker changes):

284. user (2026-07-07T16:18:50.227Z)

```
Updating Cloud Run job [rally-worker] in project [gen-lang-client-0306026826]
region [asia-southeast1]
Updating job...
Done.
Job [rally-worker] has successfully been updated.
```

```
To execute this job, use:
gcloud run jobs execute rally-worker
rally-worker
=== worker job image now ===
asia-southeast1-docker.pkg.dev/gen-lang-client-0306026826/rally/rally-
worker@sha256:8045f54ea9fcbf52426305ab987f122f4fc8bed6b13be93283d21c5cab57a655
```

285. assistant (2026-07-07T16:19:05.213Z)

```
Both the service and the job are now on the new image. Let me do a final live sanity check ?
`/api/capabilities` should confirm the control-plane still has no local weights (so the
guardrail's premise holds) ? and clean up the build worktree:
```

286. user (2026-07-07T16:19:16.426Z)

```
=== /api/capabilities (deployment + weights state) ===
compute_target: cloud_run
cloud_available: True
wasb_weights_present: False
segmenters: ['fusion_hybrid', 'gemini', 'motion', 'tracknet_hybrid', 'wasb_hybrid',
'yolo_hybrid']

=== cleanup: remove build worktree ===
buildwt removed
error: failed to delete '.git/worktrees/a1-master-worktree': Permission denied
error: failed to delete '.git/worktrees/bhi-wt-worker-reval': Permission denied
error: failed to delete '.git/worktrees/fix-worktree': Permission denied
error: failed to delete '.git/worktrees/master-compare': Permission denied
error: failed to delete '.git/worktrees/pr465': Permission denied
error: failed to delete '.git/worktrees/pr465-current-merge': Permission denied
error: failed to delete '.git/worktrees/pr465-merge': Permission denied
error: failed to delete '.git/worktrees/pr467': Permission denied
error: failed to delete '.git/worktrees/pr469': Permission denied
error: failed to delete '.git/worktrees/wt-465fix': Permission denied
error: failed to delete '.git/worktrees/wt-a1-runlog': Permission denied
error: failed to delete '.git/worktrees/wt-claudemd-r25': Permission denied
error: failed to delete '.git/worktrees/wt-cloudrun-cancel': Permission denied
error: failed to delete '.git/worktrees/wt-cloudrun-dispatch': Permission denied
error: failed to delete '.git/worktrees/wt-cp-rebuild-spec': Permission denied
error: failed to delete '.git/worktrees/wt-dockerfile-auth': Permission denied
error: failed to delete '.git/worktrees/wt-docs': Permission denied
error: failed to delete '.git/worktrees/wt-docs-reconcile': Permission denied
error: failed to delete '.git/worktrees/wt-drain-concurrency': Permission denied
error: failed to delete '.git/worktrees/wt-ffmpeg-timeouts': Permission denied
error: failed to delete '.git/worktrees/wt-gdrive-api-safety': Permission denied
error: failed to delete '.git/worktrees/wt-idempotent-process': Permission denied
error: failed to delete '.git/worktrees/wt-input-jail': Permission denied
error: failed to delete '.git/worktrees/wt-output-dir-config': Permission denied
error: failed to delete '.git/worktrees/wt-p1-genyaml': Permission denied
error: failed to delete '.git/worktrees/wt-p2-upload-stream': Permission denied
error: failed to delete '.git/worktrees/wt-p4-md5cache': Permission denied
error: failed to delete '.git/worktrees/wt-p5a-pollknob': Permission denied
error: failed to delete '.git/worktrees/wt-p6-heartbeat': Permission denied
error: failed to delete '.git/worktrees/wt-pr481-review': Permission denied
error: failed to delete '.git/worktrees/wt-retention-sweep': Permission denied
error: failed to delete '.git/worktrees/wt-review-492-mergecheck': Permission denied
error: failed to delete '.git/worktrees/wt-review-493-mergecheck': Permission denied
error: failed to delete '.git/worktrees/wt-scratch-reclaim': Permission denied
error: failed to delete '.git/worktrees/wt-signed-download': Permission denied
error: failed to delete '.git/worktrees/wt-source-cache': Permission denied
error: failed to delete '.git/worktrees/wt-upload-dir': Permission denied
error: failed to delete '.git/worktrees/wt-upload-staging': Permission denied
worktrees remaining (mine should be gone):
  mine: 0
```

287. assistant (2026-07-07T16:19:34.602Z)

Full go-live complete and verified end-to-end. The live `/api/capabilities` confirms the deployment state exactly matches the fix's premise:
- `compute_target: cloud_run`, `cloud_available: True` ? the served SPA now defaults the picker to **cloud**
- `wasb_weights_present: False` ? the #511 guardrail's condition holds, so a forced `compute=local` now fails fast with the actionable message instead of the cryptic "weights not found"

(My worktrees are all cleaned up; the `prune` permission errors are other sessions' worktrees, not mine.) Let me record the live deployment state + rollback lever in memory:

288. user (2026-07-07T16:19:37.470Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

289. user (2026-07-07T16:19:53.655Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\compute-local-hosted-guardrail.md has been updated successfully. (file state is current in your context ? no need to Read it back)

290. user (2026-07-07T16:19:58.373Z)

10   - [compute=local hosted guardrail](compute-local-hosted-guardrail.md) ? the "weights not found" trap fix, both **MERGED** (backend #511 `ec2d080`, SPA #169 `ba5f457`). LIVE still needs SPA re-vendor into `backend/ui/` + a control-plane redeploy.

291. user (2026-07-07T16:20:04.316Z)

The file C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\memory\MEMORY.md has been updated successfully. (file state is current in your context ? no need to Read it back)

292. assistant (2026-07-07T16:20:25.369Z)

? **Full go-live complete ? the fix is now live in production.**

What shipped
The confusing `compute="local"` ? "WASB weights not found" failure on the hosted control-plane is fixed end-to-end and deployed:

Change	PR	Merged	Effect
Backend fail-fast guardrail	[#511](https://github.com/Khelsutra/badminton-highlight-indexer/pull/511)	`ec2d080`	`compute=local` for a weights-backed detector on a cloud control-plane now returns *"can't run detection locally ? Choose 'Run in cloud'"* instead of the cryptic deep failure
SPA picker default?cloud	[avidullu/khelsutra#169](https://github.com/avidullu/khelsutra/pull/169)	`ba5f457`	Hosted build no longer sends `compute=local` by default
Re-vendor SPA into `backend/ui/`	[#516](https://github.com/Khelsutra/badminton-highlight-indexer/pull/516)	`f5c9dd9`	Puts the merged SPA into the engine's served bundle

Deployment
- **Image** rebuilt from master `f5c9dd9`: `rally-worker@sha256:8045f54e?` (Cloud Build `17ab46b4`, 12m48s). Bundles #511 + the re-vendored SPA + the sibling-merged #512/#513.
- **`rally-control-plane`** ? new revision **`rally-control-plane-00018-qvz`**, 100% traffic, healthy boot (`cloud-serving profile ACTIVE`, uvicorn up). **`rally-worker`** job cut to the same digest.

Live verification
- `/api/version` ? `ui.spa_commit = ba5f457?` ? the deployed control-plane serves the fixed SPA.
- `/api/capabilities` ? `compute_target=cloud_run, cloud_available=true, wasb_weights_present=false` ? exactly the state where the guardrail fires, confirming the fix's premise holds in prod.
- *Not run:* a live authenticated `POST /api/process compute=local` (needs a CF Access JWT) ? the guardrail is unit-tested and baked into the running image, so it's live regardless.

Rollback (one command)
```
gcloud run services update-traffic rally-control-plane --region asia-southeast1 \
  --to-revisions rally-control-plane-00017-qbd=100
```
(prior revision, image `sha256:09d2edf7?`; worker job rollback recorded in memory.)

Along the way
- Adversarial multi-agent review: **0 confirmed / 4 refuted** (added tracknet/fusion tests for the one substantive nit).
- Navigated a heavily contended shared checkout ? every git write went through throwaway worktrees, so sibling sessions' work was never disturbed. Rebased #511 twice (onto #513, then #512) conflict-free.
- Memory + index updated to the live state with the rollback lever.

One follow-up worth noting: sibling **[#515](https://github.com/Khelsutra/badminton-highlight-indexer/pull/515)** (`fix/cloud-run-exec-status-fastfail`) is still open ? it's **not** in this deployed image, so it'll ride the next control-plane build.